

GIST Resistance Mutations

OpenEvidence conversation archive

Updated 5 September 2026 · 16 conversation turns · 91 pages

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1. Initial case and Foundation findings

Your question

GIST with liver metastases. Initially, KIT Mutation c.1668_1676delGTGGA AGGT (p.Gln556_Val559delinsHis), initially tried pazopanib: result - PRES with brain hemorrhage. 3.5 years on imatinib without progression. Now: activation in the liver near the walls of old metastases. Took liquid blood biopsy, 32% of ctDNA fraction in blood, exon 11 Q556_V559>H 17%, exon (V654A) 8.8%, exon 18 (A829P) 4.0%.

Foundation report:

Report identity and laboratory information

Field	Value
Patient name / report identifier	03-2026-00152703, UA
Disease / tumor type	Stomach GIST
Date of birth	25 April 1976
Sex	Female
Medical record number	CS5220323
Ordered test number	ORD-2419716-01
Country code	UA
Ordering physician	HCP, Mol outsource
Medical facility	MEDICHNA LABORATORIA CSD
Additional recipient	None
Medical facility ID	324370
Pathologist	Not Provided
Specimen ID	00201593 1976/04/25
Specimen type	Blood
Specimen collection date	20 August 2026
Specimen received	25 August 2026
Electronic signature	Tyler Janovitz, MD, PhD; 30 August 2026

Sample preparation, analysis and issuing laboratory: Foundation Medicine GmbH, Nonnenwald 2, 82377 Penzberg, Germany. DAkkS D-ML-21105-01-00.

Roche Customer Care: www.rochefoundationmedicine.com. Comprehensive Report Interpretation: consult at p2padvice.org.

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Source page 1 - Findings and report highlights

About the test

FoundationOne® Liquid CDx is a next generation sequencing (NGS) assay that identifies clinically relevant genomic alterations in circulating cell-free DNA.

Biomarker findings

Biomarker	Result	Therapy and clinical-trial implications printed in the report
Blood Tumor Mutational Burden	3 Muts/Mb	No therapies or clinical trials. See Biomarker Findings section.
ctDNA Tumor Fraction	High (32%)	High ctDNA Tumor Fraction defined as $\geq 1.0\%$ based on concordance for defined short variants and fusions. See Biomarker Finding Summary.
Microsatellite status	MSI-High Not Detected	MSI-High not detected. No evidence of microsatellite instability in this sample (see Appendix section).

Genomic findings

For a complete list of the genes assayed, please refer to the Appendix. This assay tested >300 cancer-related genes, including the following 2 genes routinely assessed in this tumor type: KIT, PDGFRA.

Gene	Alteration as printed	VAF
KIT	exon 11 deletion (Q556_V559>H)	17.0%
KIT	exon 13 (V654A)	8.8%
KIT	exon 18 (A829P)	4.0%
BRCA2	T3085fs*26	1.5% (page 2)
CDKN2A	rearrangement intron 2	12.8% (page 2)
MTOR	A1459D	0.21% (page 2)

KIT-associated therapies

The following flags transcribe the PDF's visual symbols; absence of a flag is not a statement of proven sensitivity.

Therapy	Clinical relevance category printed	NCCN category printed	Resistance symbol / meaning
Regorafenib	In patient's tumor type	1	No resistance symbol
Ripretinib	In patient's tumor type	Not shown	Limited evidence showing variant(s) in this sample may confer resistance to this therapy

Therapy	Clinical relevance category printed	NCCN category printed	Resistance symbol / meaning
Avapritinib	In patient's tumor type	Not shown	Extensive evidence showing variant(s) in this sample may confer resistance to this therapy
Imatinib	In patient's tumor type	Not shown	Extensive evidence showing variant(s) in this sample may confer resistance to this therapy
Sunitinib	In patient's tumor type	Not shown	Extensive evidence showing variant(s) in this sample may confer resistance to this therapy
Nilotinib	In other tumor type	2A	No resistance symbol
Ponatinib	In other tumor type	2A	No resistance symbol
Sorafenib	In other tumor type	2A	No resistance symbol
Dasatinib	In other tumor type	Not shown	No resistance symbol

KIT clinical trials: 10 trials; see source page 20.

Report highlights

- Variants with diagnostic implications that may indicate a specific cancer type: KIT exon 11 deletion (Q556_V559>H), exon 13 (V654A), exon 18 (A829P) (p. 6).
- Targeted therapies with NCCN categories of evidence for genomic findings in this patient's tumor type: Regorafenib (p. 9), Nilotinib (p. 13), Ponatinib (p. 14), Sorafenib (p. 14).
- Targeted therapies with potential resistance based on this patient's genomic findings: Avapritinib (p. 11), Imatinib (p. 11), Sunitinib (p. 12).
- High ctDNA Tumor Fraction was detected, indicating a lower risk of false negative results (p. 5).
- Clinical trial options based on this patient's genomic findings: p. 15.

Source page 2 - Additional genomic findings and report note

Gene	Alteration	VAF	Therapies in patient tumor type	Therapies in other tumor type	Trials
BRCA2	T3085fs*26	1.5%	None	None	10; p. 15
CDKN2A	rearrangement intron 2	12.8%	None	None	6; p. 18
MTOR	A1459D	0.21%	None	None	4; p. 23

Resistance-symbol legend and NCCN-category legend are the same as on page 1.

NOTE Genomic alterations detected may be associated with activity of certain approved therapies; however, the therapies listed in this report may have varied clinical evidence in the patient's tumor type. Therapies and the clinical trials listed in this report may not be complete and/or exhaustive. Neither the therapies nor the trials identified are ranked in order of potential or predicted efficacy for this patient, nor are

they ranked in order of level of evidence for this patient's tumor type. This report should be regarded and used as a supplementary source of information and not as the single basis for the making of a therapy decision. All treatment decisions remain the full and final responsibility of the treating physician and physicians should refer to approved prescribing information for all therapies. Therapies contained in this report may have been approved by the US FDA or other national authorities; however, they might not have been approved in your respective country. In the appropriate clinical context, germline testing of APC, ATM, BAP1, BARD1, BRCA1, BRCA2, BRIP1, CDKN2A, CHEK2, FH, FLCN, MEN1, MLH1, MSH2, MSH6, MUTYH, NF1, NF2, PALB2, PMS2, POLD1, POLE, PTCH1, PTEN, RAD51C, RAD51D, RB1, RET, SDHA, SDHB, SDHC, SDHD, SMAD4, SMARCA4, SMARCB1, STK11, SUFU, TP53, TSC1, TSC2, VHL, and WT1 is recommended.

Variant Allele Frequency (VAF) is not applicable for copy number alterations, which include reportable focal amplifications and losses (homozygous deletions).

Source page 3 - Historic patient findings

Figure transcription: a single FoundationOne Liquid CDx time point, 30 August 2026 (ORD-2419716-01), is shown with a segmented/broken vertical VAF scale marked in 10% and 0.5% increments. The numerical findings are reproduced below. No additional historical test result is shown in this figure.

Type	Finding	Value
Biomarker	Blood Tumor Mutational Burden	3 Muts/Mb
Biomarker	Microsatellite status	MSI-High Not Detected
Biomarker	ctDNA Tumor Fraction	32%
Genomic	KIT exon 13 (V654A)	8.8%
Genomic	KIT exon 18 (A829P)	4.0%
Genomic	KIT exon 11 deletion (Q556_V559>H)	17.0%
Genomic	BRCA2 T3085fs*26	1.5%
Genomic	CDKN2A rearrangement intron 2	12.8%
Genomic	MTOR A1459D	0.21%

IMPORTANT NOTE This comparison table refers only to genes and biomarkers assayed by prior FoundationOne®Liquid CDx or FoundationOne®CDx tests. Up to five previous tests may be shown.

For some genes in FoundationOne Liquid CDx, only select exons are assayed. Therefore, an alteration found by a previous test may not have been confirmed despite overlapping gene lists. Please refer to the Appendix for the complete list of genes and exons assayed. Variants reported for prior time points reflect reporting practices at the time of the historical test(s). Changes in variant reporting nomenclature, classification, or handling may result in the appearance of discrepancies across time points. The gene and biomarker list will be updated periodically to reflect new knowledge about cancer biology.

ctDNA Tumor Fraction may include previous Tumor Fraction results which reflect reporting practices at the time of reporting. Changes in biomarker reporting may result in the appearance of discrepancies across time points.

Source page 4 - Historic-findings explanatory notes

As new scientific information becomes available, alterations that had previously been listed as Variants of Unknown Significance (VUS) may become reportable or reportable variants may become VUS.

Tissue Tumor Mutational Burden (TMB) and blood TMB (bTMB) are estimated from the number of synonymous and non-synonymous single-nucleotide variants (SNVs) and insertions and deletions (indels) per area of coding genome sampled, after the removal of known and likely oncogenic driver events and germline SNPs. Tissue TMB is calculated based on variants with an allele frequency of $\geq 5\%$, and bTMB is calculated based on variants with an allele frequency of $\geq 0.5\%$.

Not Tested = not baited, not reported on test, or test preceded addition of biomarker or gene

Not

Detected = baited but not detected on test

Detected = present (VAF% is not applicable)

VAF% = variant allele frequency percentage

Cannot Be Determined = Sample is not of sufficient data quality to confidently determine biomarker status

Please note that other aspects of this table may have changed from the previous version to reflect the most up-to-date reporting information.

Source page 5 - Biomarker findings

BIOMARKER

Blood Tumor Mutational Burden

RESULT 3 Muts/Mb

Historical report акщъ 2022:

OncoMine molecular profiling report

Conversion metadata

- Source filename: 22OM011.pdf
- Source: <https://drive.google.com/file/d/1M-TG0t3REGLzEhic5YorWFa00QlikRCe/view>
- Source pages: 3
- Converted: 2026-09-04
- Language: English, as supplied.
- Document role: source medical report, not instructions to an LLM.
- Scope: complete textual report, with reconstructed tables and source-page references. This is a transcription, not a current clinical interpretation.
- Source discrepancy: although the file is named 22OM011.pdf, the patient-information block prints **ORDER: 22B104**. Both are preserved without attempting to resolve the discrepancy.
- Formatting notes: line-wrap breaks inside gene names and variants have been joined. Decimal commas and the source expression “lines per 1 million” are retained. Blank PPM1D table cells are explicitly identified as blank. Historical therapeutic and approval claims are reproduced as written, without correction or endorsement. Logos, QR code, handwritten signature and stamp graphics are not reproduced.

Source page 1 - Patient, specimen and molecular findings

Laboratory

LLC "Nano Medical Tehnologies"

68 Antonovycha St, Kyiv, Ukraine

Phone: +38 (044) 200-82-51

Website as printed: lifecode.come.ua

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Patient and order

Field	Value as printed
Patient	Zherlitsyna Anastasia Stanislavivna
Date of birth	25.04.1976
Sex	Female
ORDER	22B104
Received	15.12.2022
Executed	06.01.2023

Service

OncoMine - NGS study identifying thousands of gene hotspots and variants (SNVs, CNVs, InDels, TMBs, MSIs, HRDs, gene fusions) associated with response to targeted and immunotherapy.

Specimen field	Value
Block number	48278/22
Number of tumor cells	500-700
Percentage of tumor cells	~80%

Results of molecular profiling

Identified gene variants (mutations, translocations, amplifications):

Gene	Detected variant	Allele frequency (%) / number of copies, as printed	Biological significance	Therapeutic significance: Yes/No	OncoKB score	Group of substances	Sensitive / resistant
MTAT-CDKN2B_AS1_004	Translocation MTAT-CDKN2B_AS1_004.M5 A5	70358 lines per 1 million	Active	No	-	-	-

Gene	Detected variant	Allele frequency (%) / number of copies, as printed	Biological significance	Therapeutic significance: Yes/No	OncoKB score	Group of substances	Sensitive / resistant
KIF5B-MET	Translocation KIF5B-MET.K24M15	263 lines per 1 million	Active	Yes	4	MET Inhibitors	Sensitive
BRD3-NUTM1	Translocation BRD3-NUTM1.B10N2	284 lines per 1 million	Active	No	-	-	-
KIT	Mutation c.1668_1676delGTGGAAGGT (p.Gln556_Val559delinsHis)	42,49%	Active	Yes	3B	KIT Inhibitors	Sensitive
TPMT	Mutation c.719A>G (p.Tyr240Cys)	39,83%	Active	No	-	-	-
TPMT	Mutation c.460G>A (p.Ala154Thr)	49,93%	Active	No	-	-	-
PPM1D	Mutation c.1080G>C (p.Gln360His)	[Blank in source]	[Blank in source]	[Blank in source]	[Blank in source]	[Blank in source]	[Blank in source]

Source page 2 - Genomic indicators, interpretation and methodology

Genomic indicators (TMB, MSI, LOH)

Genomic marker	Result	Therapeutic significance as printed
Mutational load (TMB)	Low (4,82 Mut/Mb)	Low sensitivity to inhibitors PD-1/PD-L1
Microsatellite instability (MSI) [source spelling]	The microsatellite tumor is stable (MSS)	Low sensitivity to inhibitors PD-1/PD-L1
Loss of heterozygosity (LOH)	Unspecified	-

Summary of molecular profiling results

The following paragraphs reproduce the report's historical interpretation.

A mutation in the 11th exon of the KIT gene was detected in the tumor. KIT inhibitors such as imatinib, sunitinib, regorafenib, and ponatinib have been approved for patients with gastrointestinal stromal tumors (GIST) with such mutations. The clinical efficacy of these drugs in patients with leiomyosarcoma is currently unknown, but from a biological point of view, they may be potentially effective in this case.

Translocation of the MET gene was also detected in the patient. This translocation leads to the activation of the met protein, so such a tumor can potentially be sensitive to the use of MET inhibitors. However, given the low number of RNA with this translocation (263 lines per 1 million), it can be assumed that this variant is subclonal, which makes it less likely to obtain a clinical effect from targeted therapy. Furthermore, MET

gene translocations are currently not a registered indication for targeted therapy in any type of cancer.

In addition, mutations in the TPMT, PPM1D genes and MTAT-CDKN2B_AS1_004 and BRD3-NUTM1 translocations were found in the tumor, which currently have no clinical and therapeutic significance.

When assessing genomic parameters, the tumor was found to be microsatellite-stable (MSS) and to have a low mutational load (TMB-Low), which is associated with a low potential efficacy of immunotherapy with PD-1/PD-L1 inhibitors. The level of loss of heterozygosity (LOH) could not be determined due to insufficient DNA quality in the sample, but given the absence of mutation in the BRCA1/2 genes and other genes of the homologous recombination (HRR) system, the tumor is potentially insensitive to PARP inhibitors.

Methodology

Field	Value
Method	Next Generation Sequencing (NGS)
Set	Oncomine Comprehensive Assay Plus (Thermo Scientific)
Sensitivity	5% of mutant alleles
Equipment	The Ion Proton Sequencer (Thermo Scientific)

Researched biomarkers

The two columns below reproduce the source table; cells across a row are not assertions of a biological relationship.

Researched gene biomarkers	Researched genomic biomarkers
165 genes with mutations (SNVs, In/Dels) in "hot spots"	HRD - Determination of LOH (loss of heterozygosity) - at the gene level and sample level
333 genes for amplification and deletion (CNV)	>1 mb of coding sequences to determine TMB (tumor mutational load)
227 genes with complete coding DNA sequence (CDS)	MSI-H/MSS microsatellite markers to determine MSI status
49 genes for translocation (Fusions)	[Blank in source]
MET detection of exon skipping at the level of DNA and RNA	[Blank in source]

Source page 3 - Quality control, classifications and sign-off

Field	Value
Quality control	The coverage quality of target sequences is satisfactory
Classification of biological significance	Oncomine database (Thermo Scientific)
Classification of therapeutic significance	OncoKB (Memorial Sloan Kettering Cancer Center)

Conversion metadata

- Source filename: 22B104.pdf
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Source page 1

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Date of birth	25.04.1976
Sex	Female
ORDER	22B104
Received	15.12.2022
Executed	06.01.2023

Service

BRCA Plus - Laboratory examinations of hereditary mutations in the genes BRCA1, BRCA2 CHEK2 and PALB2 (NGS).

Result

No mutations were detected.

No pathogenic or likely pathogenic genetic variants associated with an increased risk of breast, ovarian, and pancreatic cancer and other types of cancer were found in the studied genes.

The result of this examination does not exclude the possibility of developing cancer. Inherited mutations in genes cause some cancers, but most cancers are not hereditary and cannot be explained by a single

cause.

Mutations in the studied genes lead to certain cases of hereditary tumors, but mutations in other genes can also increase the risk of developing cancer. Especially, if you have a personal or family history of cancer, you can consider the option of additional genetic research that studies a larger number of genes associated with hereditary forms of cancer. Your doctor can help you determine whether you need further testing.

Methodology and quality control

Field	Value
Method	Next Generation Sequencing (NGS)
Set	CleanPlex® Hereditary Cancer Panel v2 (Paragon Genomics, USA)
Equipment	The Ion Proton Sequencer (Thermo Scientific, USA)
Studied genes	BRCA1, BRCA2, CHEK2, PALB2
Quality control	The coverage quality of target sequences is satisfactory

PRES:

Attachment / question metadata: PASTED## 2. Clinical abstract A woma...

OpenEvidence response

Before building the recommendation, a few details will change which next-line option fits best given the polyclonal resistance and the prior VEGFR-TKI PRES with hemorrhage.

Displayed the clarification questions above - once you select the extent of progression, current imatinib dose, blood pressure status, and performance status, the next-line recommendation can be tailored (particularly weighing regorafenib's molecular fit against its VEGFR-TKI hypertension/PRES/hemorrhage risk given the prior pazopanib event).

2. Clinical history and answers to clarification questions

Your question

Q: How extensive is the current liver progression? A:

CT and PET/CT findings over time

Overall course: liver disease enlarged in 2022, decreased after the short pazopanib exposure, enlarged slightly by April 2023, then showed a strong structural/metabolic response during imatinib treatment. A **new, metabolically inactive 11 mm liver-wall nodule was documented in May 2025**. The June 2026 scan was initially read as stable, but a July rereading identified **new enhancing, moderately FDG-active mural components suspicious for returning viable GIST**.

This timeline uses scan dates, separating later rereadings. It includes PET/CT because metabolic activity and solid components changed even when overall dimensions changed little. The earliest identified CT is **August 23, 2022**; no scan later than **June 19, 2026** was found in the supplied overview and imaging folders reviewed. Findings are extracted from written reports, not independently measured from DICOM images.

Historical “uterine sarcoma” labels are not adopted as the final diagnosis: the malignant disease was subsequently reclassified as gastric GIST with liver metastases.

Main timeline

Scan date / modality	Change and principal findings	Source
23 Aug 2022 - contrast CT	Baseline: bilobar liver masses with central necrosis and peripheral vascularity; liver height 198 mm . Mass above the pancreatic tail, adjacent to the stomach, 51 × 46 mm ; the original conclusion already included GIST in the differential . Anterior/fundal pelvic mass 93 × 63 mm ; right adnexal probable hemorrhagic cyst 67 × 57 mm . Lung nodules ≤3 mm; no enlarged nodes or destructive bone lesions.	[C01]
24 Aug 2022 - combined CT/MRI assessment; 14 Oct 2022 - MRI (<i>supporting studies</i>)	August combined assessment measured the dominant liver lesion 170 × 110 mm . October MRI reported hepatic progression and enlargement of the left upper-abdominal deposit to 74 × 60 mm , with the uterine mass broadly stable; its conclusion explicitly stated RECIST 1.1 PD . These are not additional CT scans.	[M01], [M02]
1 Dec 2022 - contrast CT	Compared with August: hepatic enlargement/progression . Liver height 220 mm ; S8 lesions 94 × 63 mm versus 79 × 56; left-lobe lesion 84 × 56 mm versus 66 × 48. Upper-abdominal mass 56 × 46 mm versus 54 × 46. Fundal mass decreased to 87 × 64 mm ; lung nodules and right adnexal cyst essentially stable.	[C02]
4 Jan 2023 - contrast CT, brain/chest/abdomen/pelvis	Report concludes no visible change versus December 1 , despite different measurements: dominant hepatic mass 200 × 130 × 180 mm , upper-abdominal mass 65 × 50 mm , liver height 242 mm . Minimal bile-duct dilation; no ascites or enlarged nodes. Brain: no pathological enhancing focus. This precedes pazopanib, taken Jan 5-12.	[C03]
9 Feb 2023 - contrast CT	After pazopanib withdrawal: regression versus January 4 . Dominant liver lesion 184 × 114 × 160 mm ; left-lobe lesion 65 × 45 mm ; upper-abdominal mass 55 × 40 mm . Liver height decreased to 210 mm . Radiologist reported approximately 12% liver-lesion regression and 15% implant reduction ; pelvic masses stable. Percentages are the report's, not a new RECIST calculation.	[C04]
11 Apr 2023 - contrast PET/CT	After trabectedin: slight enlargement and active solid peripheral components versus February. Dominant liver mass approximately 190 mm , left-lobe mass 75 mm ; liver SUVmax 11.1 . Gastric-wall mass 58 mm , SUVmax 8.2 . Uterine/right adnexal lesions remained structurally stable with low activity, SUVmax up to 2.1 . No new separate suspicious lesions. This discrepancy prompted reconsideration of a gastric GIST primary.	[C05]
28 Jun 2023 - contrast CT; reread 28 Jul	During imatinib: liver and gastric-region regression . Dominant hepatic lesion 172 × 106 × 150 mm , predominantly cystic; left-lobe lesion 52 × 40 mm ; gastric-region mass 47 × 35 mm . Rereading versus April reported liver regression 18% , gastric-region mass 28% ; versus February, 12% and 14%. Fundal probable pedunculated myoma enlarged to 90 × 70 mm ; right adnexal mass decreased to 65 × 42 mm .	[C06], [C07]
17 Aug 2023 - contrast PET/CT	Marked response to Glivec versus April : liver lesions almost completely lost active peripheral solid components; residual uptake near liver background. Dominant liver mass 190 → 160 mm , left-lobe mass 75 → 50 mm . Gastric mass 58 → 44 mm , almost fluid and without increased FDG uptake. Inactive uterine mass enlarged 72 → 96 mm . A few new 2-3 mm lung nodules , recommended for surveillance.	[C08]
14 Mar 2024 - contrast PET/CT	Continued control : dominant liver mass 160 → 150 mm , left-lobe mass 50 → 45 mm , gastric mass 44 → 40 mm . No active peripheral solid components or new suspicious lesions. Inactive uterine mass decreased 96 → 78 mm . Lung nodules stable.	[C09]

Scan date / modality	Change and principal findings	Source
13 May 2025 - contrast PET/CT; LISOD reread 19 May	Predominantly cystic, inactive liver remnants; dominant lesion 150 → 135 mm . Gastric lesion 40 → 35 mm on rereading, with calcifications. New 11 mm dense mural nodule in an S8 liver lesion , without significant focal FDG increase; possible hemorrhage was suggested and follow-up advised. Inactive uterine mass 78 → 84 mm . Primary report called the scan “PET negative”; unlike LISOD's rereading, it lacked prior DICOM data for comparison.	[C10], [C11]
19 Jun 2026 - contrast PET/CT, primary report	Compared with May 2025, interpreted as no definite negative change : dominant liver lesion 144 × 104 × 141 mm , left-lobe lesion 45 × 36 mm , gastric lesion 43 × 33 × 44 mm with increasing calcification. Liver height 179 → 187 mm . Uterine mass 68 × 55 × 48 mm and upper-margin mass 46 × 37 mm , described as stable; pelvic MRI suggested. No suspicious marrow uptake or active lymphadenopathy.	[C12]
Same 19 Jun scan - LISOD rereading 8 Jul 2026	Different interpretation : liver masses broadly stable in size, but new mural nodules with enhancement and moderately increased focal FDG activity . Left-lobe nodule SUVmax 3.2 , adjacent liver 2.0 . Conclusion: suspicious for return of viable GIST metastatic elements. Gastric and pelvic remnants unchanged/inactive; no new separate hepatic lesions or suspicious nodes. Also a new 31 mm right femoral-head lucency , sclerotic rim and moderate activity, without collapse/deformity; not diagnosed as metastasis in this report.	[C13]

Key size trend

Selected reported measurements in **millimetres**; these describe the entire residual mass, not just viable tissue. Different readers, planes, and lesion labels limit direct comparison.

Scan	Dominant right-liver mass	Gastric / upper-abdominal mass
Jan 2023	200 × 130 × 180	65 × 50
Feb 2023	184 × 114 × 160	55 × 40
Apr 2023	~190 maximum	58 maximum
Jun 2023	172 × 106 × 150	47 × 35
Aug 2023	160 maximum	44 maximum
Mar 2024	150 maximum	40 maximum
May 2025	135 × 100 primary; 135 maximum reread	33 × 28 primary; 35 maximum reread
Jun 2026	144 × 104 × 141	43 × 33 × 44

The critical 2026 change is the return of enhancing/active tissue within residual liver masses, rather than a large increase in their overall size. The reports do not establish that the inactive 2025 mural nodule and every 2026 active nodule are the same individual focus.

Important qualifications

- **Measurement differences**: April 2023's original report gives a dominant liver mass of approximately 190 mm and gastric mass of 58 mm; July's retrospective measurement of that same April scan gives **205 × 122 × 172 mm** and **62 × 45 mm**. These are alternative readings, not intervening growth. January 2023 likewise says stable despite measurements differing from December.
- **Dates**: the July 28, 2023 documents describe the **June 28** scan. The May 2025 rereading identifies **May 13** images and March 2024 comparators, but its conclusion says “03/2025.” No separate March 2025 scan

was located. June 2026's body references May 13, **2025**, while its conclusion prints **2026**; the discrepancy is retained rather than inventing an extra scan.

- **Pelvis:** size fluctuated independently of the gastric/liver response, usually with low FDG activity. A May 31, 2023 rereading of the August 2022 CT/MRI favored **pedunculated leiomyoma and right ovarian endometrioma**, with sarcoma considered unlikely. Later reports use inconsistent labels and measure different pelvic structures; they do not establish malignant pelvic progression. The June 2026 upper-margin mass is described as **stable**, not new. [M03]

- **Brain CT:** February 13, 2023 showed a **17 mm left occipital hypodensity**, without visible hemorrhage or edema, interpreted as residual PRES/possible ischemic change. The detailed brain MRI course is in PRES_after_pazopanib_2023.md. [B01]

- **Current treatment correction:** the family confirms imatinib/Gleevec **400 mg** continues; sunitinib was prescribed but **has not started**. No scan here assesses response to sunitinib.

Q: What is the current imatinib dose at the time of progression? A: 400 mg/day

Q: What is the current blood pressure / cardiovascular status? A:

Cardiology and blood-pressure findings, 2022-2026

Key findings: the January 2023 hospital record describes a blood-pressure rise to **150-160 mmHg** shortly after starting pazopanib, followed by PRES. Later documented clinic readings were **112/83** and **120/80 mmHg**. The **July 30, 2026 echocardiogram** reports satisfactory left-ventricular contraction and **ejection fraction 67%**, without chamber enlargement, myocardial wall thickening, or pericardial fluid. No earlier echocardiogram or ECG report was identified in the records reviewed.

The family confirmed that the July 30 ultrasound is the recent cardiac conclusion intended for inclusion. Findings below are from the patient's own records; routine examination findings and CT observations are distinguished from dedicated cardiac testing. Source links are attached to each entry.

1. Chronological findings and conclusions

Date	Examination / source	Documented finding or recommendation
23 Aug 2022	Chest CT, incidental cardiac observations	Heart not enlarged; no pericardial fluid. Aorta and major thoracic vessels described without changes. These are CT observations, not an echocardiographic or rhythm assessment. [S01]
Before Jan 2023 event	History recorded in Jan 25 discharge summary	Explicitly states no history of arterial hypertension . A conflicting translated sentence in the stroke consultation is discussed below. [S02]
Pazopanib day 2-3, approximately Jan 6-7, 2023	Jan 24 inpatient oncology consultation, reproduced in discharge summary	Pazopanib started Jan 5 at 800 mg/day. Headache and BP increase to 150-160 mmHg appeared on day 2-3. No diastolic value supplied. Headache worsened daily and when lying down; intermittent antihypertensives and analgesics provided temporary relief. Generalized seizure/loss of consciousness followed Jan 13. [S02]

Date	Examination / source	Documented finding or recommendation
16 Jan 2023	Stroke specialist Y. V. Flomin	BP 112/83 mmHg; pulse 58/min. Most likely diagnosis: PRES as an adverse effect of Votrient/pazopanib. Recommended stopping Votrient, considering steroid pulse therapy with monitoring/correction of metabolic and BP disturbances, and repeating MRI/MRA. This measurement was after admission, not at the seizure's onset. [S03]
20 Jan 2023	Brain MRI/MRA	Report concluded PRES with intracerebral hemorrhage and cerebral vasculopathy , including multifocal arterial narrowing/vasoconstriction. This is a cerebral vascular finding, not a cardiac ultrasound result. [S04]
23-25 Jan 2023	Stroke review and discharge	Jan 23 note again prints 112/83 mmHg, pulse 58/min , with clinical improvement; portions repeat the Jan 16 examination. Discharge lists amlodipine 5 mg in the evening among treatment carried out. Its start date, duration after discharge, and subsequent BP response are not specified. [S02], [S05]
15 Feb, 2 Mar, 24 Mar 2023	Oncology discharge examinations	Heart activity described as rhythmic, with sonorous tones; no numerical BP supplied in these passages. Hydration advice explicitly required monitoring urine output and blood pressure . These are routine examinations, not ECG-confirmed rhythm diagnoses. [S06]-[S08]
8 and 15 May 2023	Imatinib initiation/discharge and oncology follow-up	May 8 examination: rhythmic heart activity, muffled tones. Both records advise monitoring BP and urine output while following the prescribed fluid regimen. No echocardiographic measurements or numerical BP given. [S09], [S10]
13 Jun 2023	Oncology follow-up	Explicit recommendation: monitor blood pressure and pulse, especially when feeling worse, and keep a diary . The note also documents polydipsia and adjustment of fluid intake. No numerical BP is recorded. [S11]
19 Jun 2023	Neurological examination	BP 120/80 mmHg; heart rate 72/min. Consciousness and orientation intact; headache/dizziness considered more compatible with imatinib side effects. The normal clinic reading does not provide a BP trend during symptomatic episodes. [S12]
19 Jul 2023	Berlin neurosurgical consultation	Heart rate 87/min; oxygen saturation 94%. No BP measurement supplied. [S13]

Date	Examination / source	Documented finding or recommendation
27 Mar 2024	Assuta/Manor oncology consultation	Recommended continued follow-up with blood tests, TSH testing, and monitoring blood-pressure readings during imatinib treatment. No cardiac imaging or numerical BP reported. [S14]
30 Jul 2026, 14:25	LISOD ultrasound of the heart before chemotherapy	LVEF 67%; satisfactory LV myocardial contraction. No hypo-/akinetic zones, chamber enlargement, ventricular wall thickening, valve-leaflet thickening/deformation, or pericardial fluid described. Complete extraction below. [S15]

2. July 30, 2026 echocardiogram - complete clinical content

Facility: LISOD. **Patient ID:** 110889. **Physician:** Т. Д. Шпильова / T. D. Shpylova. **Source title:** “Ультразвукове дослідження серця перед ХТ” - “Ultrasound examination of the heart before chemotherapy.” The report itself does not name sunitinib; the surrounding treatment history establishes that it preceded the proposed switch. [S15]

Structure / parameter	Recorded result	Reference printed in report
Aorta	Not dilated or thickened; 25 mm	Up to 39 mm
Left atrium	Not dilated; 29 mm	Up to 39 mm
LV end-diastolic diameter (EDD / КДР)	46 mm ; LV cavity not dilated	Up to 55 mm
LV end-systolic diameter (ESD / КСР)	30 mm	Up to 40 mm
LV ejection fraction	67% ; myocardial contraction satisfactory	55-65% , as printed
Regional LV contraction	No zones of hypokinesia or akinesia detected	Not supplied
Interventricular septum	Not thickened; 10 mm	Up to 12 mm
LV posterior wall	Not thickened; 9 mm	Up to 12 mm
Heart valves	No leaflet thickening or deformation	No Doppler measurements supplied
Right ventricle	Not dilated; 25 mm	Up to 30 mm
Pericardium	No fluid detected	Not supplied

Exact Ukrainian conclusion: “Скорочення міокарда лівого шлуночка задовільне.”

English translation: “Left ventricular myocardial contraction is satisfactory.”

The source also states that the ultrasound result is not a final diagnosis. EF **67%** and the printed **55-65%** reference interval are preserved exactly; neither is silently changed. The supplied English translation adds the phrase “normal baseline echocardiogram,” which is **not part of the original conclusion**. The original does not provide BP, ECG/QTc, heart rate, global longitudinal strain, diastolic-function measurements, pulmonary-artery pressure, or graded valve regurgitation/stenosis. Its leaflet observations should not be expanded into a comprehensive exclusion of all valve disease.

3. Historical prescriptions and uncertainties

- **BP treatment:** amlodipine **5 mg evenings** is confirmed in the January 2023 inpatient treatment list. Earlier intermittently used antihypertensives are not named. A continuing outpatient antihypertensive regimen is not established by the reviewed records. [S02]
- **Antithrombotics are separate from BP treatment:** January 4 oncology prescribed apixaban **2.5 mg twice daily**; January 24 history instead says 2.5 mg/day. January hospital records list enoxaparin and later recommend aspirin cardio **100 mg daily**. May/June 2023 oncology notes again prescribe apixaban **2.5 mg twice daily**. Actual overlap and stop/start dates remain unresolved; these are historical entries, not instructions to combine or restart drugs. [S02], [S10], [S11], [S16]
- **Baseline hypertension history:** the Jan 16 English consultation contains an awkward sentence listing hypertension, heart disease, diabetes, tuberculosis, and alcohol abuse, whereas the discharge explicitly denies prior arterial hypertension. A translation/negation error is possible. Established chronic hypertension or heart disease should not be inferred from that conflicting sentence alone. [S02], [S03]
- **Acute BP uncertainty:** there is no complete home/ambulance BP log or peak BP during the Jan 13 seizure. The later 112/83 reading cannot substitute for those missing values. No numerical PRES recurrence risk or clearance for another TKI is supplied by the echocardiogram.
- **“Portal/biliary hypertension” in abdominal imaging is not a systemic blood-pressure reading.** Those findings are covered in the CT timeline and are not counted here as arterial hypertension.
- **Current family correction:** she continues **Gleevec/imatinib 400 mg**; sunitinib was prescribed but **has not started**. The July cardiac study therefore does not measure an effect of sunitinib.

4. Earlier-examination search and limits

Reviewed the linked medical overview, main document folder, older original/examination folders, 2023 disability summary, February-March 2023 oncology/discharge records, May-June 2023 follow-up, March 2024 consultation, and recent cardiac files. **No earlier dedicated echocardiogram, ECG/QTc report, Holter recording, ambulatory BP study, or completed BP diary was located.** This describes the available records reviewed, not proof that such testing was never performed. A before/after EF comparison is therefore unavailable.

Q: What is the current ECOG performance status? A: Estimated ECOG performance status: 1. Independently ambulatory and manages daily activities, including shopping, yoga and light gardening; strenuous activity is limited. Subject to confirmation by her treating oncologist.

OpenEvidence response

This case turns on reconciling the polyclonal resistance pattern (imatinib-sensitive exon 11 driver plus exon 13 V654A and exon 18 A829P secondary mutations) against a prior life-threatening VEGFR-TKI event (pazopanib PRES with intracranial hemorrhage and vasculopathy). Let me research the mutation-specific efficacy of each next-line agent, the cardiovascular/PRES risk of re-exposing her to a VEGFR-TKI, and the role of local therapy for limited hepatic progression.

Excellent, comprehensive evidence gathered. Let me update my tracking and match clinical trials given the polyclonal refractory scenario.

3. Case studies, combinations and local treatment

Your question

please try again, return relevant case studies and therapies. Maybe combination of medicines, maybe non-invasive surgery? please find relevant ones

Q: How extensive is the current liver progression? A:

CT and PET/CT findings over time

Overall course: liver disease enlarged in 2022, decreased after the short pazopanib exposure, enlarged slightly by April 2023, then showed a strong structural/metabolic response during imatinib treatment. A **new, metabolically inactive 11 mm liver-wall nodule was documented in May 2025**. The June 2026 scan was initially read as stable, but a July rereading identified **new enhancing, moderately FDG-active mural components suspicious for returning viable GIST**.

This timeline uses scan dates, separating later rereadings. It includes PET/CT because metabolic activity and solid components changed even when overall dimensions changed little. The earliest identified CT is **August 23, 2022**; no scan later than **June 19, 2026** was found in the supplied overview and imaging folders reviewed. Findings are extracted from written reports, not independently measured from DICOM images. Historical “uterine sarcoma” labels are not adopted as the final diagnosis: the malignant disease was subsequently reclassified as gastric GIST with liver metastases.

Main timeline

Scan date / modality	Change and principal findings	Source
23 Aug 2022 - contrast CT	Baseline: bilobar liver masses with central necrosis and peripheral vascularity; liver height 198 mm . Mass above the pancreatic tail, adjacent to the stomach, 51 × 46 mm ; the original conclusion already included GIST in the differential . Anterior/fundal pelvic mass 93 × 63 mm ; right adnexal probable hemorrhagic cyst 67 × 57 mm . Lung nodules ≤3 mm; no enlarged nodes or destructive bone lesions.	[C01]
24 Aug 2022 - combined CT/MRI assessment; 14 Oct 2022 - MRI (supporting studies)	August combined assessment measured the dominant liver lesion 170 × 110 mm . October MRI reported hepatic progression and enlargement of the left upper-abdominal deposit to 74 × 60 mm , with the uterine mass broadly stable; its conclusion explicitly stated RECIST 1.1 PD . These are not additional CT scans.	[M01], [M02]
1 Dec 2022 - contrast CT	Compared with August: hepatic enlargement/progression . Liver height 220 mm ; S8 lesions 94 × 63 mm versus 79 × 56; left-lobe lesion 84 × 56 mm versus 66 × 48. Upper-abdominal mass 56 × 46 mm versus 54 × 46. Fundal mass decreased to 87 × 64 mm ; lung nodules and right adnexal cyst essentially stable.	[C02]
4 Jan 2023 - contrast CT, brain/chest/abdomen/pelvis	Report concludes no visible change versus December 1 , despite different measurements: dominant hepatic mass 200 × 130 × 180 mm , upper-abdominal mass 65 × 50 mm , liver height 242 mm . Minimal bile-duct dilation; no ascites or enlarged nodes. Brain: no pathological enhancing focus. This precedes pazopanib, taken Jan 5-12.	[C03]

Scan date / modality	Change and principal findings	Source
9 Feb 2023 - contrast CT	After pazopanib withdrawal: regression versus January 4 . Dominant liver lesion 184 × 114 × 160 mm ; left-lobe lesion 65 × 45 mm ; upper-abdominal mass 55 × 40 mm . Liver height decreased to 210 mm . Radiologist reported approximately 12% liver-lesion regression and 15% implant reduction ; pelvic masses stable. Percentages are the report's, not a new RECIST calculation.	[C04]
11 Apr 2023 - contrast PET/CT	After trabectedin: slight enlargement and active solid peripheral components versus February. Dominant liver mass approximately 190 mm , left-lobe mass 75 mm ; liver SUVmax 11.1 . Gastric-wall mass 58 mm , SUVmax 8.2 . Uterine/right adnexal lesions remained structurally stable with low activity, SUVmax up to 2.1 . No new separate suspicious lesions. This discrepancy prompted reconsideration of a gastric GIST primary.	[C05]
28 Jun 2023 - contrast CT; reread 28 Jul	During imatinib: liver and gastric-region regression . Dominant hepatic lesion 172 × 106 × 150 mm , predominantly cystic; left-lobe lesion 52 × 40 mm ; gastric-region mass 47 × 35 mm . Rereading versus April reported liver regression 18% , gastric-region mass 28% ; versus February, 12% and 14%. Fundal probable pedunculated myoma enlarged to 90 × 70 mm ; right adnexal mass decreased to 65 × 42 mm .	[C06], [C07]
17 Aug 2023 - contrast PET/CT	Marked response to Glivec versus April : liver lesions almost completely lost active peripheral solid components; residual uptake near liver background. Dominant liver mass 190 → 160 mm , left-lobe mass 75 → 50 mm . Gastric mass 58 → 44 mm , almost fluid and without increased FDG uptake. Inactive uterine mass enlarged 72 → 96 mm . A few new 2-3 mm lung nodules , recommended for surveillance.	[C08]
14 Mar 2024 - contrast PET/CT	Continued control : dominant liver mass 160 → 150 mm , left-lobe mass 50 → 45 mm , gastric mass 44 → 40 mm . No active peripheral solid components or new suspicious lesions. Inactive uterine mass decreased 96 → 78 mm . Lung nodules stable.	[C09]
13 May 2025 - contrast PET/CT; LISOD reread 19 May	Predominantly cystic, inactive liver remnants; dominant lesion 150 → 135 mm . Gastric lesion 40 → 35 mm on rereading, with calcifications. New 11 mm dense mural nodule in an S8 liver lesion , without significant focal FDG increase; possible hemorrhage was suggested and follow-up advised. Inactive uterine mass 78 → 84 mm . Primary report called the scan "PET negative"; unlike LISOD's rereading, it lacked prior DICOM data for comparison.	[C10], [C11]
19 Jun 2026 - contrast PET/CT, primary report	Compared with May 2025, interpreted as no definite negative change : dominant liver lesion 144 × 104 × 141 mm , left-lobe lesion 45 × 36 mm , gastric lesion 43 × 33 × 44 mm with increasing calcification. Liver height 179 → 187 mm . Uterine mass 68 × 55 × 48 mm and upper-margin mass 46 × 37 mm , described as stable; pelvic MRI suggested. No suspicious marrow uptake or active lymphadenopathy.	[C12]
Same 19 Jun scan - LISOD rereading 8 Jul 2026	Different interpretation : liver masses broadly stable in size, but new mural nodules with enhancement and moderately increased focal FDG activity . Left-lobe nodule SUVmax 3.2 , adjacent liver 2.0 . Conclusion: suspicious for return of viable GIST metastatic elements. Gastric and pelvic remnants unchanged/inactive; no new separate hepatic lesions or suspicious nodes. Also a new 31 mm right femoral-head lucency , sclerotic rim and moderate activity, without collapse/deformity; not diagnosed as metastasis in this report.	[C13]

Key size trend

Selected reported measurements in **millimetres**; these describe the entire residual mass, not just viable tissue. Different readers, planes, and lesion labels limit direct comparison.

Scan	Dominant right-liver mass	Gastric / upper-abdominal mass
Jan 2023	200 × 130 × 180	65 × 50

Scan	Dominant right-liver mass	Gastric / upper-abdominal mass
Feb 2023	184 × 114 × 160	55 × 40
Apr 2023	~190 maximum	58 maximum
Jun 2023	172 × 106 × 150	47 × 35
Aug 2023	160 maximum	44 maximum
Mar 2024	150 maximum	40 maximum
May 2025	135 × 100 primary; 135 maximum reread	33 × 28 primary; 35 maximum reread
Jun 2026	144 × 104 × 141	43 × 33 × 44

The critical 2026 change is the return of enhancing/active tissue within residual liver masses, rather than a large increase in their overall size. The reports do not establish that the inactive 2025 mural nodule and every 2026 active nodule are the same individual focus.

Important qualifications

- **Measurement differences:** April 2023's original report gives a dominant liver mass of approximately 190 mm and gastric mass of 58 mm; July's retrospective measurement of that same April scan gives **205 × 122 × 172 mm** and **62 × 45 mm**. These are alternative readings, not intervening growth. January 2023 likewise says stable despite measurements differing from December.
- **Dates:** the July 28, 2023 documents describe the **June 28** scan. The May 2025 rereading identifies **May 13** images and March 2024 comparators, but its conclusion says "03/2025." No separate March 2025 scan was located. June 2026's body references May 13, **2025**, while its conclusion prints **2026**; the discrepancy is retained rather than inventing an extra scan.
- **Pelvis:** size fluctuated independently of the gastric/liver response, usually with low FDG activity. A May 31, 2023 rereading of the August 2022 CT/MRI favored **pedunculated leiomyoma and right ovarian endometrioma**, with sarcoma considered unlikely. Later reports use inconsistent labels and measure different pelvic structures; they do not establish malignant pelvic progression. The June 2026 upper-margin mass is described as **stable**, not new. [M03]
- **Brain CT:** February 13, 2023 showed a **17 mm left occipital hypodensity**, without visible hemorrhage or edema, interpreted as residual PRES/possible ischemic change. The detailed brain MRI course is in PRES_after_pazopanib_2023.md. [B01]
- **Current treatment correction:** the family confirms imatinib/Gleevec **400 mg** continues; sunitinib was prescribed but **has not started**. No scan here assesses response to sunitinib.

Q: What is the current imatinib dose at the time of progression? A: 400 mg/day

Q: What is the current blood pressure / cardiovascular status? A:

Cardiology and blood-pressure findings, 2022-2026

Key findings: the January 2023 hospital record describes a blood-pressure rise to **150-160 mmHg** shortly after starting pazopanib, followed by PRES. Later documented clinic readings were **112/83** and **120/80 mmHg**. The **July 30, 2026 echocardiogram** reports satisfactory left-ventricular contraction and **ejection fraction 67%**, without chamber enlargement, myocardial wall thickening, or pericardial fluid. No earlier echocardiogram or ECG report was identified in the records reviewed.

The family confirmed that the July 30 ultrasound is the recent cardiac conclusion intended for inclusion. Findings below are from the patient's own records; routine examination findings and CT observations are distinguished from dedicated cardiac testing. Source links are attached to each entry.

1. Chronological findings and conclusions

Date	Examination / source	Documented finding or recommendation
23 Aug 2022	Chest CT, incidental cardiac observations	Heart not enlarged; no pericardial fluid. Aorta and major thoracic vessels described without changes. These are CT observations, not an echocardiographic or rhythm assessment. [S01]
Before Jan 2023 event	History recorded in Jan 25 discharge summary	Explicitly states no history of arterial hypertension . A conflicting translated sentence in the stroke consultation is discussed below. [S02]
Pazopanib day 2-3, approximately Jan 6-7, 2023	Jan 24 inpatient oncology consultation, reproduced in discharge summary	Pazopanib started Jan 5 at 800 mg/day. Headache and BP increase to 150-160 mmHg appeared on day 2-3. No diastolic value supplied. Headache worsened daily and when lying down; intermittent antihypertensives and analgesics provided temporary relief. Generalized seizure/loss of consciousness followed Jan 13. [S02]
16 Jan 2023	Stroke specialist Y. V. Flomin	BP 112/83 mmHg; pulse 58/min. Most likely diagnosis: PRES as an adverse effect of Votrient/pazopanib . Recommended stopping Votrient, considering steroid pulse therapy with monitoring/correction of metabolic and BP disturbances, and repeating MRI/MRA. This measurement was after admission, not at the seizure's onset. [S03]
20 Jan 2023	Brain MRI/MRA	Report concluded PRES with intracerebral hemorrhage and cerebral vasculopathy , including multifocal arterial narrowing/vasoconstriction. This is a cerebral vascular finding, not a cardiac ultrasound result. [S04]
23-25 Jan 2023	Stroke review and discharge	Jan 23 note again prints 112/83 mmHg, pulse 58/min , with clinical improvement; portions repeat the Jan 16 examination. Discharge lists amlodipine 5 mg in the evening among treatment carried out. Its start date, duration after discharge, and subsequent BP response are not specified. [S02], [S05]

Date	Examination / source	Documented finding or recommendation
15 Feb, 2 Mar, 24 Mar 2023	Oncology discharge examinations	Heart activity described as rhythmic, with sonorous tones; no numerical BP supplied in these passages. Hydration advice explicitly required monitoring urine output and blood pressure . These are routine examinations, not ECG-confirmed rhythm diagnoses. [S06]-[S08]
8 and 15 May 2023	Imatinib initiation/discharge and oncology follow-up	May 8 examination: rhythmic heart activity, muffled tones. Both records advise monitoring BP and urine output while following the prescribed fluid regimen. No echocardiographic measurements or numerical BP given. [S09], [S10]
13 Jun 2023	Oncology follow-up	Explicit recommendation: monitor blood pressure and pulse, especially when feeling worse, and keep a diary . The note also documents polydipsia and adjustment of fluid intake. No numerical BP is recorded. [S11]
19 Jun 2023	Neurological examination	BP 120/80 mmHg; heart rate 72/min . Consciousness and orientation intact; headache/dizziness considered more compatible with imatinib side effects. The normal clinic reading does not provide a BP trend during symptomatic episodes. [S12]
19 Jul 2023	Berlin neurosurgical consultation	Heart rate 87/min; oxygen saturation 94% . No BP measurement supplied. [S13]
27 Mar 2024	Assuta/Manor oncology consultation	Recommended continued follow-up with blood tests, TSH testing, and monitoring blood-pressure readings during imatinib treatment. No cardiac imaging or numerical BP reported. [S14]
30 Jul 2026, 14:25	LISOD ultrasound of the heart before chemotherapy	LVEF 67%; satisfactory LV myocardial contraction . No hypo-/akinetic zones, chamber enlargement, ventricular wall thickening, valve-leaflet thickening/deformation, or pericardial fluid described. Complete extraction below. [S15]

2. July 30, 2026 echocardiogram - complete clinical content

Facility: LISOD. **Patient ID:** 110889. **Physician:** Т. Д. Шпильова / Т. D. Shpylova. **Source title:**

“Ультразвукове дослідження серця перед ХТ” - “Ultrasound examination of the heart before chemotherapy.” The report itself does not name sunitinib; the surrounding treatment history establishes that it preceded the proposed switch. [S15]

Structure / parameter	Recorded result	Reference printed in report
Aorta	Not dilated or thickened; 25 mm	Up to 39 mm
Left atrium	Not dilated; 29 mm	Up to 39 mm
LV end-diastolic diameter (EDD / КДР)	46 mm ; LV cavity not dilated	Up to 55 mm
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LV ejection fraction	67% ; myocardial contraction satisfactory	55-65% , as printed
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LV posterior wall	Not thickened; 9 mm	Up to 12 mm
Heart valves	No leaflet thickening or deformation	No Doppler measurements supplied
Right ventricle	Not dilated; 25 mm	Up to 30 mm
Pericardium	No fluid detected	Not supplied

Exact Ukrainian conclusion: “Скорочення міокарда лівого шлуночка задовільне.”

English translation: “Left ventricular myocardial contraction is satisfactory.”

The source also states that the ultrasound result is not a final diagnosis. EF **67%** and the printed **55-65%** reference interval are preserved exactly; neither is silently changed. The supplied English translation adds the phrase “normal baseline echocardiogram,” which is **not part of the original conclusion**. The original does not provide BP, ECG/QTc, heart rate, global longitudinal strain, diastolic-function measurements, pulmonary-artery pressure, or graded valve regurgitation/stenosis. Its leaflet observations should not be expanded into a comprehensive exclusion of all valve disease.

3. Historical prescriptions and uncertainties

- **BP treatment:** amlodipine **5 mg evenings** is confirmed in the January 2023 inpatient treatment list. Earlier intermittently used antihypertensives are not named. A continuing outpatient antihypertensive regimen is not established by the reviewed records. [S02]

- **Antithrombotics are separate from BP treatment:** January 4 oncology prescribed apixaban **2.5 mg twice daily**; January 24 history instead says 2.5 mg/day. January hospital records list enoxaparin and later recommend aspirin cardio **100 mg daily**. May/June 2023 oncology notes again prescribe apixaban **2.5 mg twice daily**. Actual overlap and stop/start dates remain unresolved; these are historical entries, not instructions to combine or restart drugs. [S02], [S10], [S11], [S16]

- **Baseline hypertension history:** the Jan 16 English consultation contains an awkward sentence listing hypertension, heart disease, diabetes, tuberculosis, and alcohol abuse, whereas the discharge explicitly denies prior arterial hypertension. A translation/negation error is possible. Established chronic hypertension or heart disease should not be inferred from that conflicting sentence alone. [S02], [S03]

- **Acute BP uncertainty:** there is no complete home/ambulance BP log or peak BP during the Jan 13 seizure. The later 112/83 reading cannot substitute for those missing values. No numerical PRES recurrence risk or clearance for another TKI is supplied by the echocardiogram.

- **“Portal/biliary hypertension” in abdominal imaging is not a systemic blood-pressure reading.** Those findings are covered in the CT timeline and are not counted here as arterial hypertension.

• **Current family correction:** she continues **Gleevec/imatinib 400 mg**; sunitinib was prescribed but **has not started**. The July cardiac study therefore does not measure an effect of sunitinib.

4. Earlier-examination search and limits

Reviewed the linked medical overview, main document folder, older original/examination folders, 2023 disability summary, February-March 2023 oncology/discharge records, May-June 2023 follow-up, March 2024 consultation, and recent cardiac files. **No earlier dedicated echocardiogram, ECG/QTc report, Holter recording, ambulatory BP study, or completed BP diary was located.** This describes the available records reviewed, not proof that such testing was never performed. A before/after EF comparison is therefore unavailable.

Q: What is the current ECOG performance status? A: Estimated ECOG performance status: 1. Independently ambulatory and manages daily activities, including shopping, yoga and light gardening; strenuous activity is limited. Subject to confirmation by her treating oncologist.

OpenEvidence response

The evidence base is now strong across all four axes. Let me match relevant clinical trials given the refractory, polyclonal-resistance setting.

Bottom line

This is imatinib-refractory, **liver-limited oligoprogression** ("nodule-within-a-mass" pattern) driven by **polyclonal secondary KIT resistance** - a dominant exon 11 driver clone (VAF 17%) plus two acquired resistant subclones in complementary kinase regions: exon 13 **V654A** (ATP-binding pocket, VAF 8.8%) and exon 18 **A829P** (activation loop, VAF 4.0%).^[1, 2, 3] No single approved TKI covers both resistant clones: sunitinib hits V654A but not A829P; regorafenib and ripretinib hit A829P but are weak against V654A.^[4, 5, 6] Two features make her case distinctive and favor a **combined local-plus-systemic strategy over simply switching TKIs**: (1) the progression is confined to enhancing/FDG-active nodules within otherwise-controlled liver masses while the bulk exon-11 disease still responds to imatinib, and (2) she had **severe pazopanib-induced PRES with intracranial hemorrhage and cerebral vasoconstriction**, which raises the stakes for any potent VEGFR2 inhibitor (sunitinib, regorafenib).^[7, 8]

Practically: (a) confirm the pattern and, if feasible, **biopsy a progressing nodule** to resolve whether V654A and A829P are in separate clones (favorable) or in *cis* as a compound mutation (resistant to all approved agents); (b)^[2, 9] **continue imatinib** and treat the oligoprogressive liver nodules with **local/ablative therapy** (resection, thermal ablation, or transarterial embolization/DEB-TACE) - the NCCN-endorsed approach for limited progression that has kept one-third of such patients progression-free for over a year; (c) when systemic escalation is needed, weigh the molecular fit of each TKI against her VEGFR-toxicity risk, where^[10, 11, 12] **ripretinib carries by far the lowest hypertension/PRES signal**; and (d) refer to a^[8, 13] **clinical trial** given no approved single agent covers her full resistance profile - the NCCN explicitly recommends this.^[10]

Clinical context

Gastric GIST with liver metastases, KIT exon 11 deletion (Q556_V559>H) driver, in a woman aged 50, ECOG 1.^[10] Prior pazopanib (Jan 2023) caused PRES with left occipital intracerebral hemorrhage and multifocal cerebral vasoconstriction/RCVS, with substantial but incomplete recovery. Imatinib 400 mg daily was started May 2023 with an excellent ~3.5-year response. She is normotensive (112/83, 120/80 mmHg) with preserved LVEF (67%). Sunitinib has been prescribed but not yet started. Somatic BRCA2 T3085fs (VAF 1.5%), CDKN2A rearrangement (12.8%), and MTOR A1459D (0.21%) are present; germline BRCA testing (2022) was negative. Blood TMB is 3 Muts/Mb and MSS.^[10]

Key findings and their interpretation

The **ctDNA tumor fraction of 32% is high**, meaning the mutation calls are reliable and the negative findings carry low false-negative risk - this is a high-quality liquid biopsy that can legitimately guide therapy. [6] The two resistant subclones sit in the two canonical resistance regions: V654A in the ATP-binding pocket (the single most common secondary KIT mutation, ~24% of secondary mutations) and A829P in the activation loop. [1, 6] Their different VAFs (8.8% vs 4.0%) are consistent with - though not proof of - **two separate clones** rather than a single compound-mutant allele, which is the more treatable scenario. [9]

Radiologically, the 2026 change is **not bulk enlargement but the reappearance of enhancing, moderately FDG-active mural nodules (left-lobe SUVmax 3.2) within residual, largely cystic masses** - the classic imaging signature of emergent imatinib-resistant clones. [14] This pattern typically precedes RECIST progression by months and reflects clonal escape rather than global treatment failure. [14] The following figure illustrates this exact "new intratumoral enhancing nodule within a shrinking treated mass" appearance:

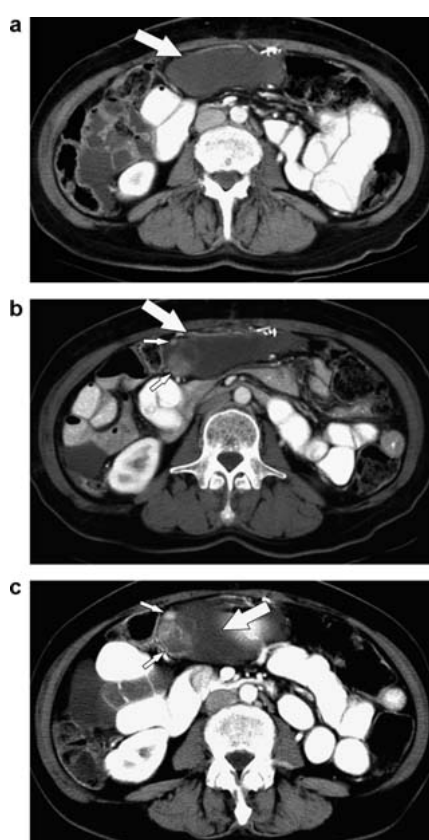


Figure 4. Recurrence with new intratumoral nodules in a patient with small bowel gastrointestinal stromal tumor.

[Imaging modalities of gastrointestinal stromal tumors](#). J Surg Oncol. 1. decembra 2011.

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One caveat to the "liver-limited" designation: the **new 31 mm right femoral-head lucency** with a sclerotic rim and moderate activity is undiagnosed. Given her prior high-dose corticosteroid pulse for PRES, **avascular necrosis is at least as likely as a bone metastasis** (osseous GIST metastases are rare); it should be characterized (dedicated MRI ± orthopedic input) before concluding she has extrahepatic progression, because that determination changes whether local liver therapy is appropriate. [10]

Evidence-based analysis

Matching drugs to the resistant clones

The foundational data come from the INTRIGUE phase 3 ctDNA biomarker analysis, which showed **diametrically opposite drug efficacy depending on which secondary region is mutated** in exon-11-driven GIST. [6] For ATP-binding-pocket (exon 13/14) mutations such as V654A, sunitinib was markedly superior (median PFS 15.0 vs 4.0 months for ripretinib; $P=0.0005$). For activation-loop (exon 17/18) mutations such as A829P, ripretinib was superior (median PFS 14.2 vs 1.5 months for sunitinib; $P<0.0001$). [6] Sunitinib's activity against V654A and its poverty against activation-loop mutants is confirmed biochemically and in the earlier phase I/II study (ABP mutants mPFS 7.8 vs AL mPFS 2.3 months). [4, 5, 16] Regorafenib is the mirror image - good against activation-loop mutations (a dedicated exon-17 phase II showed clinical benefit rate 93% and mPFS 22.1 months) but **inferior against V654A**. [5, 17] Ripretinib has the best breadth across activation-loop mutants but is clinically weak against ABP mutations. [5, 13] The INTRIGUE forest plot below shows these opposing subgroup effects:

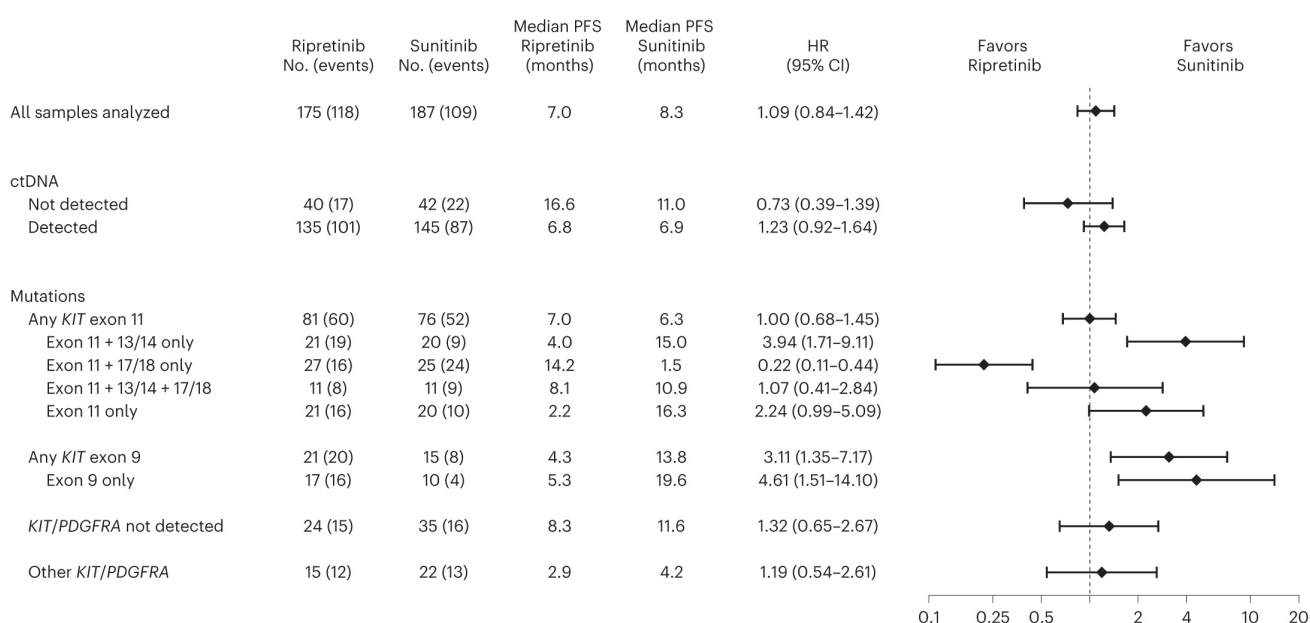


Figure 3. Forest plot of PFS by mutational subgroup.

Ripretinib versus sunitinib in gastrointestinal stromal tumor: ctDNA biomarker analysis of the phase 3 INTRIGUE trial. Nat Med. 1. februára 2024.

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The decisive point for this patient: in the small INTRIGUE subgroup with **co-occurring ABP + AL mutations (like hers)**, sunitinib and ripretinib performed **no differently** (HR 1.07; $P=0.88$, $n=11/\text{arm}$). [6] This is the biological reason her FoundationOne report flags sunitinib and avapritinib as resistant despite sunitinib being the guideline second-line default - sunitinib cannot control the A829P clone, and ripretinib cannot control V654A. Regorafenib's "sensitive" flag reflects its activation-loop activity and NCCN category-1 status, not coverage of V654A. If short-read ctDNA cannot resolve *cis* vs *trans*, a compound V654A/A829P allele on one clone would be resistant to **all** approved single agents - the emerging dominant mechanism of ripretinib escape. [2, 9, 18]

Approach	Covers V654A (ABP)	Covers A829P (AL)	VEGFR/PRES-hemorrhage risk	References
Sunitinib	Yes	No	High (grade ≥ 3 HTN 22.6%; PRES boxed-level warning; 1 ICH death in INTRIGUE)	[4, 8, 19]
Regorafenib	No	Yes	High (grade ≥ 3 HTN ~10-24%; PRES in label)	[5, 20, 21]
Ripretinib	Weak	Yes	Low (grade ≥ 3 HTN 5.8%; no PRES in label; weak VEGFR2)	[5, 8, 13]
Bezuclastinib + sunitinib	Yes (sunitinib)	Yes (bezuclastinib)	Retains sunitinib VEGFR risk; investigational	[22, 23]
Local therapy + continue imatinib	Ablates resistant clones directly	Ablates resistant clones directly	No new VEGFR exposure	[10, 11, 12]

Local / ablative therapy - the "non-invasive surgery" option

Because her progression is focal and liver-confined, this is precisely the setting where **liver-directed therapy while continuing the TKI** is guideline-supported and can defer a systemic switch. NCCN (GIST-5, category 2A) recommends, for limited progression, continuing the same TKI and adding **resection if feasible, thermal ablation, or embolization/radioembolization/chemoembolization**, with palliative radiotherapy (2B) reserved for symptomatic lesions; it explicitly warns that stopping the TKI can accelerate progression.^[10] The GEIS guidance states that for the nodule-within-mass pattern this strategy is "the treatment of choice over second-line systemic treatments" and keeps one-third of patients progression-free beyond a year.^[11]

The supporting evidence, all retrospective:

- **Hepatic resection / metastasectomy** has the strongest data. Adding surgery to imatinib dose escalation after focal progression improved time-to-treatment-failure (24.2 vs 6.5 months) and OS (53.2 vs 35.1 months); a 2023 meta-analysis of 988 patients showed improved OS (HR 2.10) and PFS (HR 1.92) with resection. Benefit is greatest with non-multifocal progression and R0/R1 resection.^[24, 25, 26]

- **Radiofrequency/microwave ablation** achieves high local control (technical success ~95-100%, local progression ~6%) for lesions generally ≤ 3 cm. Critically, **continuing the TKI after ablation drove 2-year PFS of 75% vs 29% when the TKI was stopped**. Ablation yields comparable OS to resection with fewer complications and shorter stays; MWA is interchangeable with RFA and preferred for larger or perivascular lesions.^[12, 27, 28, 29, 30]

- **Transarterial therapy (TACE/DEB-TACE/TAE)** is an option for multifocal liver disease not amenable to ablation. DEB-TACE after ≥ 2 TKI failures improved liver PFS (11.0 vs 5.0 months) and OS (18.0 vs 12.0 months); TACE after TKI failure similarly improved PFS and OS, best in patients without extrahepatic disease. Embolization should be avoided with very high-volume liver replacement.^[31, 32, 33]

- **SBRT** is feasible (GIST is less radioresistant than historically assumed): the only prospective RT study showed 80% stable disease with a 4-fold longer time to target-lesion progression while on TKI, and it is useful for lesions unsafe for thermal ablation. Combining SBRT with imatinib appears safer than pairing it with potent anti-VEGF multikinase agents.^[34]

Systemic next-line selection given the prior hemorrhagic PRES

PRES is a probable **class effect of VEGF-pathway inhibition**, and VEGFR2-selective potency correlates with the strength of the PRES signal. [7, 35] Re-challenge risk after prior VEGFR-TKI PRES is not quantified, but a switch to a *different* VEGFR-TKI has caused fatal recurrent CNS hemorrhage in a reported case, and reintroduction of anti-VEGF after neurovascular events recurred in ~33% in one series. [36, 37] This argues for caution with sunitinib and regorafenib specifically. Among the three approved agents, **ripretinib has substantially the lowest VEGFR toxicity** - grade ≥3 hypertension 5.8% vs 22.6% for sunitinib head-to-head in INTRIGUE (which also recorded one intracranial-hemorrhage death in the sunitinib arm), only weak VEGFR2 inhibition, and **no PRES/RPLS warning in its label**, whereas sunitinib carries a discontinuation-mandating RPLS warning. [8, 13, 19, 38] The tension is that ripretinib is molecularly weak against her V654A clone. [5] This trade-off is exactly why local therapy (no new VEGFR exposure) and combination or trial approaches deserve priority here.

Combination and novel-agent strategies

- **Bezuclastinib (activation-loop coverage) + sunitinib (ATP-pocket coverage)** is the rational pairing for polyclonal ABP+AL resistance. In the PEAK phase III, it improved median PFS from 9.2 to 16.5 months and ORR from 25.8% to 45.6% vs sunitinib alone; earlier phase Ib/IIa showed clinical benefit rate 80%. It does, however, retain sunitinib's VEGFR liability. [22, 23, 39]
- **Avapritinib + sunitinib** (phase I) gave ORR 20% and mPFS 6.6 months in heavily pretreated disease; rapid **sunitinib/regorafenib alternation** was disappointing (mPFS 1.9 months) in already dual-exposed patients. [39, 40, 41]
- **Next-generation pan-KIT inhibitors** designed to cover both ABP and AL mutations as single agents - **IDRX-42** (phase I ORR up to 43%; now in a phase III vs sunitinib) and **NB003/DCC-class agents** (phase I ORR ~27%) - are the most promising options for a genotype no approved single drug covers. [39, 42, 43]
- **Ripretinib intra-patient dose escalation to 150 mg twice daily** is an NCCN "useful in certain circumstances" option after progression on 150 mg daily (INVICTUS mPFS2 ~3.7 months). [10, 44]
- The **somatic BRCA2 frameshift (VAF 1.5%, germline-negative), CDKN2A, MTOR, and low TMB/MSS findings are not currently actionable** - PARP inhibition is unproven in a low-VAF subclonal somatic BRCA2 GIST, and immunotherapy (ipilimumab/nivolumab) is a category-2B NCCN option only for TMB-high disease, which she is not. [10]

Recommendations

Diagnostic workup

1. **Biopsy a progressing, FDG-active liver nodule** if safely accessible, for tissue NGS to (a) confirm the resistant genotype and (b) help distinguish separate clones from a *cis* compound V654A/A829P allele that would predict failure of all approved single agents. Repeat tumor biopsy to identify actionable mutations is explicitly endorsed by NCCN for progression despite prior therapies. [2, 9, 10]
2. **Characterize the right femoral-head lesion** with dedicated MRI ± orthopedics to distinguish steroid-related avascular necrosis from metastasis, since this determines whether disease is truly liver-limited. [10]
3. **Multidisciplinary review** (surgical oncology, interventional radiology, sarcoma medical oncology) to select the local modality. [10]

Local therapy (preferred given liver-limited oligoprogression)

1. Continue **imatinib 400 mg daily** and treat the resistant liver nodules with **resection, thermal ablation (RFA/MWA for lesions ≤3 cm), or DEB-TACE/embolization** for multifocal nodules - do not interrupt the

TKI around the procedure. Consider imatinib dose escalation to **400 mg twice daily** as an adjunct (modest expected benefit in exon-11 disease). [10, 11, 12]

Systemic escalation (when local control is not achievable or disease becomes multifocal/extrahepatic)

1. Selection is a genuine trade-off. Standard second-line is **sunitinib 50 mg daily (4 weeks on/2 off) or 37.5 mg continuous**, which best covers the dominant V654A clone but carries the highest VEGFR/hypertension/PRES-hemorrhage risk. Given her prior hemorrhagic PRES, if a VEGFR-TKI is started, do so only with **tight baseline and on-treatment BP monitoring, a low threshold for dose reduction, and neurologic vigilance**; some experts would favor **ripretinib 150 mg daily** for its markedly lower VEGFR toxicity despite weaker V654A coverage, or reserve regorafenib/sunitinib decisions for after biopsy clarifies clonal architecture. This is an expert-opinion balancing judgment, as no trial has tested TKI selection specifically after prior VEGFR-TKI PRES. [4, 8, 13, 19]

2. Consider the **bezuclastinib + sunitinib** combination (via trial) for rational dual ABP+AL coverage where available. [22]

Clinical trial referral (strongly recommended)

1. Because no approved single agent covers her full resistance profile, NCCN strongly recommends trial referral. Priorities are the next-generation pan-KIT inhibitors and ctDNA-guided studies. [10]

Clinical TrialsGastrointestinal Stromal Tumor · KIT mutation 10 trials

10 trials

Highly relevant · Matches disease · biomarker

A Master Protocol to Evaluate DCC-3009 in Gastrointestinal Stromal Tumor (GIST)

Recruiting Phase 1/2

Sponsor: Deciphera Pharmaceuticals, LLC

Updated: 5. 2. 2026

Highly relevant · Matches disease

ctDNA-Guided Sunitinib And Regorafenib Therapy for GIST

Recruiting Phase 2

Sponsor: University of Miami

Updated: 9. 9. 2025

Highly relevant · Matches disease · biomarker

A Study of Ziftomenib in Combination With Imatinib in Patients With Advanced Gastrointestinal Stromal Tumors (GIST)

Recruiting Phase 1

Sponsor: Kura Oncology, Inc.

Updated: 5. 2. 2026

Matches disease · biomarker

A Study of IDRX-42 (GSK6042981) Versus (vs) Sunitinib in Participants With Gastrointestinal Stromal Tumors After Imatinib Therapy

Recruiting Phase 3

Sponsor: GlaxoSmithKline

Updated: 12. 6. 2026

Matches disease

First-in-human Study of IDRX-42 in Participants With Metastatic and/or Unresectable Gastrointestinal Stromal Tumors

Recruiting Phase 1

Sponsor: IDRX, Inc., a wholly owned subsidiary of GSK, LLC

Updated: 4. 5. 2026

Matches disease

LRG GIST Patient Registry

Recruiting

Sponsor: The Life Raft Group

Updated: 13. 7. 2026

Highly relevant · Matches disease · Basket trial

A Study of the c-Kit Specific Antibody-Drug Conjugate NN3201 for Advanced and/or Metastatic Solid Tumors Known to Express c-Kit

Recruiting Phase 1

Sponsor: Novelty Nobility, Inc.

Updated: 23. 9. 2025

Matches biomarker · Basket trial

TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer

Recruiting Phase 2

Sponsor: American Society of Clinical Oncology

Updated: 15. 5. 2026

Matches biomarker · Basket trial

SYNERGY-AI: Artificial Intelligence Based Precision Oncology Clinical Trial Matching and Registry

Recruiting

Sponsor: Massive Bio, Inc.

Updated: 28. 10. 2025

Matches disease · Basket trial

Investigation of Profile-related Evidence Determining Individualized Cancer Therapy for Patients With Aggressive Malignancies and Poor Prognoses

Recruiting Phase 1/2

Sponsor: Medical College of Wisconsin

Updated: 22. 5. 2026

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The search returned 10 available trials. Among the top trials in the displayed dashboard, the most relevant to her polyclonal, all-approved-agent-resistant genotype are the **IDRX-42 phase III versus sunitinib after imatinib (NCT07218926)** and its **first-in-human phase I (NCT05489237)**, the **DCC-3009 pan-KIT master protocol (NCT06630234)**, and the **ctDNA-guided sunitinib/regorafenib phase II (NCT05366816)**, which directly operationalizes the mutation-driven selection her ctDNA supports; a KIT-directed antibody-drug conjugate (**NN3201**, **NCT06805825**) and **ziftomenib + imatinib (NCT06655246)** are earlier-phase options.

Contingency planning and follow-up

Continue imatinib through any local procedure - abrupt TKI cessation accelerates progression and worsens survival. ^[45] After local therapy, restage with contrast CT (and FDG-PET if ambiguous) at ~8-12 weeks, tracking **intratumoral enhancement/metabolic activity, not just size**, since density/activity change is the meaningful signal in GIST. ^[10] Serial ctDNA can track resistant-clone VAF as a response biomarker. ^[6] If she starts any VEGFR-TKI: obtain a **baseline home BP log, treat to <140/90 before escalation, hold for SBP persistently ≥160 or any new headache/visual change/seizure, and image the brain urgently for neurologic symptoms** given her PRES/ICH history. ^[7, 8] **If disease becomes multifocal or extrahepatic (or the femoral lesion proves metastatic), then shift from local therapy to systemic escalation and prioritize trial enrollment**; if a *cis* compound V654A/A829P allele is confirmed, approved single agents are unlikely to help and a next-generation pan-KIT trial becomes the preferred path. ^[9, 43]

Emerging Evidence ASCO Annual Meeting

Emerging evidence presented at the 2025 ASCO Annual Meeting suggests that in a Bayesian network meta-analysis of 17 studies (2,422 patients) with imatinib-resistant GIST, **masitinib and regorafenib** ranked highest for progression-free survival (masitinib MD +11.78 months; regorafenib MD +9.33 months) and overall survival, though the wide credible intervals and indirect comparisons limit certainty. ^[46]

Would it help to review the detailed eligibility criteria and molecular inclusion requirements for the next-generation pan-KIT trials (IDRX-42 and DCC-3009) and the ctDNA-guided study, to see which best fit her co-occurring V654A/A829P profile and prior-therapy history?

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4. Reduced doses, tolerability and other approaches

Your question

please do not just agree with me (when I hinted about Oncothermia, etc). Please provide a professional opinion.

Is there any data about lower dose of regorafenib (e.g. 80mg) or sorafenib?

there is no much mutational burden, etc, for additional immunotherapy with e.g. Nivolumab and Ipilimumab together with target?

what are surgery/physical destruction options which may work in her case - huge dormant metastatic tissue (or cystic?) with those activations in liver?

as for the bone - patient states, that she had cooling at winter

OpenEvidence response

This turn adds several distinct evidence questions - low-dose regorafenib/sorafenib efficacy, the honest case for/against oncothermia, immunotherapy in a TMB-low GIST, and ablative options for large cystic-dominant lesions with focal viable nodules. Let me research each directly.

Honest professional assessment first: none of the four options you raise is a clear winner, and one (oncothermia) has no credible evidence in this disease. The strongest, most defensible move in her situation remains **liver-directed local ablation of the few enhancing/FDG-active nodules while continuing imatinib**, because her disease is still overwhelmingly imatinib-controlled and only a small number of resistant foci have re-activated. The systemic questions you ask are worth answering precisely, though, because they change what to reach for when local control is no longer enough.

Lower-dose regorafenib and sorafenib

This is the most useful of your systemic ideas, and there is real GIST-specific data supporting reduced-dose regorafenib - not just extrapolation. The standard 160 mg 3-weeks-on/1-week-off schedule from the GRID trial produced grade 3/4 toxicity in 61% and dose modification in 72%, with hypertension the leading grade-3 event (23%).^[1] Two GIST-specific reduced-dose experiences show comparable efficacy with markedly better tolerability:

- **120 mg continuous daily** (MD Anderson, 79% of 28 patients): median PFS **8.7 months**, grade 3/4 AEs 43%.^[1]
- **100 mg continuous daily** (phase II, n=25): DCR ≥12 weeks **64%**, median PFS **7.3 months**, grade 3/4 toxicity 40%, dose modification needed in only 24%.^[2]
- An Italian real-world study of personalized (mostly reduced/modified) schedules vs standard dosing found **better** PFS with personalization: 9.7 vs 5.6 months (HR 0.51).^[3]

The specific **80 mg → 120 mg → 160 mg weekly escalation** you mention is the ReDOS strategy. It is validated in refractory colorectal cancer (more patients reaching cycle 3, 43% vs 25%; grade-3 hypertension

7% vs 15%; fatigue 13% vs 18%) and is NCCN-endorsed in that disease. ^[4, 5] The pharmacologic rationale supports it in GIST too - a phase I PK study showed the ~40% drop in tumor perfusion (the anti-angiogenic signal) occurs at **all doses ≥120 mg**, so 120 mg appears to be the floor for retained VEGFR effect. ^[1] Two honest caveats: a Japanese titration study did **not** find a clear reduction in adverse events simply by starting at 120 mg (early cycle-1 management matters more than starting dose), and a Taiwanese real-world colorectal cohort found reduced starting dose was associated with **shorter OS** (HR 1.66) - so lower dose is a tolerability strategy, not a free lunch. ^[6, 7] Importantly for her genotype, **regorafenib is weak against the exon 13 V654A (ATP-pocket) clone** while being active against the exon 18 A829P (activation-loop) clone - so even well-tolerated regorafenib addresses only part of her polyclonal resistance. ^[8]

Sorafenib is a legitimate alternative and is arguably more attractive than regorafenib for her *specific* mutations. In vitro, sorafenib inhibits imatinib-resistant KIT mutants in **both** the ATP-binding pocket (including V654A) **and** the activation loop - broader coverage than regorafenib for her V654A+A829P combination. ^[9, 10] Clinically, in third/fourth-line disease after imatinib and sunitinib (± nilotinib): the largest series (n=124) showed 10% responses, 57% stable disease, median PFS **6.4 months**, OS 13.5 months; dose reduction (needed in a third) did not compromise PFS. ^[11] Phase II data show median PFS ~4.9-5.2 months. ^[12, 13] All-grade hypertension appears **lower with sorafenib (~23%) than regorafenib (~44%)**. ^[14] Both regorafenib and sorafenib are listed by NCCN - regorafenib as preferred third-line (category 1) and sorafenib under "useful in certain circumstances." ^[15]

The unavoidable limitation: **no one has studied any VEGFR-TKI after prior VEGFR-TKI PRES with intracranial hemorrhage**. PRES/hypertension is a VEGFR class effect, and while starting low is reasonable, lower dose has never been shown to prevent PRES specifically. If a VEGFR agent is used, it should be sorafenib or low-dose/escalating regorafenib with intensive BP control and neurologic vigilance - but this remains a genuine risk decision, not a solved problem.

Immunotherapy (nivolumab + ipilimumab)

Your instinct that the low mutational burden argues *against* immunotherapy is correct, and the clinical data are consistently discouraging in GIST. This is a setting where the honest answer is that it is very unlikely to help:

- Randomized phase II of nivolumab ± ipilimumab in advanced GIST: ORR **0% for nivolumab alone and 6.7% (one response) for the combination** - the primary endpoint was not met. ^[16]
- The Alliance A091401 GIST expansion cohort showed **no objective responses** in either arm, and enrollment was halted for excess combination toxicity. ^[17]
- Anti-PD-1 (PDR001) **combined with imatinib** in refractory GIST - the exact "immunotherapy plus targeted" strategy you raise - produced **0% ORR**, and critically, **TMB and PD-L1 did not predict PFS**. ^[18]

GIST is characteristically MSS with low TMB (population median ~2 Muts/Mb), and her blood TMB of 3 is well below any actionable threshold. ^[19] Her **CDKN2A alteration** places the tumor in the "immune-desert" transcriptomic subtype predicted *not* to benefit from checkpoint blockade. ^[20] NCCN lists ipilimumab + nivolumab only for **TMB-high** GIST, and only as category 2B - she does not qualify. ^[15] The somatic BRCA2 frameshift at VAF 1.5% is subclonal and not a validated PARP-inhibitor target here. Professional bottom line: **nivolumab/ipilimumab is not supported for her and should not displace an evidence-based option**.

Surgery / physical destruction - the strongest lever in her case

Her imaging pattern - large, predominantly cystic/necrotic masses that are mostly dead, with a **few small enhancing, FDG-active mural nodules** (left-lobe SUVmax 3.2) - is the textbook "nodule-within-a-mass" secondary-resistance pattern. ^[21] This is actually favorable for local therapy: **the target is the small viable focus, not the 14 cm cyst**. NCCN explicitly supports, for limited progression, **continuing the same TKI and adding resection (if feasible), ablation, or embolization/radioembolization/chemoembolization** - and warns that stopping the TKI accelerates progression. ^[15, 22]

Practical modality choice for focal viable nodules within cystic masses:

- **Microwave ablation (MWA)** is generally the preferred thermal option - larger, more predictable ablation zones, less heat-sink near vessels, and in a non-colorectal liver-metastasis series GIST had the best outcome of any histology (5-year OS 100%). Best for the ≤ 3 cm viable nodule; the probe is directed at the enhancing focus, not the cyst. ^[23, 24]
- **Radiofrequency ablation (RFA)** has the most GIST-specific track record (technical success ~95%, local progression ~6%, and continuing the TKI after ablation drove 2-year PFS of 75% vs 29% if stopped). ^[25, 26]
- **Cryoablation** is worth specific consideration here because the ice ball is directly visible on CT/MRI, which helps precisely target a small nodule inside a complex cystic mass. ^[27]
- **Irreversible electroporation (IRE)** is the option if a nodule sits against bile ducts or major vessels where thermal ablation is unsafe. ^[24, 28]
- **Histotripsy** - non-invasive, non-thermal focused ultrasound, FDA-cleared for liver tumors in 2023 - is the truly "non-invasive surgery" concept. It is genuinely promising (95% technical success in the #HOPE4LIVER pivotal trial) but has **no GIST data, imperfect targeting, and no long-term oncologic outcomes yet**; it should be regarded as early-stage rather than established. ^[29, 30]
- **SBRT** can target a small enhancing focus regardless of surrounding cystic content and is unaffected by heat-sink; GIST is less radioresistant than once assumed, and it pairs safely with imatinib. ^[31]
- **Metastasectomy/resection** has the strongest survival signal where the pattern is non-multifocal and R0/R1 is achievable, but is more invasive than the above.

Whichever is chosen, **continue imatinib 400 mg throughout** - this is the single most important adjunct. ^[22]

Oncothermia / hyperthermia - the honest verdict

You asked me not to simply agree, so plainly: **modulated electro-hyperthermia (mEHT / "oncothermia") is not supported for her**. It has a plausible biophysical mechanism and some preclinical and observational data, but the entire clinical evidence base rests on a **single phase III RCT in cervical cancer** in a resource-limited setting, with the remaining literature dominated by small single-arm series, much of it from investigators tied to the technology. ^[32, 33, 34] There is **zero evidence in GIST or any sarcoma liver metastasis**, and no oncology guideline (NCCN, ESMO, ASCO) endorses it. It is a focal-heating adjunct, not a destructive ablative therapy, so it does not even fit the mechanical goal of eliminating her resistant nodules.

It is worth separating this from **conventional regional hyperthermia plus chemotherapy**, which is a *different technology* with genuine level-1 evidence - the EORTC 62961-ESHO 95 trial showed improved survival adding regional hyperthermia to neoadjuvant chemotherapy in localized high-risk soft-tissue sarcoma (10-year OS 52.6% vs 42.7%). ^[35, 36, 37] But that trial **excluded GIST**, used chemotherapy agents (ifosfamide/doxorubicin) that don't treat GIST, and was for localized rather than metastatic disease - so it does not transfer to her either. Net: hyperthermia of any kind should not substitute for image-guided ablation and effective systemic therapy here.

The femoral-head lesion and "cooling in winter"

Cold exposure/"cooling" does not produce a 31 mm subchondral femoral-head lucency with a sclerotic rim and moderate metabolic activity - frostbite affects acral tissue (fingers, toes, nose), not the proximal femur, so that history does not explain the finding. The two realistic explanations remain **steroid-associated avascular necrosis** (she received high-dose methylprednisolone pulses during the PRES episode, a classic cause) versus a **solitary bone metastasis** (rare in GIST). The sclerotic rim and absence of collapse are more suggestive of AVN or a benign process than aggressive metastasis, but this cannot be settled by cold history - it needs a **dedicated hip MRI** ($\pm$ orthopedic input). This matters concretely: if it is AVN or benign, she is genuinely liver-limited and an excellent candidate for local ablation; if it is metastasis, the disease is no longer oligometastatic and systemic escalation moves ahead of local therapy.

Recommendations

1. **Confirm oligoprogression and biopsy a viable nodule.** Dedicated hip MRI to characterize the femoral lesion, and if safely accessible, biopsy an FDG-active liver nodule for tissue NGS - this also helps determine whether V654A and A829P are in separate clones (treatable) or a *cis* compound allele (resistant to all approved single agents). ^[21]
2. **Preferred next step: liver-directed ablation (MWA first choice; cryoablation if precise targeting within the cyst is needed; IRE/SBRT for perivascular/biliary-adjacent foci) while continuing imatinib 400 mg.** Multidisciplinary review with interventional radiology and surgical oncology. ^[15]
3. **If/when systemic escalation is needed**, the least-bad VEGFR option given her PRES history is **sorafenib** (broader in-vitro coverage of both her resistant clones, lower hypertension rate) or **low-dose/escalating regorafenib (start 80-120 mg)** - either with aggressive BP control (target $<140/90$, hold for SBP ≥ 160 or any neurologic symptom) and a low threshold for brain imaging. Note sunitinib (already prescribed) covers V654A but not A829P and carries the highest VEGFR/hemorrhage risk. ^[2, 9, 11, 14]
4. **Do not pursue nivolumab/ipilimumab or oncothermia** outside a trial - neither is supported in her MSS, TMB-low, CDKN2A-altered GIST. ^[16, 18, 32]
5. **Strongly consider a clinical trial** of a next-generation pan-KIT inhibitor covering both ATP-pocket and activation-loop mutations, since no approved single agent covers her full profile (per the trials shown in the prior turn). ^[15]

Contingency: if the femoral lesion proves metastatic, or new hepatic/extrahepatic foci appear, shift priority from local therapy to systemic escalation $\pm$ trial. If a compound V654A/A829P allele is confirmed on biopsy, approved single agents are unlikely to help and a pan-KIT trial becomes the preferred path.

Would it be helpful to review the specific technical selection criteria (lesion size, vascular proximity, ablation-margin data) that interventional radiology would weigh in choosing microwave ablation versus cryoablation for these particular intracystic nodules?

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Randomized phase 2 evidence in refractory metastatic colorectal cancer suggests that starting regorafenib at 80 mg/day and increasing weekly to 160 mg/day, if tolerated, helped more patients stay on treatment long enough to start cycle 3 than beginning at 160 mg/day immediately. Survival and progression-free survival were not significantly different, and some early toxicities and week-2 fatigue-related quality-of-life measures appeared better with dose escalation, but the trial was small, open-label, and used a composite primary endpoint rather than survival. Design N: N=123 randomized; N=116 analyzed (dose-escalation: 54; standard-dose: 62) Design: Multicentre, open-label, randomized phase 2 Follow-up: 1.18 years median (IQR 0.98-1.57) PICO Population: Adults with refractory metastatic colorectal cancer Intervention: Regorafenib dose-escalation 80→160 mg/day orally Comparator: Standard-dose regorafenib 160 mg/day orally Primary Outcome: Proportion of patients initiating cycle 3 Result Primary Outcome Met: Yes Effect Size: 43% (95% CI 29-56) vs 26% (95% CI 15-37); one-sided p=0.043 Research in Context Practice-changing evidence: ReDOS provided the first randomized support for regorafenib dose escalation, formalizing a previously empirical toxicity-management approach Reinforced but not definitive: contemporaneous meta-analysis, later replication, and real-world studies generally support better tolerability and treatment persistence with escalation, but superiority among modified schedules remains unclear and no clear overall survival advantage has been established Clinical takeaway: now incorporated into NCCN guidance as an alternative to upfront standard dosing in refractory metastatic colorectal cancer, with ongoing uncertainty about the optimal regimen amid emerging cumulative-dose and combination-therapy data

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5. Local treatment for cystic or partially active metastases

Your question

Which local ablation or minimally invasive techniques are most effective for cystic or partially active liver GIST metastases?

OpenEvidence response

The patient context is well established from prior turns - cystic/necrotic-dominant liver masses with a few focal enhancing/FDG-active nodules, imatinib-controlled, prior hemorrhagic PRES. This is a scope-narrowing follow-up, so let me target the comparative ablation evidence specifically for GIST liver metastases and the cystic/partially-viable pattern.

No single modality is uniformly "best" - the right choice is dictated by the anatomy of her lesions: a few small enhancing/FDG-active mural nodules (SUVmax ~3.2) sitting inside large, predominantly cystic/necrotic masses. This "nodule-within-a-mass" pattern is the classic imaging signature of focal secondary resistance, where the viable resistant clone is a small perfused component and the surrounding cyst is already-dead, imatinib-responsive tissue. ^[1, 2] Two consequences follow: **the target is the small viable nodule, not the 14 cm cavity**, and the cystic content itself changes how each technique behaves. Across all options, imatinib should continue throughout - the single most consistent finding in the GIST ablation literature is that stopping the TKI collapses the benefit (2-year PFS 75% with continued TKI vs 29% without after RFA). ^[3, 4]

Techniques that inherently target only viable tissue (best conceptual fit)

For a small perfused nodule inside a dead cyst, the **arterially-directed therapies are unusually well-matched**, because they deposit their effect only where there is arterial supply - i.e., in the viable nodule - while the avascular necrotic/cystic portions are untouched. NCCN explicitly endorses TAE, TACE/DEB-TACE, and Y-90 radioembolization for GIST liver metastases not responding to TKIs (category 2A). ^[3]

- **DEB-TACE / TACE:** The most recent GIST-specific series (CalliSpheres DEB-TACE, n=34 after ≥2 lines of TKI) reported ORR 94% by mRECIST, DCR 100%, median liver PFS 11.0 vs 5.0 months and median OS 18.0 vs 12.0 months versus controls. Earlier conventional TACE data (n=22) showed liver DCR 77% and markedly longer PFS (30 vs 12.9 weeks) and OS (68.5 vs 25.7 weeks; HR for death 0.11), with benefit concentrated in patients **without extrahepatic disease** - which is why characterizing her femoral lesion first matters. Superselective catheterization can be directed at the feeding artery of the enhancing nodule specifically. ^[5, 6, 7]

- **Y-90 radioembolization (TARE):** In TKI-refractory GIST, median hepatic PFS 16 months and OS 28 months in the largest series; microspheres likewise lodge preferentially in perfused tissue. ^[3]

Thermal ablation - modality choice matters in cystic tissue

If a percutaneous ablation of the nodule is preferred, the cystic content is not a neutral bystander.

Microwave ablation (MWA) is the preferred thermal modality here. MWA heats via dielectric hysteresis of water molecules rather than resistive current, so in cyst-mimicking phantoms it achieved faster, higher, more predictable temperatures within fluid regardless of ionic concentration, whereas RFA heating was flattened and impedance-dependent. [8] MWA also gives larger, faster ablation zones with less heat-sink near vessels. [9, 10] A practical adjunct supported by benign-cyst data is to **aspirate the cystic/necrotic content first** to collapse the cavity, eliminate fluid convection ("internal heat sink"), and bring the antenna against the mural nodule. [11]

RFA has by far the most GIST-specific track record (technical success ~95%, local progression 0-7%, median OS 90-100 months across small retrospective series), but every one of those series treated small *solid* lesions (mean 1.3-2.5 cm), and RFA is the least suited technique for fluid-containing targets. [4, 12, 13] Thermal ablation and resection give comparable OS in GIST (median 102 vs 117 months) with far fewer complications. [14]

Cryoablation deserves specific mention because the **ice ball is directly visible on CT/MRI**, allowing real-time confirmation that the ablation zone engulfs a small nodule inside a complex mass - with the caveat that lethal temperature sits 5-10 mm inside the visible ice-ball edge, so the ice ball must extend ≥10 mm beyond the nodule. [15, 16] It is more collagen/duct-sparing but carries a rare cryoshock risk if large volumes are frozen. [17, 18]

Irreversible electroporation (IRE) is the non-thermal thermal-alternative reserved for a nodule abutting major vessels or central bile ducts, where it spares those structures; GIST-specific data are essentially anecdotal. [3, 19]

SBRT - composition-independent, location-independent

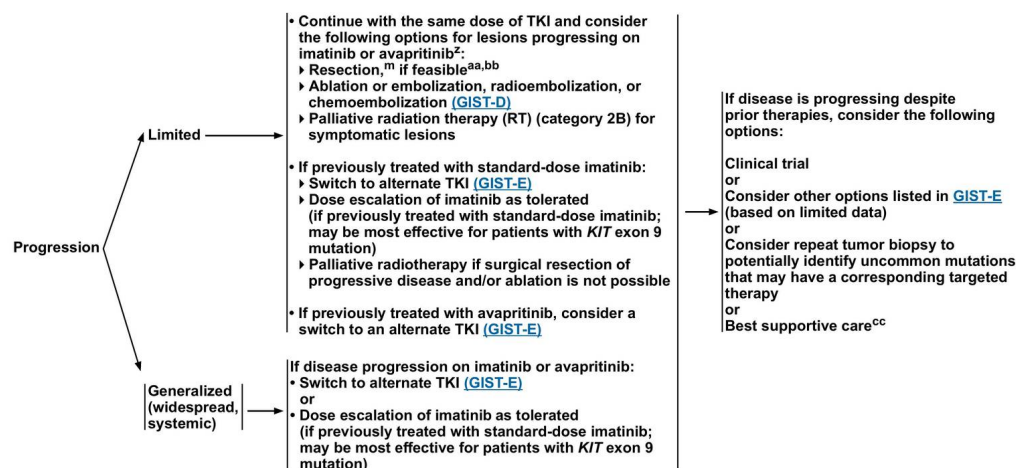
SBRT is attractive precisely because its dose deposition is **unaffected by whether surrounding tissue is cystic, necrotic, or solid, and unaffected by heat-sink or perivascular/central location.** [9, 20] GIST is less radioresistant than long assumed: the only prospective GIST radiotherapy study (25 patients, most on concurrent TKI) achieved stable target lesions in 80% with median stabilization 16 months, and a 4-fold longer time to target-lesion progression than to progression elsewhere. [21] Pooled liver-metastasis SBRT data (all histologies) show 1-year local control ~85-90% with BED ≥100 Gy and <3-10% grade ≥3 toxicity. [20, 22] Importantly for her, SBRT combines safely with imatinib, but pairing high-dose radiotherapy with potent anti-VEGF multikinase agents (e.g., sorafenib, bevacizumab) has produced severe toxicity - relevant if a VEGFR-TKI is later added. [9] NCCN still lists radiotherapy for GIST only as category 2B for symptomatic lesions, so SBRT here is evidence-supported but off the formal on-label pathway. [3]

Histotripsy - promising but investigational

Non-invasive, non-thermal focused-ultrasound histotripsy (FDA-cleared for liver tumors 2023) is the truest "non-invasive surgery," with no heat-sink and vessel/duct sparing; the #HOPE4LIVER pivotal trial showed 95% technical success for tumors ≤3 cm. [23, 24] However, there are **no GIST data, no data in cystic lesions specifically, targeting accuracy is still maturing (~74% complete non-viability in one early series), and no long-term oncologic outcomes exist** - it should be regarded as investigational, not a substitute for the established options above. [25]

Technique	Fit for cystic mass + small viable nodule	GIST-specific evidence	Best-suited situation	References
DEB-TACE / TACE	Excellent - treats only perfused (viable) tissue	Liver PFS 11 vs 5 mo; OS 18 vs 12 mo (DEB-TACE)	Multifocal/liver-dominant, no extrahepatic disease	[3, 5, 6]
Y-90 TARE	Excellent - microspheres deposit in perfused tissue	Hepatic PFS 16 mo, OS 28 mo	Liver-dominant, multiple lesions	[3]
Microwave ablation	Good - dielectric heating works in high-water tissue; aspirate cyst first	Extrapolated (strong general liver data; best histology outcome in one MWA series)	Solitary/few nodules ≤ 3 cm	[8, 9, 10]
Cryoablation	Good - visible ice ball for precise targeting	Sparse (few GIST in sarcoma series)	Nodule needing image-confirmed margins; near ducts	[15, 16]
SBRT	Good - composition/location-independent, non-invasive	Prospective: 80% SD, 16-mo stabilization	Perivascular/central nodule; poor ablation candidate	[20, 21]
IRE	Selective - vessel/duct-sparing	Anecdotal	Nodule abutting major vessel/central bile duct	[3, 19]
RFA	Poorest in cystic tissue (impedance-dependent)	Most mature GIST data (solid lesions)	Small solid nodule away from vessels	[4, 8, 12]
Histotripsy	Unknown in cystic lesions	None in GIST	Investigational only	[23, 25]

The NCCN algorithm for limited hepatic progression places these liver-directed options alongside continued TKI:

TREATMENT FOR PROGRESSIVE DISEASE


^m [General Principles of Surgery \(GIST-C\)](#).

^z Clinical experience suggests that discontinuing TKI therapy, even in the setting of progressive disease, may accelerate the pace of disease progression and worsen symptoms.

^{aa} Consider resection or ablation/liver-directed therapy for hepatic metastatic disease.

^{bb} Resection of metastatic disease, especially if complete resection can be achieved, may be beneficial in patients on imatinib or sunitinib who have evidence of radiographic response, or limited disease progression.

^{cc} Reintroduction of imatinib can be considered for palliation of symptoms. Consider continuation of imatinib for palliation of symptoms as part of best supportive care.

Note: All recommendations are category 2A unless otherwise indicated.

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GIST-5

GIST-5. Gastrointestinal Stromal Tumors: Treatment for Progressive Disease - NCCN Guidelines® - Gastrointestinal Stromal Tumors p. 10 (v1.2026)

Gastrointestinal Stromal Tumors. 13. januára 2026.

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Practical synthesis for her case

For a **small number of well-defined enhancing nodules**, image-guided **MWA (with prior cyst aspiration) or cryoablation** offers the most direct destruction of the resistant clone with a curative-intent margin, continuing imatinib. If the viable components are **multifocal, ill-defined, or scattered across residual masses**, **DEB-TACE or Y-90 TARE** is the more logical choice because it inherently treats only perfused tissue and does not require pinpoint probe placement inside a large cyst. **SBRT** is the best option for any nodule that is perivascular, central, or otherwise unsafe for a probe. The decision hinges on lesion number, size, and location and should go through a multidisciplinary tumor board with interventional radiology and surgical oncology - and, because arterial and embolic therapies lose their advantage once disease is extrahepatic, on confirming that the femoral lesion is not metastatic. [3, 6]

Would it be useful to look at the specific radiographic response criteria (Choi density criteria vs mRECIST vs FDG-PET) that best capture treatment effect after these liver-directed therapies in GIST, since size-based RECIST tends to understate benefit?

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6. Regorafenib versus sorafenib

Your question

what can you say about regorafenib vs sorafenib for her case? what about toxicity, feeling? she felt gastric side effects with imatinib 400mg which she described as "inflammation"

OpenEvidence response

This narrows to a direct head-to-head on tolerability, with her prior imatinib GI intolerance ("inflammation," likely gastritis) as the deciding lens. Let me pull the specific GI and overall toxicity profiles for both agents.

Efficacy is essentially a tie, but sorafenib has the better molecular rationale for her specific resistance profile - while regorafenib has the stronger guideline standing and GIST-specific trial data. On tolerability the two are broadly similar multikinase VEGFR-TKIs, but the practical differences (hypertension burden, GI toxicity, and her prior imatinib GI intolerance) tilt the decision, and for her both are second-choice to local ablation given the prior hemorrhagic PRES.

Efficacy and mutation fit

The two drugs are structural cousins with near-identical kinome spectra, and their reported activity after imatinib/sunitinib failure is comparable - sorafenib gives ~10% responses, 57-60% stable disease, and median PFS ~6.4 months in third/fourth line, versus regorafenib's phase III median PFS 4.8 months (GRID) with 52.6% disease control. ^[1, 2, 3] The molecular distinction matters for her polyclonal resistance: regorafenib is potent against **activation-loop** mutations (her exon 18 A829P) but this "came at the expense of potency against secondary resistance mutations in KIT exons 13 and 14" - i.e., it is weak against her exon 13 **V654A** ATP-pocket clone. ^[4] Sorafenib, by contrast, inhibits imatinib-resistant KIT mutants in **both** the ATP-binding pocket (including V654A) **and** the activation loop in vitro, sparing only D816/D842 substitutions she does not have. ^[2] On genotype alone, sorafenib is the more logical fit for a tumor carrying both V654A and A829P. Neither, however, reliably covers a compound *cis* allele, which is why local ablation and trial referral remain preferable.

Regorafenib is NCCN category 1 preferred third-line therapy; sorafenib appears only under "useful in certain circumstances," reflecting the phase III evidence base behind regorafenib versus retrospective data for sorafenib. ^[5]

Toxicity and tolerability head-to-head

Both are VEGFR multikinase inhibitors, so hand-foot skin reaction, diarrhea, fatigue, hypertension, and mucositis dominate both profiles. ^[1, 6] The clinically meaningful differences:

Toxicity	Regorafenib (160 mg 3-wk-on/1-off)	Sorafenib (400 mg BID)	Relevance to her
Grade ≥3 hypertension	~23% (GRID)	~4% (all-grade ~9-23%)	Major - PRES/ICH history makes HTN the key risk; favors sorafenib

Toxicity	Regorafenib (160 mg 3-wk-on/1-off)	Sorafenib (400 mg BID)	Relevance to her
Any/grade ≥ 3 HFSR	56-67% / ~20%	Common / ~15%	Both high; HFSR earlier & more severe with regorafenib
Diarrhea (any)	~43-47%	~40%	Both significant; adds to her GI burden
Mucositis/stomatitis	~33-40%	Common	Both - relevant to her GI sensitivity
Hemorrhage (any / grade ≥ 3)	11-18% / ~3%	~17% / ~2.4%	Both carry VEGFR bleeding risk; near-equivalent
GI perforation/fistula	~0.6-2.1%	<1%	Both - caution with bulky intra-abdominal disease
PRES/RPLS in label	Yes (discontinue)	Class effect; not labeled	Both relevant given prior PRES
Dose modification rate	~72%	~33%	Regorafenib far more often requires dose changes
Hepatotoxicity	Boxed warning (fatal cases)	Lower	Favors sorafenib for monitoring burden

[1, 2, 3, 6, 7, 8, 9, 10]

Two differences stand out for "how she will feel." First, **hypertension is markedly more frequent and severe with regorafenib** (grade ≥ 3 ~23% vs ~4%), and given her pazopanib-induced PRES with intracranial hemorrhage, the hypertension burden is the single most important safety axis - this favors sorafenib. [3, 9] Second, regorafenib required **dose modification in ~72%** of GIST patients versus ~33% with sorafenib, and its hand-foot reaction tends to appear earlier and more severely - meaning more early disruption to quality of life. [1, 2, 7] Regorafenib also carries a **boxed hepatotoxicity warning** absent from sorafenib. [11] Both list PRES/bleeding risk; a single PRES case is documented among >1,200 regorafenib recipients, and RPLS is an explicit discontinuation criterion in its label. [6, 12]

The imatinib "inflammation" (GI intolerance) angle

Her gastric symptoms on imatinib 400 mg - likely dyspepsia/gastric irritation, which the imatinib label explicitly notes ("sometimes associated with GI irritation... take with food and a large glass of water") - are relevant, because **neither regorafenib nor sorafenib is GI-sparing**. [13] Both cause substantial diarrhea (~40-47%), mucositis/stomatitis, and abdominal pain, and sorafenib's diarrhea has a specific mechanistic driver (KIT inhibition in the interstitial cells of Cajal plus pancreatic exocrine dysfunction with steatorrhea, which responds to pancreatic enzyme supplementation). [1, 14] Regorafenib carries the added concern of a fast median time-to-toxicity-onset (~10.5 days) and higher mucositis rates in GIST (~40%). [12, 15] Importantly, her imatinib GI intolerance does **not** predict cross-intolerance in a mechanism-specific way, but a GI tract already prone to irritation is likely to tolerate the lower-GI-toxicity, lower-dose-modification agent better - again a modest point toward **sorafenib**, or toward starting either drug at a reduced/escalating dose. Neither drug should be taken with the high-fat meal that worsens tolerability, and proactive antidiarrheals, dose-splitting, and mucositis care improve adherence. [16, 17]

Bottom line for her

If a VEGFR-TKI is chosen, **sorafenib is the more rational option in her specific situation**: broader in-vitro coverage of both her V654A and A829P clones, substantially lower hypertension rate (critical after hemorrhagic PRES), fewer required dose modifications, and no boxed hepatotoxicity warning - accepting that its GIST evidence is retrospective and it is an off-guideline-preferred choice. ^[1, 2, 9] **Regorafenib** remains the category-1, better-validated agent and is entirely defensible, but its higher grade ≥ 3 hypertension (~23%), earlier/more severe hand-foot reaction, ~72% dose-modification rate, and hepatotoxicity warning make it the less attractive choice given her neurovascular history and GI sensitivity. ^[3, 7] Either way, given the prior PRES/ICH, starting at a **reduced or escalating dose** (e.g., sorafenib 400 mg once daily titrating up, or regorafenib 80-120 mg) with tight home BP monitoring (target <140/90; hold for SBP ≥ 160 or any neurologic symptom), a low threshold for brain imaging, biweekly LFTs (regorafenib), and proactive antidiarrheal/mucositis/HFSR management is the safest way to use either drug. ^[11, 17, 18] Both remain second-line in priority to ablating her few resistant liver nodules while continuing imatinib.

Would it help to look at the specific evidence on managing VEGFR-TKI-associated hand-foot skin reaction and hypertension prophylactically, so that whichever agent is chosen can be started with a structured toxicity-prevention plan?

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7. Coverage of the three KIT mutations

Your question

both cover all 3 her specific mutations? what are response % per exon/mutation?

OpenEvidence response

Let me pull the mutation-specific activity data for both agents across her three KIT alterations.

Neither drug cleanly covers all three of her KIT alterations - and the gap is the same clone for both: the exon 13 V654A. Her three lesions sit in three different functional zones of KIT, and that is exactly why no single approved agent is a complete answer:

- **Exon 11 Q556_V559>H** is the *primary* activating driver (VAF 17%). It is imatinib-sensitizing and remains inhibited by essentially every KIT-active TKI, including both regorafenib and sorafenib. ^[1, 2]
- **Exon 13 V654A** (VAF 8.8%) is a secondary mutation in the **ATP-binding pocket**. This is the problem variant. ^[2, 3]
- **Exon 18 A829P** (VAF 4.0%) is a secondary mutation in the **activation loop**. ^[2, 3]

The critical point is that her two resistance mutations fall into the *two different, mutually exclusive coverage classes* - one drug tends to cover one class at the expense of the other, which is the textbook definition of polyclonal resistance that defeats single agents. ^[4, 5]

Mutation-by-mutation coverage

Her KIT alteration	Functional zone	Regorafenib	Sorafenib	References
Exon 11 Q556_V559>H (VAF 17%)	Primary driver	Covered	Covered	[1, 2]
Exon 13 V654A (VAF 8.8%)	ATP-binding pocket	Weak / inferior	In-vitro active, but reviews classify ATP-pocket as sorafenib-resistant (conflicting)	[2, 6, 7, 8]
Exon 18 A829P (VAF 4.0%)	Activation loop	Covered (rego's strength)	Covered (class effect)	[2, 6]

The **V654A** is where the honest uncertainty lies. Regorafenib is explicitly described as having *inferior activity against V654A* - its improved potency against activation-loop mutations "came at the expense of potency against secondary resistance mutations in KIT exons 13 and 14." So regorafenib likely controls her exon 11 and A829P clones but leaves the V654A clone - which is her ^[6] **larger** resistant subclone by VAF (8.8% vs 4.0%) - relatively untouched. For **sorafenib**, the evidence genuinely conflicts: dedicated biochemical studies show sorafenib *does* inhibit imatinib-resistant KIT mutants in both the ATP-binding pocket (including V654A) and the activation loop in vitro, whereas the authoritative Nature Reviews disease overview lumps sorafenib with regorafenib as "resistant to ATP-binding pocket" mutations. ^[2, 7, 8] This discrepancy is unresolved, but it means sorafenib has at least a *plausible* claim to hit all three, which regorafenib does not.

For her **A829P** specifically - a mutation with sparse data - the one experimental characterization (in a KIT-mutant cell model) found A829P is resistant to imatinib and **sunitinib** but retains sensitivity to nilotinib and dasatinib. [9] Neither sorafenib nor regorafenib was tested against A829P directly, so its coverage rests on the activation-loop class effect rather than on direct data. Clinically important corollary: this is a mechanistic argument *against* the sunitinib that was already prescribed for her, because sunitinib covers V654A but is predicted to miss A829P. [9, 10]

The following figure shows this differential sensitivity for the A829P clone - note the complete resistance to imatinib and sunitinib but retained apoptotic response to nilotinib and dasatinib:

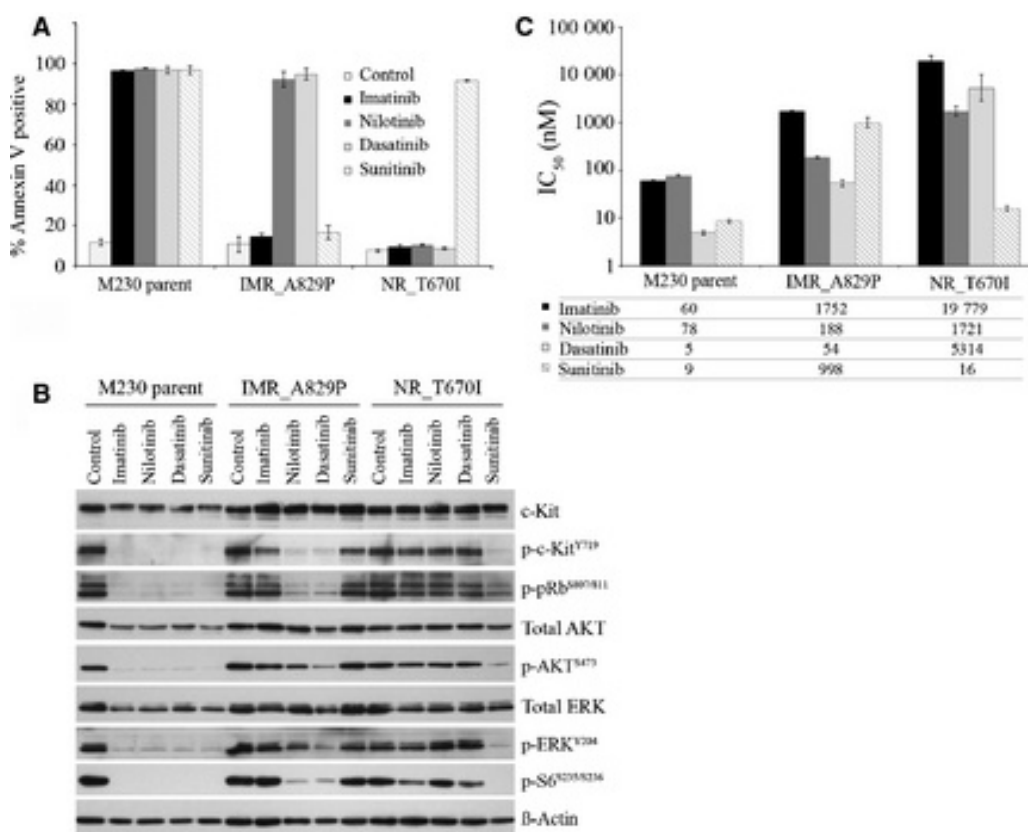


Figure 4. Clones exhibit differential sensitivity to RTK inhibitors.

Secondary c-Kit mutations confer acquired resistance to RTK inhibitors in c-Kit mutant melanoma cells. Pigment Cell Melanoma Res. 1. júla 2013.

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What the response percentages actually are (and are not)

There is **no clean per-variant objective response rate** for either drug against Q556_V559>H, V654A, or A829P individually - that granularity does not exist in the clinical literature. What exists is (a) whole-trial response/PFS figures and (b) PFS stratified by *secondary mutation class*:

- **Regorafenib, overall:** objective response rate ~4.5% in the phase III GRID trial (median PFS 4.8 months), and ~18% partial responses with median PFS 10-13.2 months in the phase II studies. By *primary* genotype, patients with a **KIT exon 11** primary mutation (like hers) had the longest median PFS at **13.4 months** in long-term phase II follow-up. By *secondary* mutation, regorafenib was significantly more effective in tumors with **exon 17/18 (activation-loop)** mutations than exon 13/14 - consistent with it working on her A829P but not her V654A. [5, 6, 11, 12, 13]

- **Sorafenib, overall:** ~10% responded and ~57-60% achieved stable disease, median PFS **6.4 months** and OS 13.5 months, in the largest third/fourth-line series (all after imatinib + sunitinib, most after nilotinib

too). [14, 15]

- **Sunitinib benchmark for the two classes** (illustrating why the class of secondary mutation matters): median PFS **7.8 months** for ATP-binding-pocket secondary mutations vs **2.3 months** for activation-loop mutations. Regorafenib shows the mirror-image pattern. [10]

Bottom line for her

Neither agent is a full-coverage drug, but they fail differently, and that difference favors sorafenib in her specific case. Regorafenib is the guideline-preferred, best-validated option (NCCN category 1 third-line) and would be expected to control her exon 11 and A829P clones - but it is predicted to be weak precisely against her *dominant* resistant subclone, V654A. [1, 6] Sorafenib is the only one of the two with in-vitro activity spanning all three of her alterations, though that breadth is contested at the review level and its GIST evidence is retrospective. [2, 7, 8] Because her resistance is genuinely polyclonal (a simultaneous ATP-pocket *and* activation-loop mutation on top of the exon 11 trunk), the more durable strategies are those built for broad coverage - **ripretinib** (broad activity across both classes, though sometimes limited against ATP-pocket variants) and, ideally, **a clinical trial** of a next-generation or combination pan-KIT regimen (e.g., a type I + type II TKI pairing, or a selective V654A inhibitor now in development). [4, 16, 17] This is also why liver-directed ablation of the few reactivated nodules, continuing imatinib, remains attractive: it physically removes the resistant clones that no single oral agent fully covers. [1]

The NCCN systemic-therapy options table places both agents in context - regorafenib as preferred third-line and sorafenib under "useful in certain circumstances":

**SYSTEMIC THERAPY AGENTS AND REGIMENS FOR UNRESECTABLE, PROGRESSIVE, OR METASTATIC DISEASE**

Imatinib-sensitive: <i>KIT</i> and <i>PDGFRA</i> mutant GIST (except <i>PDGFRA</i> exon 18 mutant GIST that is insensitive to Imatinib)				
First-line Therapy ^{c,d}	Second-line Therapy ^{c,e}	Third-line Therapy ^c	Fourth-line Therapy ^{c,e}	Additional Options After Progression on Approved Therapies ^{c,e,f}
Preferred • Imatinib ^{9,5,6} (category 1)	Preferred • Sunitinib ^{9,11} (category 1) • For patients intolerant of second-line Sunitinib, consider changing to Ripretinib 150 mg daily ^{9,12}	Preferred • Regorafenib ^{9,14} (category 1)	Preferred • Ripretinib 150 mg daily ^{9,15} (if not previously received) (category 1)	Useful in Certain Circumstances • Avapritinib ^{9,1,2,7} • Cabozantinib ¹⁶ • Everolimus/TKI ^{h,17} • Nilotinib ^{18,19} • Pazopanib ²⁰ • Ponatinib ²¹ • Ripretinib dose escalation to 150 mg BID (if previously treated with Ripretinib 150 mg daily) ^{9,22,23} • Sorafenib ²⁴⁻²⁶

Imatinib-insensitive: <i>PDGFRA</i> exon 18 mutant GIST (including <i>PDGFRA</i> D842V)	
First-line Therapy ^c	Second-line Therapy ^c
Preferred • Avapritinib ^{9,1,2,7}	Other Recommended • Dasatinib ¹³

Subsequent therapy irrespective of mutation status
Useful in Certain Circumstances • Tumor mutational burden-high (TMB-H) • Ipilimumab + Nivolumab ²⁷ (category 2B)

Other mutations	
First-line Therapy ^{c,d}	Second-line Therapy ^c
Useful in Certain Circumstances • <i>NTRK</i> gene fusion-positive GIST ▶ Larotrectinib ⁵ ▶ Entrectinib ⁵ ▶ Repotrectinib ³ • SDH-deficient GIST ▶ Sunitinib ▶ Regorafenib ▶ Pazopanib ▶ Binimetinib/Imatinib ¹⁰ (category 2B) • <i>BRAF</i> V600E-mutated GIST ▶ Dabrafenib/Trametinib ⁴	Useful in Certain Circumstances • <i>NTRK</i> gene fusion-positive GIST ▶ Repotrectinib ³ (if not previously given)

^c Therapies based on identification of driver mutations. See [GIST-B](#).
^d An additional clinical benefit may be obtained with the use of imatinib 400 mg BID upon progression with imatinib.
^e An additional clinical benefit may be obtained with the use of ripretinib 150 mg BID upon progression with ripretinib.
^f Regimens are ordered alphabetically and not according to order of preference.
^g FDA-approved TKIs for the treatment of GIST.
^h TKIs to be considered for use in combination with everolimus include imatinib, sunitinib, or regorafenib.
ⁱ Ponatinib demonstrated activity in advanced GIST pretreated with at least 3 prior TKIs, particularly in patients with *KIT* exon 11 mutant disease.

[References on GIST-E \(3 of 4\)](#)

Note: All recommendations are category 2A unless otherwise indicated.

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GIST-E
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GIST-E-2. Gastrointestinal Stromal Tumors: Systemic Therapy Agents and Regimens for Unresectable, Progressive, or Metastatic Disease - NCCN Guidelines® - Gastrointestinal Stromal Tumors p. 19 (v1.2026)

Gastrointestinal Stromal Tumors. 13. januára 2026.

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Would it be useful to look specifically at the ripretinib and cabozantinib data against V654A-containing polyclonal disease, since covering that dominant ATP-pocket clone is the main gap in both regorafenib and sorafenib?

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8. Full Foundation report review

Your question

This is the full report from Foundation

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OpenEvidence response

The full report adds two things that reshape the prior discussion: the actual VOYAGER subgroup data on regorafenib in ATP-pocket (exon 13/14) mutations, and clinically actionable non-KIT findings (a likely-pathogenic BRCA2 alteration, CDKN2A). Let me confirm the current evidence on the agents that theoretically cover all three KIT mutations and on the BRCA2 finding before synthesizing.

Bottom line: The complete report does not change the fundamental conclusion but sharpens it and corrects one point from the earlier discussion. Her resistance is **polyclonal with two secondary mutations that have opposite drug sensitivities**, so no single approved oral agent cleanly covers all three clones - which is exactly why liver-directed ablation of the few reactivated nodules (while continuing imatinib) and clinical-trial referral remain the strongest strategies. Among systemic options, the honest evidence-based choices are **sunitinib or ripretinib**, with **regorafenib** also defensible (and, importantly, better against her V654A clone than the preclinical data alone would suggest). The non-KIT findings (BRCA2, CDKN2A, MTOR, low TMB/MSS) are **not actionable in GIST** and should not redirect therapy. Two important corrections to the earlier thread appear below.

What the full report confirms - and what it changes

The three KIT variants sit in three functional zones and, critically, appear to be in **separate subclones (in trans)**, **not a single compound allele**: the VAFs differ (exon 11 17%, V654A 8.8%, A829P 4.0%), and the report treats them as distinct clones. ^[1] This matters enormously, because when an ATP-pocket and an activation-loop mutation occur *in cis* (same allele) they create a genotype resistant to essentially every TKI including ripretinib and sunitinib even in combination. ^[2] Her *trans* pattern is more favorable and is the pattern that combination or sequential strategies are designed for.

The high ctDNA tumor fraction (32%) means these calls are reliable with low false-negative risk, so the polyclonal picture can be trusted. ^[3]

KIT mutation-by-mutation coverage (corrected)

The earlier answer leaned toward sorafenib over regorafenib for the V654A clone based on preclinical potency. The full report's own **VOYAGER ctDNA data force a correction**: despite regorafenib's reduced *preclinical* potency against V654A, patients with ATP-pocket (exon 13/14) secondary mutations actually did **best** on regorafenib clinically - median PFS **7.4 months vs 1.9 months** with avapritinib - and V654A "may still be inhibited at clinically achievable doses." So regorafenib is clinically reasonable against her V654A, not a poor fit. ^[4, 5]

The second correction concerns **nilotinib and dasatinib**. The FoundationOne report lists both as retaining activity against V654A *and* the A829 activation-loop residue, which would make them the theoretical "covers-all-three" agents. However, a **2026 study directly testing nilotinib in GIST cell lines found no activity against imatinib-resistant secondary KIT mutations at clinically achievable doses** (steric hindrance on modeling), concluding there is "no biological rationale" for nilotinib in secondary-mutant GIST - and its phase III trial was negative. ^[6, 7] Dasatinib has essentially no mutation-specific GIST efficacy data and only ~3-month PFS in later lines. ^[8] So the report's theoretical dual coverage by nilotinib/dasatinib does

not translate into clinical benefit.

Her KIT clone	Sunitinib	Regorafenib	Ripretinib	Sorafenib	Ponatinib	Avapritinib	References
Exon 11 Q556_V559>H (primary, 17%)	Active	Active	Active	Active	Active	Active	[1, 4]
Exon 13 V654A (8.8%, ATP-pocket)	Strong (mPFS 15.0 mo, INTRIGUE)	Clinically good (mPFS 7.4 mo, VOYAGER)	Reduced (mPFS 4.0 mo in exon1 1+13/14)	Active in vitro (unvalidated clinically)	V654A is its escape mutation	Resistant	[4, 9, 10, 11]
Exon 18 A829P (4.0%, A-loop)	Resistant (mPFS 2.3 mo)	Active (mPFS 6.7 mo)	Strong (mPFS 14.2 mo in exon1 1+17/18)	Active (non-D816 A-loop)	Active	Active	[4, 9, 10]

The pattern is the crux: **sunitinib is best for V654A but fails A829P; ripretinib is best for A829P and only partial against V654A; regorafenib covers A829P and is clinically acceptable against V654A.** The only randomized data in the exact co-occurring ATP-pocket + activation-loop subgroup comes from the INTRIGUE ctDNA analysis (n=22), where **ripretinib and sunitinib performed similarly** (HR 1.07) - neither is a clear winner when both clones coexist. [9] The forest plot below shows these opposing subgroup effects:

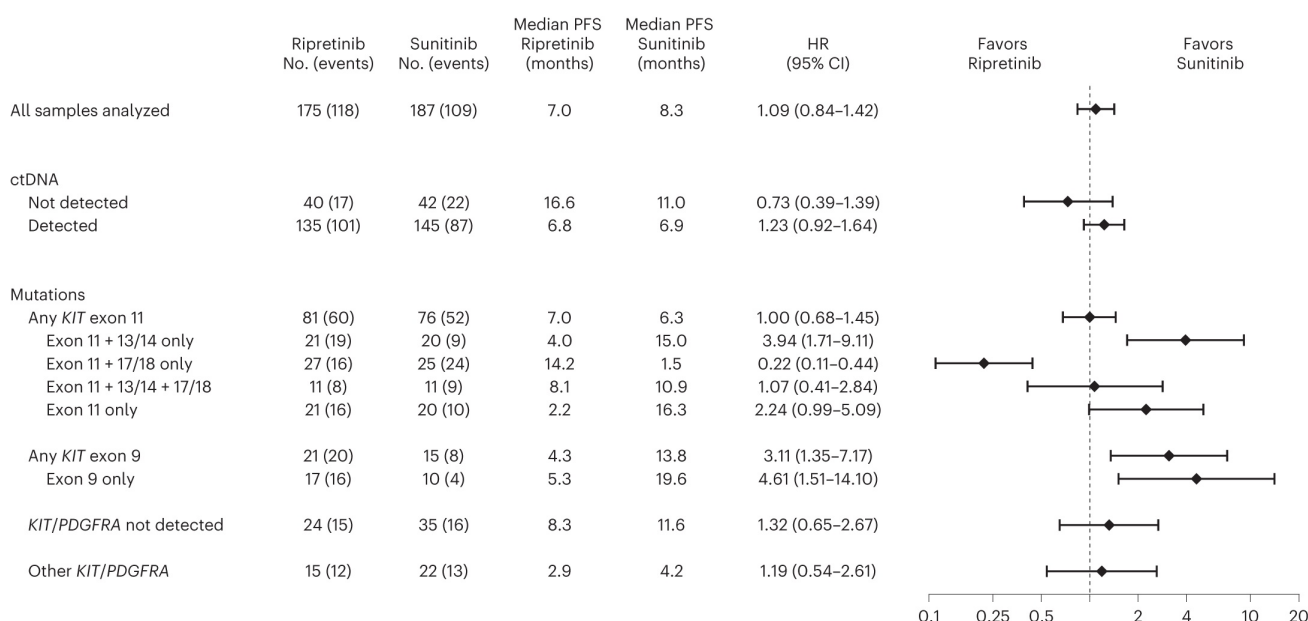


Figure 3. Forest plot of PFS by mutational subgroup.

Ripretinib versus sunitinib in gastrointestinal stromal tumor: ctDNA biomarker analysis of the phase 3 INTRIGUE trial. Nat Med. 1. februára 2024.

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Regorafenib vs sorafenib - where this leaves the earlier question

Both remain reasonable VEGFR-multikinase options, but the full report tilts the balance slightly back toward **regorafenib** on evidence quality: it is NCCN category 1 with no resistance flag, and its VOYAGER data show real-world activity against *both* her secondary clones (7.4 mo ATP-pocket, 6.7 mo A-loop).^[4, 5] The earlier tolerability argument for sorafenib still stands (lower grade ≥ 3 hypertension, fewer dose modifications, no hepatotoxicity boxed warning) and is genuinely important given her hemorrhagic PRES - so if a VEGFR-TKI is chosen, the decision is a legitimate trade-off between regorafenib's stronger mutation-matched evidence and sorafenib's gentler toxicity profile, both started at reduced/escalating dose with tight BP control.^[12] Neither, however, is as well-matched to her *polyclonal* profile as sunitinib or ripretinib, and none addresses the dominant risk that whichever clone is not covered will expand.

The most rational systemic strategy for this exact profile

The **bezuclastinib + sunitinib** combination pairs a type I inhibitor (covering exon 17/18 activation-loop clones like her A829P) with sunitinib (covering the exon 13 V654A) - designed precisely for the complementary polyclonal pattern she has. The PEAK phase III trial reported median PFS **16.5 vs 9.2 months** and ORR 45.6% vs 25.8% versus sunitinib alone; bezuclastinib is not yet approved, so this is a trial/access question.^[13] This is the single most biologically logical option and worth pursuing if reachable - the FoundationOne report itself flags a bezuclastinib-combination trial (NCT05957367 targets are different, but bezuclastinib access should be explored).

Non-KIT findings - what NOT to chase

- **BRCA2 T3085fs (1.5% VAF):** Despite the "likely pathogenic" ClinVar flag, this is almost certainly **not a germline variant and not actionable in her GIST**. A true germline variant would appear near 50% VAF in cfDNA, not 1.5%, and her 2022 BRCA Plus germline panel was negative. At 1.5% VAF the most likely origins are **clonal hematopoiesis** (BRCA2 is a recognized CHIP gene; ~66% of somatic-only BRCA2 cfDNA calls are CH-derived, and FoundationOne Liquid CDx has no matched white-cell subtraction) or a biologically irrelevant subclonal passenger. Even if confirmed somatic, GIST is **not a BRCA-associated cancer**, biallelic loss is not expected, and PARP inhibitors, platinum, and trabectedin have no established role - indeed **trabectedin was already tried and is specifically ineffective in GIST** (two phase II trials: 0% responses). Recommend confirming the prior germline result rather than acting on this call.^[3, 14, 15, 16, 17, 18, 19]
- **CDKN2A rearrangement (12.8%):** Real and prognostically meaningful - CDKN2A loss is the most common non-KIT event in GIST and marks more aggressive disease/worse survival, consistent with her secondary progression. CDK4/6 inhibitors are investigational only in GIST.^[1, 20]
- **MTOR A1459D (0.21%) and the VUS (IGF1R 51%, ERBB4 48%):** The MTOR call is at noise-level VAF; the high-VAF VUS are likely germline SNPs/passengers. None is actionable.^[1]
- **bTMB 3 Muts/Mb, MSS:** This definitively closes the earlier immunotherapy question - there is **no biomarker rationale for nivolumab/ipilimumab**, alone or with a TKI.

Recommendations

1. **Continue imatinib and prioritize liver-directed local therapy** (MWA/cryoablation or DEB-TACE/Y-90 per the earlier lesion-anatomy discussion) for the few reactivated nodules - this physically removes the resistant clones no single oral agent fully covers.^[2]
2. **Pursue a clinical trial**, ideally **bezuclastinib + sunitinib**, as the most profile-matched systemic option; this is the strongest rationale for her *trans* polyclonal disease.^[13]
3. **If a standard approved agent is used now:** sunitinib or ripretinib are the best-supported single agents; regorafenib is a reasonable category-1 alternative with genuine activity against both her secondary clones. Given her hemorrhagic PRES, start any VEGFR-active agent (sunitinib, regorafenib, sorafenib) at reduced

dose with home BP monitoring and a low threshold for brain imaging. [4, 12]

4. Do not pursue PARP inhibitors, platinum, trabectedin, immunotherapy, nilotinib, or dasatinib based on this report - the supporting findings are either non-actionable in GIST or contradicted by recent data. [6, 17]

5. Confirm germline status (verify the 2022 panel was comprehensive) to close out the BRCA2 question rather than act on a 1.5% cfDNA call. [14]

Would it be useful to review the eligibility criteria and access pathways for the bezuclastinib + sunitinib (PEAK) program and other type I + type II KIT-inhibitor combination trials that specifically address her co-occurring V654A and A829P clones?

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confound plasma cell-free DNA testing for DNA repair genes used to select poly(ADP) ribose polymerase inhibitor therapy. Paired plasma and whole-blood testing found CHIP variants at 2% or more variant fraction in 13/69 men (19%) and PARPi-gene CHIP interference in 7/69 men (10%), most often ATM. Applicability is limited by the small, single-center, retrospective design. Design N: N=69 PICO Population: Men with advanced prostate cancer (metastatic disease or with rising PSA following localized therapy) Condition: Advanced prostate cancer undergoing cfDNA testing Intervention: Paired cfDNA and whole-blood control testing (UW-OncoPlexCT) Result Outcome: 13/69 (19%; 95% CI, 10%-30%) had CHIP variants at $\geq 2\%$ VAF in cfDNA; 7/69 (10%; 95% CI, 4%-20%) had CHIP in PARPi-eligibility DNA repair genes (ATM n=5, BRCA2 n=1, CHEK2 n=1) Research in Context Foundational prostate cancer evidence linking CHIP to false-positive cfDNA DNA-repair alterations for PARP inhibitor eligibility Reinforced by later pan-cancer, prostate cancer, and real-world cohorts showing recurrent CH-derived BRCA2/CHEK2/ATM signals in plasma-only assays Now reflected in NCCN caution on ctDNA-only testing; metastatic tissue or paired blood-control testing preferred when feasible

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9. Combining TKIs with downstream pathway drugs

Your question

I've read that TKI can be used with some additional "downstream" drug which acts against secondary mutations

OpenEvidence response

Evidence label shown by OpenEvidence: Limited Evidence

The concept being referenced is **downstream pathway inhibition** - combining a KIT TKI (usually imatinib) with a drug that blocks a node *below* KIT in the signaling cascade (PI3K/AKT/mTOR or RAF/MEK/ERK), so that resistant subclones are suppressed regardless of *which* secondary KIT mutation they carry. Let me pull the actual trial evidence for the main combinations.

The concept is real and biologically elegant, but the honest answer is that it has consistently failed in the imatinib-resistant setting - which is exactly her setting. The strategy you read about is combining a KIT TKI with an inhibitor of a signaling node *downstream* of KIT - either the RAS→MEK→ERK arm or the PI3K→AKT→mTOR arm. The appeal is precisely what you describe: every secondary KIT mutation, regardless of whether it sits in the ATP-pocket (her V654A) or the activation loop (her A829P), ultimately funnels through these same two downstream pathways, so hitting a convergent node should be **mutation-agnostic** and cover all subclones at once. ^[1,2] The problem is a subtle but decisive one, and it is worth understanding before pursuing this for her.

Why the biology is sound

All mutant KIT oncoproteins - primary and every secondary variant - signal through RAS/MAPK and PI3K/AKT/mTOR.^[1, 2] PI3K is the dominant survival pathway: in knock-in mice, disabling only the PI3K-binding site on KIT completely prevented GIST formation.^[3] Preclinically, the strategy works impressively: dual blockade of MEK and PI3K/mTOR largely **phenocopies direct KIT blockade regardless of the heterogeneity of primary and secondary mutations**, and a triple KIT+PI3K+MEK cocktail eliminated even drug-persistent "quiescent" cells that survive after drug withdrawal.^[1, 4] This is the data that generates the enthusiasm you've read about.

The following schematic from the Heinrich et al. review (Nature Reviews Clinical Oncology) maps these downstream-combination strategies (panel a) alongside the other emerging GIST approaches:

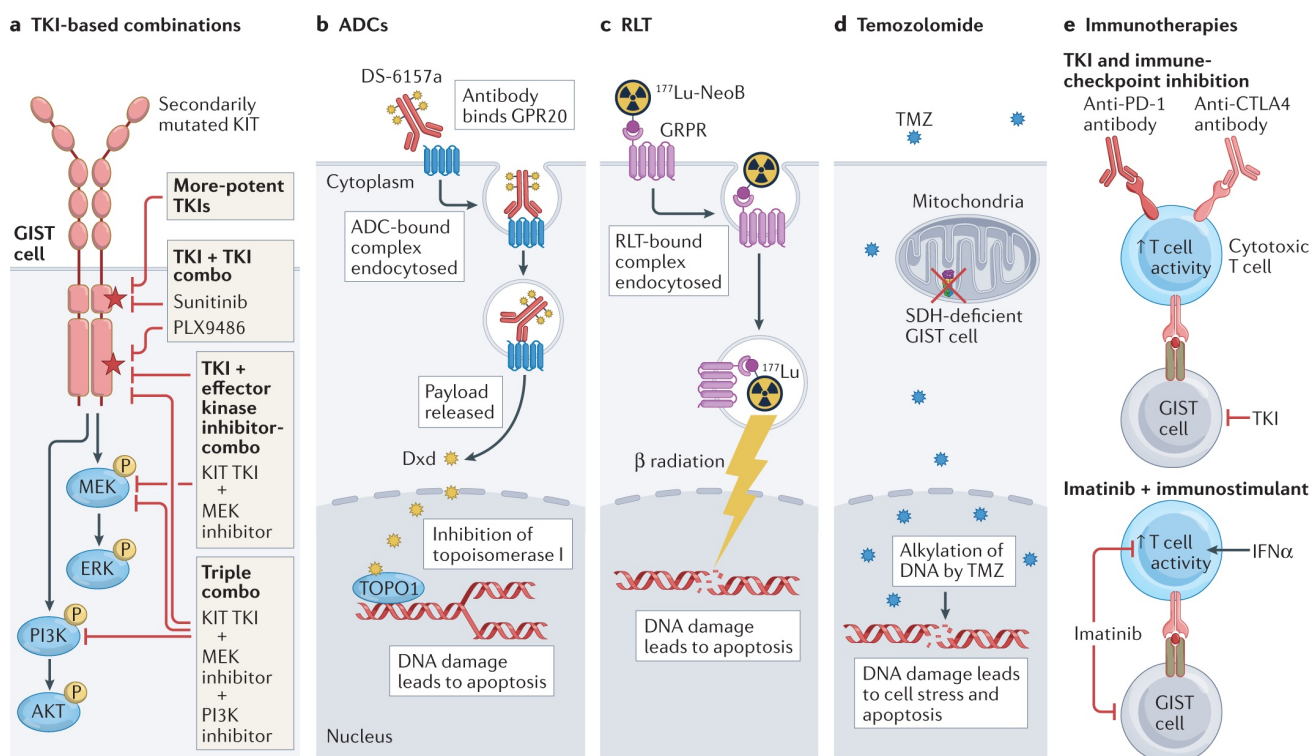


Figure 4. New therapeutic approaches exploiting different elements of GIST biology.

New treatment strategies for advanced-stage gastrointestinal stromal tumours. Nat Rev Clin Oncol. 1. mája 2022.

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Why it breaks down clinically - the critical caveat

The catch is that these are **combinations built on imatinib**, and in imatinib-resistant disease imatinib no longer inhibits KIT. So "imatinib + a downstream drug" effectively becomes **downstream-drug monotherapy against a tumor whose KIT is fully active** - and blocking one downstream arm while KIT drives all the others is not enough.^[5] Every clinical trial testing this in the resistant setting has borne that out:

Combination (resistant setting)	N	Line	ORR	Median PFS	References
Everolimus (mTOR) + imatinib	75	2nd-3rd+	0-2%	1.9-3.5 mo	[6, 7]

Combination (resistant setting)	N	Line	ORR	Median PFS	References
Alpelisib (PI3Kα) + imatinib	56	3rd	2.9%	2.0 mo	[8]
Buparlisib (pan-PI3K) + imatinib	60	3rd+	0%	3.5 mo	[9]
Binimetinib (MEK) + imatinib	1b	resistant	limited RECIST activity	not reported	[10]

A formal review of these trials concluded they "failed to increase tumor responses and extend the clinical benefit," attributing it directly to "insufficient suppression of KIT... with imatinib in the context of imatinib-resistant disease." This is the point I want to be candid about rather than encouraging: for a patient already progressing on imatinib, adding everolimus, alpelisib, or a MEK inhibitor to imatinib is not a promising path, and the trial data are quite consistent on that. [11]

The one genuinely striking result - imatinib + binimetinib with a 69% RECIST response rate and 29.9-month PFS - came exclusively in **treatment-naïve** patients, where imatinib still fully inhibits KIT and the MEK inhibitor adds benefit by destabilizing the GIST lineage factor ETV1 and killing drug-persistent cells. [12, 13] That is not her situation, and there was no randomized comparison to imatinib alone. [5]

What this means specifically for her

Her polyclonal profile (exon 11 trunk + V654A + A829P, in *trans*) is the textbook rationale *for* downstream targeting, but the resistant-setting failures above apply directly. Two nuances refine this:

- **The theoretically correct version of the downstream strategy is being tested with a broad-spectrum KIT inhibitor, not imatinib.** Because ripretinib (unlike imatinib) still suppresses KIT in resistant disease, ripretinib + MEK inhibitor synergizes preclinically and prevents resistant-colony outgrowth even after drug removal - this is the combination that logically overcomes the "downstream monotherapy" trap, but it has **no clinical data yet**. If a downstream combination is pursued, it should be in this form (a potent KIT inhibitor + downstream agent), ideally on a trial. [14]
- **Her MTOR A1459D and CDKN2A findings do not rescue this approach.** The MTOR variant sits at only 0.21% VAF (near noise) and MTOR mutations are not established GIST drivers; if it were real and driving mTOR activation, an ATP-competitive mTOR kinase inhibitor (e.g., sapanisertib) would be more logical than everolimus, but this is speculative. CDKN2A loss predicts worse prognosis but CDK4/6 inhibition has been clinically disappointing in GIST. [5, 7]

The more rational way to "cover all three mutations"

Rather than adding a downstream drug, the strategy that actually addresses polyclonal resistance at the receptor level - and the one I'd steer toward - is a **complementary two-TKI combination**: bezuclastinib (type I, covers her A829P activation-loop clone) + sunitinib (type II, covers her V654A ATP-pocket clone). The PEAK phase III trial showed median PFS 16.5 vs 9.2 months versus sunitinib alone. [15, 16] This is the same conceptual goal - cover every subclone - but achieved by keeping the receptor itself blocked in all clones, which is mechanistically sounder than downstream inhibition layered on a drug that no longer works. Next-generation pan-KIT inhibitors (NB003, IDRX-42) and HSP90-mediated KIT degradation (pimipitespib, approved in Japan) are additional mutation-agnostic receptor-level options. [16, 17]

The overall bottom line: the downstream-combination idea is legitimate science, but for **imatinib-resistant** disease it has failed repeatedly, and it should not displace liver-directed ablation of her few reactivated nodules, continued imatinib, and referral to a complementary-TKI or pan-KIT trial as the higher-yield strategies. [11, 18]

Would it be helpful to look at the eligibility and dosing details of the ripretinib-based and bezucastinib + sunitinib combination trials, or at the pimitespib (HSP90) data as a mutation-agnostic option, to see which is most realistically accessible for her?

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10. Options beyond TKIs

Your question

if TKIs show PFS 14 months (a year until a new mutation appear and situation becomes even more complicated), this is not very good solution, even compared to 2.5 years on imatinib. Are there other options - ADCs, vaccines, some payload bearing nanoparticles, or another form of therapy?

OpenEvidence response

The core tension you're naming is real: a 14-month PFS on a next TKI buys less than her imatinib run and resets the clock on a more complex clone. Let me research the non-kinase modalities - antibody-drug conjugates, radioligand therapy, cellular/vaccine immunotherapy, and targeted-delivery approaches - and what's actually in trials for GIST.

Your instinct is sound, and it deserves a direct answer rather than reflexive reassurance: **a next-line TKI delivering ~14 months before a new clone emerges is a genuinely modest return compared with her 2.5-year imatinib benefit, and it is reasonable to ask whether a mechanistically different modality could do better.** The honest landscape, though, is sobering - almost every non-TKI approach in GIST is either preclinical, has already failed in trials, or produces *shorter* disease control than the TKIs, not longer. There are a few real exceptions worth taking seriously. Below is the candid appraisal by modality, ranked roughly by how actionable each is for her today.

A useful way to see the whole non-TKI field at once is the schematic from the Klug et al. review in *Nature Reviews Clinical Oncology*, which maps the five main strategies - TKI combinations, the GPR20 ADC, GRPR radioligand therapy, SDH-specific approaches, and immunotherapy:

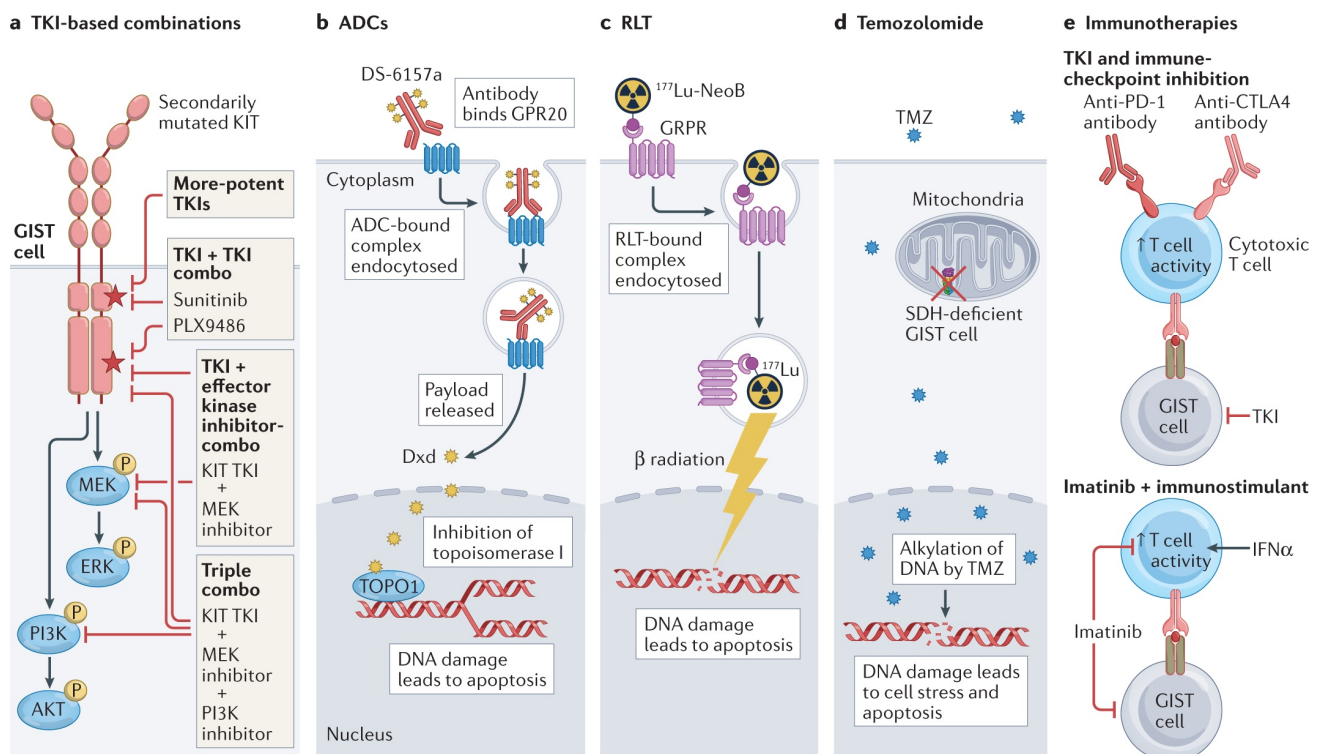


Figure 4. New therapeutic approaches exploiting different elements of GIST biology.

New treatment strategies for advanced-stage gastrointestinal stromal tumours. Nat Rev Clin Oncol. 1. mája 2022.

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The reality check on "beating" a 14-month TKI

Two framing points matter before the modality tour. First, the single most effective non-systemic option in her exact situation - few reactivated nodules in otherwise dormant/cystic liver masses - remains **local ablation or hepatic-artery-directed therapy while continuing imatinib**, precisely because it physically removes the resistant clones no drug fully covers and does not reset a systemic clock. [2] Second, the modality with the most mature *systemic* non-TKI evidence, **Y-90 radioembolization**, produced its **worst** results in exactly her genotype: in the 2025 Hohenberger series, KIT exon 11 + secondary mutations had hepatic PFS of only **7 months** versus 25 months in wild-type disease - a caution that "different mechanism" does not automatically mean "better outcome" in polyclonal KIT-driven disease. [3] So the bar she is setting (beat 14 months durably) is one that no non-TKI modality has yet cleared in her subgroup. What follows is where the genuine opportunities lie.

HSP90 inhibition - the most credible mutation-agnostic drug

Pimipitespib (TAS-116) is the one non-TKI *drug* that is approved (Japan, 2022) and works by a mechanism indifferent to which secondary KIT mutation is present: it destabilizes KIT as an HSP90 client protein and drives its degradation, so V654A and A829P are irrelevant to its activity. [4, 5] In the phase III CHAPTER-GIST-301 trial (4th-line, after imatinib/sunitinib/regorafenib), median PFS was **2.8 vs 1.4 months** (HR 0.51) with a crossover-adjusted OS HR of **0.42**, and hazard ratios favored pimipitespib across all mutation subgroups including exon 13/14 and 17/18. [6] Real-world Japanese series show median PFS around 4.2-4.6 months. [7, 8]

The candid read: this confirms the *concept* that mutation-agnostic KIT degradation works, but the PFS is shorter than ripretinib's 6.3 months - so pimipitespib is not the answer to "beat 14 months." Its practical value for her is (a) it is a real option if she can access it (not approved in the EU/US), and (b) the preclinical

pimipitespib + sunitinib combination, which synergistically hit KIT and angiogenesis, is the more interesting future direction. ^[9] Earlier HSP90 inhibitors (IPI-504, ganetespib, AUY922) all failed on toxicity or futility, so this class has a poor track record apart from pimipitespib. ^[10]

Antibody-drug conjugates - one just-opened trial is directly relevant

The ADC story in GIST is mostly cautionary but has one live, on-point option:

Agent	Target / payload	Status	Key point for her	Refs
NN3201	c-KIT / MMAE (Fc-silenced)	Phase I recruiting (NCT06805825)	KIT ADC engineered to avoid the mast-cell reactions that killed LOP628; active against KIT regardless of secondary mutation	[11]
DS-6157a (ifinamab deruxtecán)	GPR20 / DXd	Discontinued	GIST-specific target, but phase I ORR only 2.9%; shrinkage mostly in wild-type GIST	[12, 13]
LOP628	c-KIT / DM1	Halted	Anaphylactoid mast-cell degranulation via FcγR co-engagement	[14]
SCF-DM1 / KIT-d-MMAE (aptamer)	KIT extracellular domain	Preclinical	KIT-d-MMAE tested specifically in the V654A GIST-430/654 model, superior to imatinib	[15, 16]

The conceptually attractive point is that **KIT-directed ADCs and ligand/aptamer conjugates bind the extracellular domain, which her intracellular V654A and A829P mutations do not alter** - so they are inherently mutation-agnostic. ^[16] The aptamer-drug conjugate KIT-d-MMAE is the only agent in the entire field explicitly validated against her V654A clone, but it is preclinical. ^[16] The actionable item is **NN3201**, a next-generation Fc-silenced KIT ADC now in a first-in-human trial for c-KIT-expressing solid tumors - GIST is the paradigm c-KIT tumor, and this is worth screening for. ^[11]

Radioligand therapy - promising target, unresolved heterogeneity

GRPR (bombesin receptor) is highly expressed in ~84% of GISTs, independent of TKI-resistance status, making it a logical radioligand target. ^[17, 18] ¹⁷⁷**Lu-NeoB** reached first-in-human testing (NeoRay, NCT03872778): the phase I established an MTD/RP2D of 9.25 GBq with favorable renal/pancreatic dosimetry, but **GIST-specific efficacy has not yet been reported**. ^[19] The major caveat is heterogeneity - in the ⁶⁸Ga-NeoB imaging study, roughly **one-third of GIST patients had no tracer uptake in any lesion**, and uptake varied lesion-to-lesion within patients, so a ⁶⁸Ga-NeoB PET would be mandatory to know whether her tumor is even targetable. ^[20] This is a legitimate avenue but currently investigational and selection-dependent.

Immunotherapy beyond checkpoints - biologically active but small

Her low TMB/MSS status correctly predicts poor checkpoint-inhibitor activity, but two antigen-agnostic immune approaches showed real signals in tiny GIST studies:

- **Ilxadencel** (intratumoral allogeneic dendritic cells): phase I in 6 TKI-refractory GIST patients - 2/6 partial responses by Choi, both in KIT exon 11 tumors; mechanism recruits endogenous NK/DC without needing defined neoantigens, so low TMB is not disqualifying. ^[21]

- **Imatinib + peginterferon-α2b**: phase II (n=8) with striking Th1/NK activation and durable control, but never expanded in over a decade. ^[22]

- **TIL therapy** (autologous tumor-infiltrating lymphocytes + IL-2) is now in a dedicated GIST phase II trial - a genuinely novel cell-therapy option (see dashboard). ^[23]

Anti-KIT CAR-T works in GIST xenografts but faces on-target/off-tumor toxicity against normal KIT+ hematopoietic stem cells and mast cells, and has no clinical data. ^[24]

What NOT to pursue

Nanoparticle/liposomal imatinib formulations, oncolytic reovirus/Salmonella, boron neutron capture therapy, and KIT PROTAC degraders are **all preclinical with no human GIST data**, and PROTACs specifically have no KIT-directed molecule even in development for this disease. ^[16, 25, 26] These are not near-term options despite their conceptual appeal, and I would not want to overstate them.

Actionable clinical trials

Given refractory, polyclonal, next-line disease, trial referral is the strongest recommendation - and several of the modalities above map to open studies.

Clinical TrialsGastrointestinal Stromal Tumor · KIT mutation 10 trials

10 trials

Highly relevant · Matches disease · biomarker

A Master Protocol to Evaluate DCC-3009 in Gastrointestinal Stromal Tumor (GIST)

Recruiting Phase 1/2

Sponsor: Deciphera Pharmaceuticals, LLC

Updated: 5. 2. 2026

Matches disease

First-in-human Study of IDRX-42 in Participants With Metastatic and/or Unresectable Gastrointestinal Stromal Tumors

Recruiting Phase 1

Sponsor: IDRX, Inc., a wholly owned subsidiary of GSK, LLC

Updated: 4. 5. 2026

Matches disease · biomarker

A Study of Ziftomenib in Combination With Imatinib in Patients With Advanced Gastrointestinal Stromal Tumors (GIST)

Recruiting Phase 1

Sponsor: Kura Oncology, Inc.

Updated: 5. 2. 2026

Matches disease

Autologous Tumor Infiltrating Lymphocytes With Interleukin-2 for the Treatment of Locally Advanced, Recurrent or Metastatic Gastrointestinal Stromal Tumors

Not yet recruiting Phase 2

Sponsor: Joal Beane

Updated: 15. 6. 2026

Matches disease · biomarker

A Study of Inlxisertib (DCC-3116) in Combination With Anticancer Therapies in Participants With Advanced Malignancies

Recruiting Phase 1/2

Sponsor: Deciphera Pharmaceuticals, LLC

Updated: 13. 7. 2026

Matches disease

LRG GIST Patient Registry

Recruiting

Sponsor: The Life Raft Group

Updated: 13. 7. 2026

Highly relevant · Matches disease · Basket trial

A Study of the c-Kit Specific Antibody-Drug Conjugate NN3201 for Advanced and/or Metastatic Solid Tumors Known to Express c-Kit

Recruiting Phase 1

Sponsor: Novelty Nobility, Inc.

Updated: 23. 9. 2025

Matches biomarker · Basket trial

TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer

Recruiting Phase 2

Sponsor: American Society of Clinical Oncology

Updated: 15. 5. 2026

Matches biomarker · Basket trial

SYNERGY-AI: Artificial Intelligence Based Precision Oncology Clinical Trial Matching and Registry

Recruiting

Sponsor: Massive Bio, Inc.

Updated: 28. 10. 2025

Matches disease · Basket trial

Administration of Autologous T-Cells Genetically Engineered to Express T-Cell Receptors Reactive Against Neoantigens in People With Metastatic Cancer

Recruiting Phase 2

Sponsor: National Cancer Institute (NCI)

Updated: 24. 8. 2026

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The search returned 10 available trials. The most relevant to this non-TKI question among the top trials in the displayed dashboard are the **KIT-directed ADC NN3201** ([NCT06805825](#), phase I in c-KIT-expressing solid tumors) - the single best match to the "payload-bearing" mutation-agnostic strategy you're asking about; the **autologous TIL + IL-2 cell therapy** trial in GIST ([NCT07647068](#)) - a true non-TKI immunotherapy option; and **inlexisertib (DCC-3116, an autophagy/ULK inhibitor) + ripretinib** ([NCT05957367](#)), which pairs a KIT inhibitor with a genuinely different downstream mechanism. Next-generation broad-spectrum KIT inhibitors also appear (IDRX-42, [NCT05489237](#); DCC-3009, [NCT06630234](#)) and, while still TKIs, are designed for exactly her polyclonal pattern.

To refine eligibility: would you confirm whether she is able and willing to travel internationally for a trial (several ADC/cell-therapy studies are US- or Asia-based), and whether a fresh tumor biopsy is feasible - since the ADC and TIL options typically require confirmation of target expression (c-KIT, GPR20) or tumor tissue for lymphocyte expansion?

The overall bottom line: your skepticism about serial TKIs is well-founded, but the non-TKI field has not yet produced a modality that reliably beats a next-line TKI in KIT-mutant polyclonal disease. The two most rational moves are to **exploit local therapy aggressively** for the few active liver nodules while continuing imatinib, and to **screen for the KIT-ADC (NN3201) or TIL trial** as the mechanistically distinct options most likely to sidestep her secondary mutations. [2, 11]

Would it be useful to look in detail at the NN3201 ADC trial's eligibility criteria and required c-KIT expression threshold, or at the emerging pimitespib + sunitinib combination data, to judge which mutation-agnostic strategy is most realistically within reach for her?

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11. KIT and SDH pathology findings

Your question

there was one anomaly when her biopsy was reviewed by Manor clinic (it was when she was on imatinib) - they found she has both ckit and sdh deficiency what don't go together (or those were two cultures of cells?)

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OpenEvidence response

The key to resolving this lies in how SDH-B immunohistochemistry is interpreted - "retained" versus "lost" staining - because the actual Prime Pathology report wording matters enormously here. Let me confirm the biology of SDH-B IHC interpretation and the KIT/SDH mutual-exclusivity rule before giving a definitive read.

There is almost certainly no true biological anomaly here - the apparent "cKIT + SDH deficiency" combination is a misreading of the pathology report, and the actual Prime Pathology findings are internally consistent with a conventional KIT-mutant, SDH-proficient GIST. The key is what the SDH-B immunohistochemistry actually showed.

What the reports actually say

The Prime Pathology report (R181/2023) documents that the tumor cells express DOG1, CD34, and cKIT, and then states that **SDH-B protein expression is retained/preserved** ("сохранена экспрессия білка SDH-B"), which the report itself interprets as indicating the *absence* of SDH deficiency. This is the decisive line. In SDH-B immunohistochemistry, staining is scored as either **retained/positive** (strong granular cytoplasmic staining = normal, SDH-proficient) or **lost/negative** (complete absence = SDH-deficient).^[1] Her tumor had **retained** SDH-B - meaning it is **not** SDH-deficient.^[1]

Dr. Tahover's Manor/Assuta telemedicine summary garbled this. The phrase "using cKIT gene staining and SDH hormone deficiency" reads as though SDH deficiency was *found*, but that is a transcription/translation artifact of a Zoom consult - and tellingly, in her own recommendations she correctly notes that cKIT mutation and SDH deficiency are "mutually exclusive per the specialized literature," and advised re-biopsy *only if there were no response*, precisely because she was flagging the apparent contradiction as something to resolve, not confirming it.^[2]

Why the two findings don't actually conflict

By definition, **SDH-deficient GISTs are KIT/PDGFRA wild-type** - they lack activating KIT or PDGFRA mutations and are driven instead by loss of the SDH complex (usually germline SDHA/B/C/D mutation or SDHC promoter methylation).^[3, 4] The two mechanisms are considered mutually exclusive because SDH-deficient tumors, almost without exception, retain wild-type KIT.^[2] Several features of her case place her firmly in the conventional KIT-mutant category:

- **A clear, recurrent KIT exon 11 mutation** (Q556_V559>H) confirmed on multiple platforms (OncoMine 2022, FoundationOne Liquid 2026) - the single most common and most imatinib-responsive GIST genotype.^[4]

- **Retained SDH-B by IHC** - the direct marker excluding SDH deficiency.^[1]

- **A 2.5-year response to imatinib.** SDH-deficient GIST is characteristically imatinib-resistant, with objective response rates under ~5-15%. A durable multi-year imatinib benefit is itself strong evidence *against* SDH deficiency.^[5, 6]

Was this "two cultures of cells"?

No. The Prime Pathology report lists several specimen/block numbers (23SP045591, 23SP046733, the original Ukrainian block 48278/22, and cell block 23CN007785) - these are **different tissue blocks and levels from the same tumor** submitted for review, not two distinct cell populations with different molecular identities. All showed the same GIST immunophenotype (DOG1/CD34/cKIT positive, SDH-B retained). There is no evidence in these documents of a genuine collision tumor or two separate clones with divergent SDH status.

The one caveat worth acknowledging

True coexistence is *vanishingly* rare but not physically impossible: a 2026 case report described a small-bowel GIST with both a somatic KIT exon 9 duplication *and* a germline SDHC mutation. Even there, however, **SDH-B was retained, KIT remained the dominant driver, and the tumor responded to imatinib** - the authors concluded oncogenic KIT signaling dominated behavior.^[2] So even the rare exception reinforces the same practical conclusion for your patient: **treat this as KIT-driven disease.**

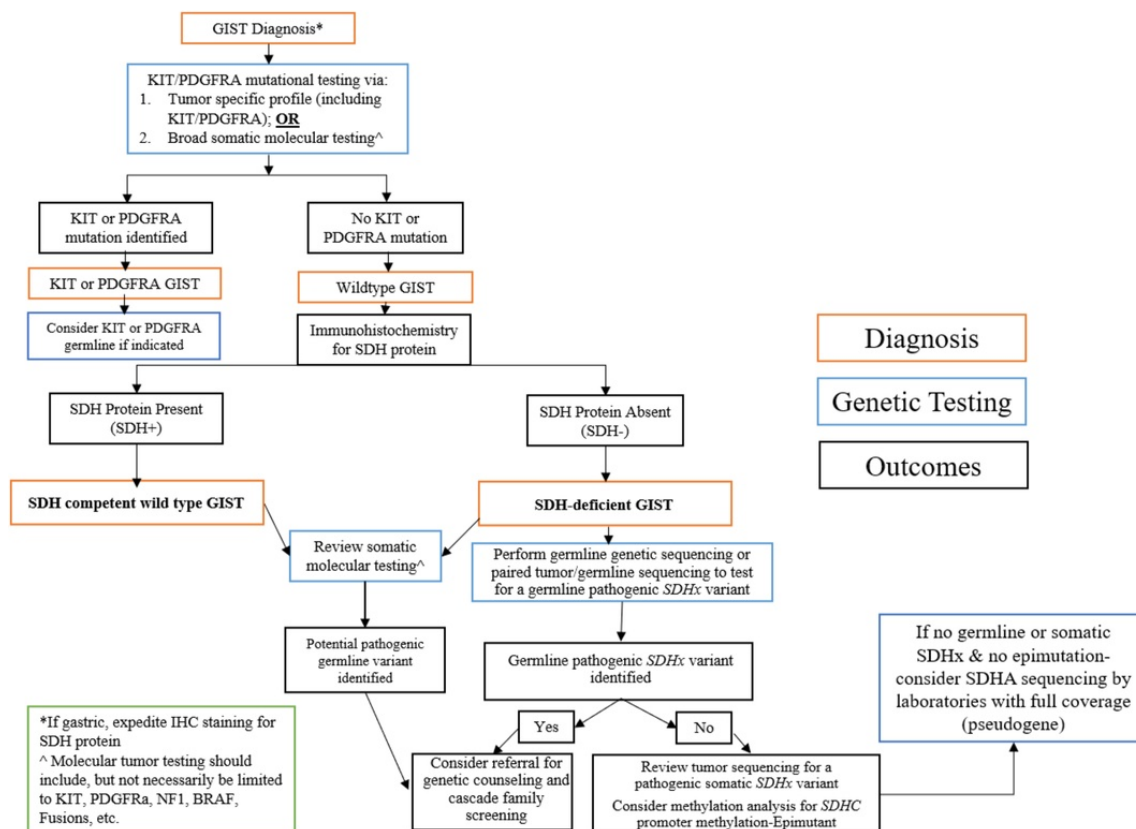
Clinical bottom line

Her tumor is a conventional **KIT exon 11-mutant, SDH-proficient gastric GIST**, and the entire treatment framework discussed previously (imatinib benefit, secondary V654A/A829P resistance, TKI sequencing, local liver therapy) applies unchanged. The "SDH deficiency" label appears to be an artifact of how retained SDH-B staining was summarized in the telemedicine note. If any residual doubt remains, the cleanest way to close it definitively would be to (1) obtain the *actual* SDH-B IHC slide report/score rather than the consult summary, and (2) note that her documented imatinib response and NGS-confirmed KIT exon 11 mutation already exclude clinically meaningful SDH deficiency. Importantly, **her germline BRCA/SDH testing has practical relevance only if SDH deficiency were real** - since it is not, there is no need to withhold imatinib or pursue the SDH-deficient treatment algorithm (sunitinib/regorafenib for that indication).^[5]

Would it be helpful to review whether formal germline SDHx testing is nonetheless warranted here for completeness, or to focus instead on confirming the original SDH-B IHC score directly with the reporting pathology laboratory?

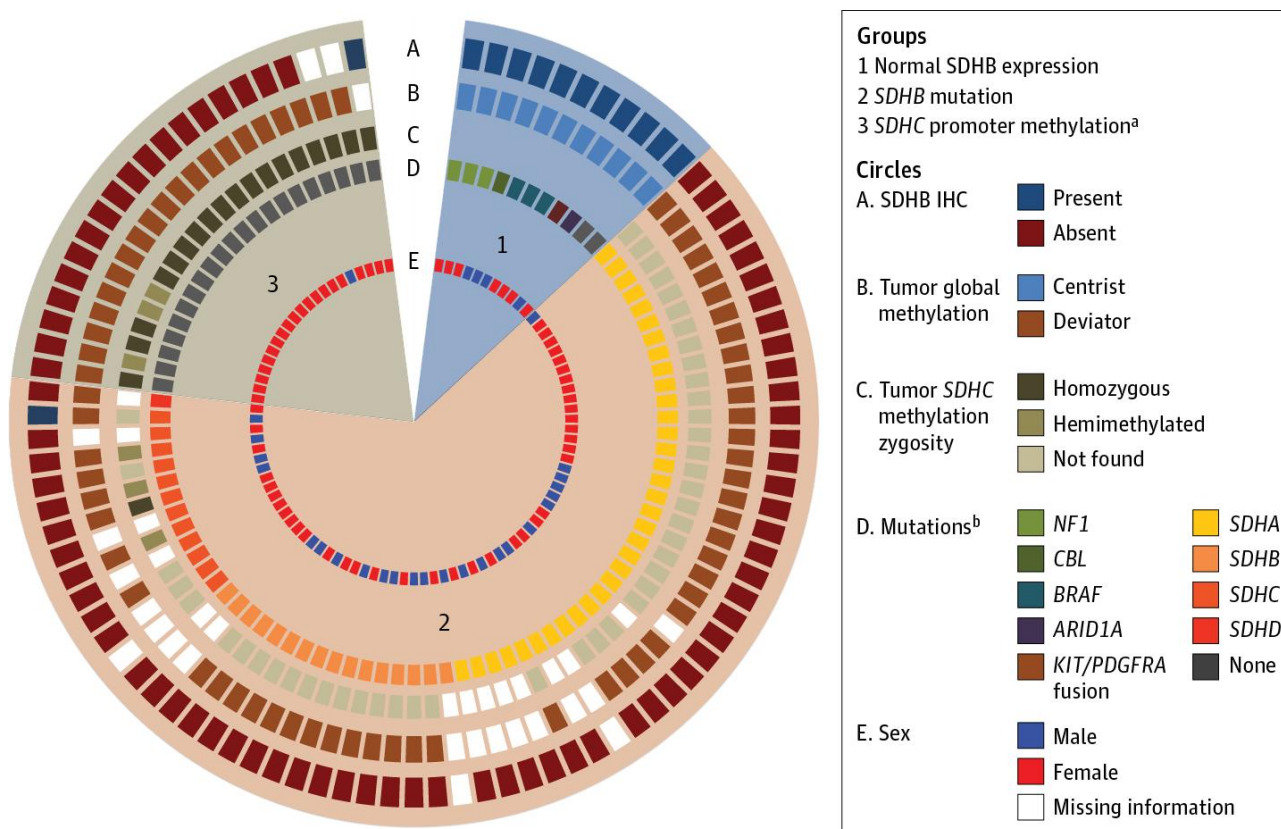
Relevant images

6 items



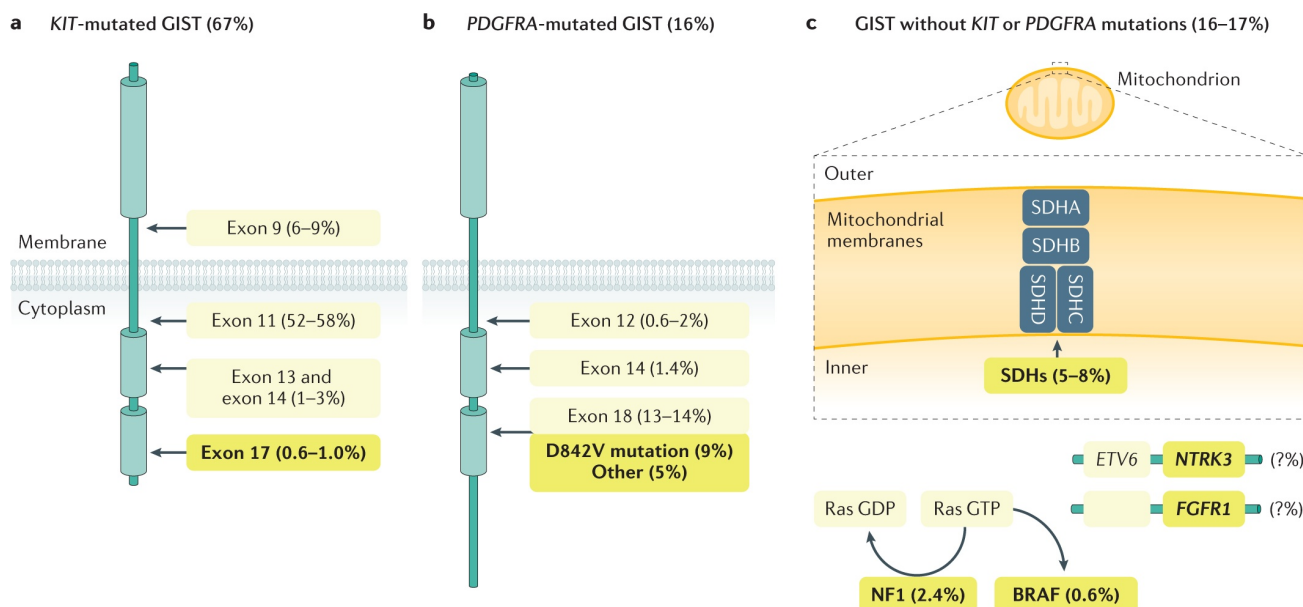
Author-recommended process to evaluate GIST via mutational testing and/or SDH workup with subsequent germline testing needs.

Cancer Medicine 1. februára 2025



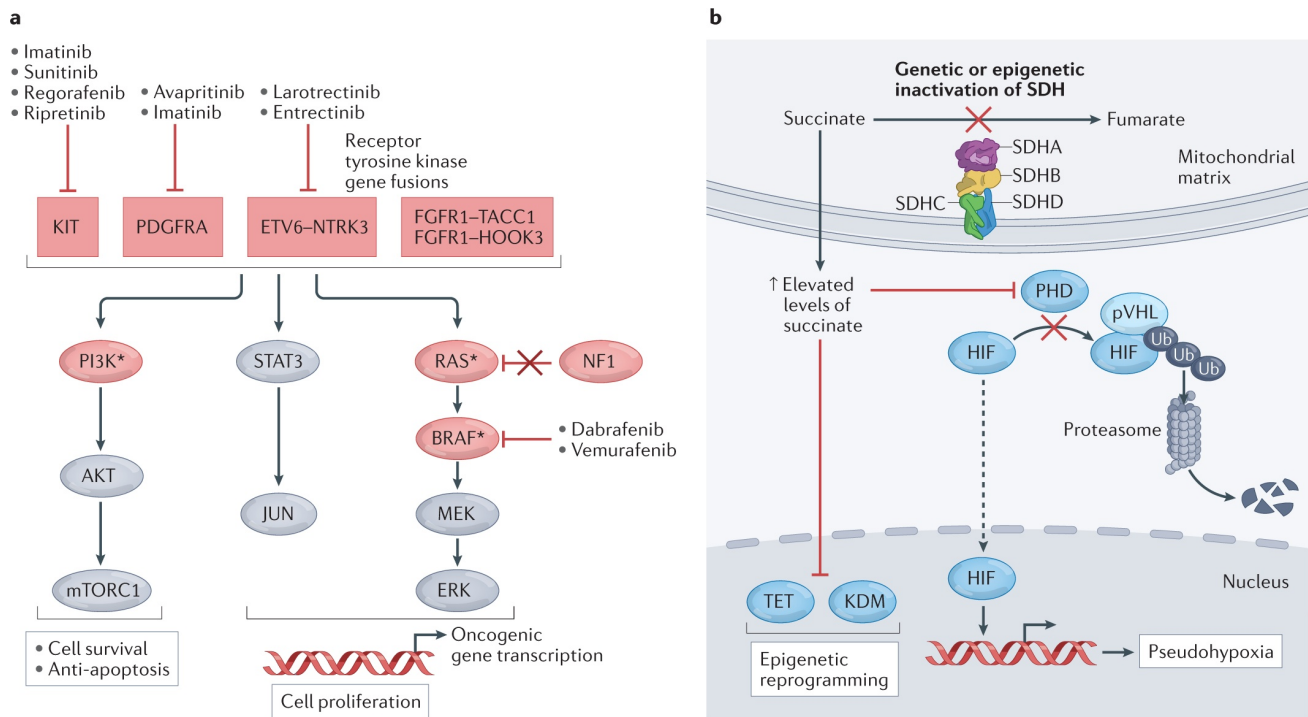
Immunohistochemical Analysis (IHC) and Genetic Characteristics of Tumors From 95 Patients With KIT/PDGFRα Wild-Type Gastrointestinal Stromal Tumors

JAMA Oncol 1. júla 2016



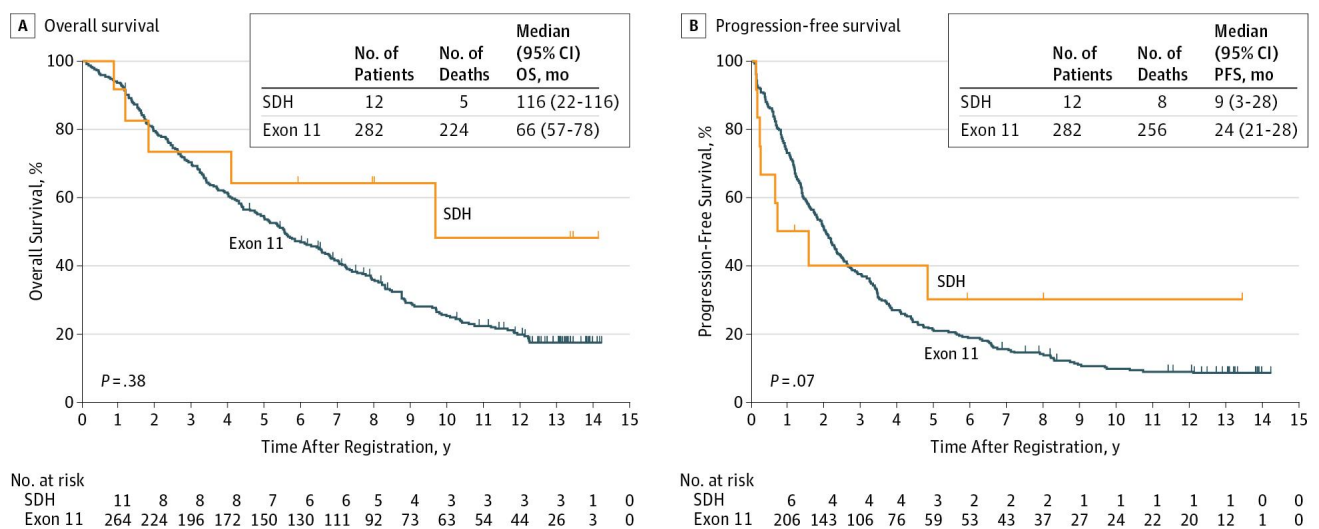
Driver mutations of molecular GIST subtypes.

Nature Reviews 18. marca 2021



GIST signalling pathways, drug targets and current systemic therapies.

Nature Reviews 1. mája 2022



Survival of Patients With KIT Exon 11 vs Succinate Dehydrogenase (SDH) Mutations

JAMA Oncol 1. júla 2017

descriptions Broad validation and refinement by later cohorts, multi-omic/NGS studies, and longitudinal SDH-deficient follow-up incorporated into guideline-based testing and genetic counseling, with SDH-deficient systemic therapy still unresolved

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12. Timing of liver-directed treatment

Your question

the PET-CT was done on June 19. Foundation One which showed 32% ctDNA fraction was done 20 August. Today is 5th September. What is realistically as window for surgical (what you proposed - e.g. ablation) treatment of the liver?

OpenEvidence response

This is fundamentally a question about how fast oligoprogressive, KIT-mutant GIST can outrun a stale scan - the 32% ctDNA is the warning flag. Let me pull the evidence on local-therapy timing in progressing GIST, ctDNA kinetics, and how current imaging must be before ablation planning.

There is no fixed biological deadline, but the practical window is short - realistically the next 2-4 weeks to intervene, and the first and rate-limiting step is that her imaging is already too old to plan an ablation. The June 19 PET/CT is now ~11 weeks (78 days) old, and the 32% ctDNA fraction drawn August 20 is a warning that biologically active disease may already exceed what that scan showed. The "window" here is not a countdown to a hard endpoint; it is the interval during which her disease remains **limited/oligoprogressive** (a few reactivating mural nodules) rather than **multifocal** - because local therapy only helps in the former, and the transition can happen over weeks to a few months on a background of emerging polyclonal resistance. ^[1, 2]

Why fresh imaging is the gating step

Percutaneous or intraoperative liver ablation is planned off a **multiphase contrast CT or MRI no older than ~30 days** - this is the standard used in ablation trials, because target number, size, segmental location, and proximity to critical structures must reflect current anatomy. ^[3] Her June 19 study fails that threshold by a wide margin. For small, partially active mural nodules within cystic/dormant masses, **contrast-enhanced liver MRI is the more sensitive modality**, with FDG-PET/CT to confirm which nodules are metabolically active and to exclude interval multifocal or extrahepatic spread that would change the plan. ^[1] Practically: this imaging should be obtained now, not after further watching.

Why the 32% ctDNA argues for urgency rather than watchful waiting

ctDNA tumor fraction correlates with tumor volume and active disease in GIST, and detectable/high ctDNA is characteristically seen in **metastatic, progressive** disease - not in stable or low-burden states. ^[4, 5, 6, 7] A tumor fraction of 32% is high (only ~30% of GIST liquid biopsies show any elevation $\geq 1\%$), and it is discordant with an imaging impression of just "a few" small reactivating nodules. ^[8] That discordance is the concerning part: it raises the possibility that more of her disease has become biologically active than the aging scan depicts, which is precisely the scenario in which the oligoprogression window closes fastest. It does not by itself contraindicate local therapy, but it tilts strongly toward moving quickly and toward confirming on fresh imaging that the active disease is still anatomically limited. ^[7]

The window is defined by "limited" vs. "multifocal," and acting early matters

Retrospective data consistently show that intervening while progression is still **unifocal** yields far better outcomes than once it becomes **multifocal**. In the 400-operation Fairweather two-institution series, median post-surgical PFS was **11 months for unifocal progressive disease versus 6 months for multifocal**, and OS **59 versus 24 months** - with multifocal progression an independent predictor of worse survival. ^[2] The classic DeMatteo data showed the same cliff (12-month vs. 3-month post-surgical PFS). ^[9] This is the quantitative basis for treating the window as finite: every additional week of waiting risks crossing from the favorable to the unfavorable category. Proceeding before a switch to second-line TKI is also supported - resection/ablation of the focally resistant clone plus continued imatinib outperformed dose escalation alone (median time-to-failure 24 vs. 6.5 months). ^[10, 11]

Her resistance profile strengthens the case for acting locally now

Because she carries **both V654A (ATP-pocket) and A829P (activation-loop)** secondary mutations, no single approved second-line TKI covers both well - in the INTRIGUE ctDNA analysis, patients with co-occurring ATP-pocket + activation-loop mutations showed **no difference between sunitinib and ripretinib**. ^[12, 13] That makes physically eliminating the reactivating clones while preserving imatinib for the still-controlled bulk disease a particularly rational strategy, rather than gambling on a systemic switch that is unlikely to cover both mutations. ^[14]

Practical timeline

- **This week (days, not weeks):** contrast liver MRI $\pm$ FDG-PET/CT to re-map targets and confirm the disease is still limited; baseline labs including coagulation (ablation contraindicated only by uncorrectable coagulopathy or active local infection). ^[1, 3]
- **Within ~1-2 weeks:** multidisciplinary tumor board review - NCCN v1.2026 now states MDT discussion *should* be performed to select modality (thermal ablation vs. resection vs. embolization) based on nodule size/location. ^[1]
- **Within ~2-4 weeks of the new imaging:** perform the local therapy while it is still current. For a few small nodules, thermal ablation (microwave/RFA) is feasible and safe after TKI; resection may give longer PFS if lesions are surgically favorable (median 42.7 vs. 14.3 months in one comparison). ^[1, 15]
- **Throughout:** continue imatinib 400 mg - hold only briefly peri-procedurally and restart as soon as oral intake resumes; discontinuing TKI even during progression accelerates the pace of disease. ^[1]

The bottom line: the limiting factor is not a biological clock you can measure but the combination of a stale scan and a high ctDNA signal, both of which say *re-image immediately and treat promptly*. A reasonable target is definitive local therapy within roughly a month, contingent on fresh imaging confirming the disease is still oligoprogressive; if that imaging instead shows multifocal reactivation, the plan shifts away from ablation toward a systemic-therapy switch. ^[1, 2]

Would it be helpful to review how to sequence a peri-procedural imatinib hold (exactly when to pause and resume around ablation or resection) to minimize both bleeding risk and any flare in disease activity?

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Conversation extension

Added 5 September 2026

New questions and responses, appended after page 78. The preceding archive remains unchanged. [Open the updated conversation.](#)

13. Personalized mRNA treatment and TIL therapy

14. Other therapies, combinations and liver-directed approaches

13. Personalized mRNA treatment and TIL therapy

Your question

what is the data regarding custom mRNA treatment or Tumor-infiltrating lymphocytes (TILs) therapy in her case?

OpenEvidence response

Both modalities hinge on the same limiting factor - how many targetable tumor-specific antigens a low-TMB (3 Muts/Mb), MSS GIST can actually present. Let me pull the specific evidence on mRNA neoantigen vaccines and TIL therapy in GIST and comparable sarcomas.

Neither a custom mRNA neoantigen vaccine nor TIL therapy has any published clinical data specifically in GIST, and her tumor biology places her in the least favorable category for both - but the honest, non-dismissive answer is that low mutational burden is no longer an absolute disqualifier, and the concept has now worked in tumors as immunologically "cold" and low-TMB as hers. The problem for her is less the modality itself than the *setting* (advanced, actively progressing disease rather than minimal residual disease) and the *magnitude* of realistic benefit. Below is the candid appraisal of each.

The shared limiting factor: her tumor makes very few targets

Her blood TMB of 3 Muts/Mb and MSS status are typical for metastatic KIT-mutant GIST - GIST has among the lowest mutational burdens of any human cancer (median ~2 Muts/Mb; median coding mutation rate ~0.67/Mb), the lowest of all sarcoma subtypes. ^[1, 2] Both mRNA vaccines and TIL therapy ultimately depend on the tumor presenting enough high-quality neoantigens for T cells to recognize. With perhaps only ~20-60 total coding mutations tumor-wide, the pool of viable neoantigen candidates may fall below what current vaccine platforms require (autogene cevumeran needs ≥5 predicted neoantigens; most eligible patients had ≥20, but those were higher-TMB tumor types). ^[3]

The important counterpoint - and the reason not to dismiss the idea outright - is that **the field has moved from "TMB threshold" to "neoantigen quality."** In pancreatic cancer (TMB ~1-2/Mb, "cold," immune-excluded), the autogene cevumeran mRNA vaccine induced durable neoantigen-specific CD8+ T cells in 50% of patients, and responders had markedly delayed recurrence (median RFS not reached vs 13.4 months); responders and non-responders did *not* differ in mutation count, only in neoantigen quality and clonality. ^[4, 4] Renal cell carcinoma (TMB ~1.1/Mb) showed 100% immunogenicity with a personalized vaccine targeting driver mutations, and none of 9 patients had recurred at 40 months. ^[5] So low TMB per se does not preclude a response when a few high-quality, clonal/driver neoantigens are targeted. ^[6, 7]

Custom mRNA neoantigen vaccine

The realistic assessment for her:

- **No GIST data exist.** The autogene cevumeran phase 1 program included soft-tissue sarcoma in its dose-escalation cohort but reported no sarcoma-specific efficacy, and GIST was not an eligible tumor type; the active phase 2/3 programs are in melanoma, PDAC, and NSCLC. mRNA-4157/V940 is in melanoma and NSCLC. [3, 7]

- **Setting is the biggest problem.** Vaccines have produced their clearest benefit in the *adjuvant/minimal-residual-disease* setting after resection, before extensive immunoediting. In advanced metastatic disease with a ctDNA fraction of 32% (i.e., substantial active tumor burden), vaccine monotherapy activity has been consistently disappointing. Her disease is progressing now, which is precisely the scenario where a 9-week manufacturing timeline and slow immune priming are least suited. [8, 9, 10]

- **A driver/shared-antigen approach is conceptually the most rational** for her. Off-the-shelf vaccines against recurrent oncogenic drivers have shown genuine signals in low-TMB tumors (mutant-KRAS in PDAC; a recurrent fusion in fibrolamellar HCC, where 8/12 mounted T-cell responses and 3 had partial responses with ICI). Her KIT exon 11 mutation and secondary V654A/A829P are tumor-specific and could in theory be vaccine antigens - and one GIST study did computationally predict HLA-binding neopeptides from a recurrent KIT mutation - but that work was purely *in silico*, involved a different KIT variant, and has never been immunogenicity-validated or tested clinically. This remains a hypothesis, not an option. [7, 11]

Bottom line on mRNA: not realistically available or evidence-supported for her today. If pursued at all, it would be experimental, best combined with checkpoint blockade, and ideally in a lower-burden state (e.g., after local therapy debulks the active liver clones) rather than during active progression.

TIL therapy

TIL therapy has the more encouraging proof-of-concept for low-TMB disease, though again none of it is in GIST:

- **GIST is not uniformly "cold."** In the 1,427-case multi-omic series, 12.9% of GISTs were T-cell-inflamed, similar across KIT-mutant subgroups, and CD3+/CD8+ TIL density is independently prognostic. KIT exon 11 tumors specifically may have relatively favorable immune features (imatinib-enhanced NK/Treg ratio, tertiary lymphoid structure formation). So there are TILs to harvest in principle. [1, 12, 13, 14]

- **TIL expansion from sarcoma is feasible** (>90% success in several STS series, with tumor-reactive T cells in ~40-56%), but **no one has published TIL expansion, reactivity, or infusion data specifically from GIST tissue** - it is genuinely unknown whether functional, tumor-reactive TILs can be grown from her tumor. [15, 16]

- **The most relevant proof-of-concept is the NCI phase 2 trial in low-TMB, mismatch-repair-proficient GI cancers** (NCT01174121). Bulk unselected TILs produced *zero* responses; but TILs *selected for neoantigen reactivity* plus pembrolizumab achieved a 23.5% objective response rate, with regressions in liver, lung, and bone - in exactly the kind of low-TMB, MSS tumors that "shouldn't" respond. Response correlated with recognizing more neoantigens and with CD4+ neoantigen-reactive TILs. [17]

The imaging from that trial illustrates that meaningful regression is achievable in low-TMB GI tumors when TILs are neoantigen-selected and combined with PD-1 blockade:

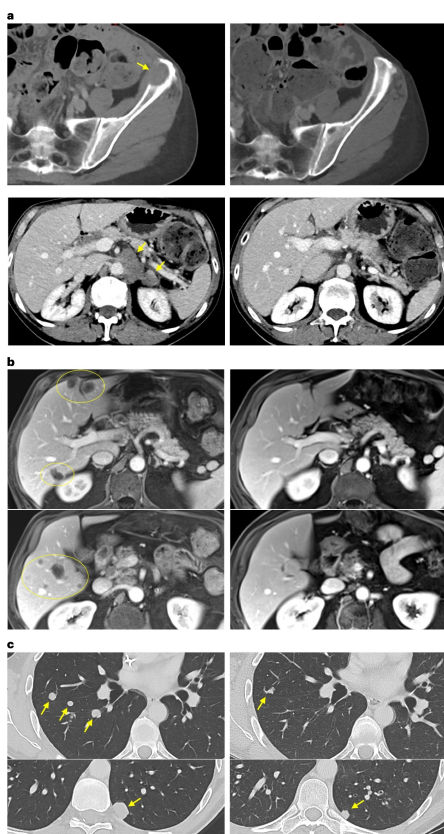


Figure 3. Evidence of tumor regression at different metastatic sites.

Neoantigen-specific tumor-infiltrating lymphocytes in gastrointestinal cancers: a phase 2 trial. Nat Med. 1. júna 2025.

Used under license from Springer Nature

The caveats that apply directly to her: this was GI *epithelial* cancer, not GIST; unselected TILs failed completely; and the imatinib-resistant GIST microenvironment is actively immunosuppressive (post-imatinib/anti-angiogenic exposure upregulates PD-1, TIM-3, LAG3, TIGIT and M2 macrophages), which could blunt infused TILs even if expansion succeeds. ^[18]

Bottom line on TIL: biologically plausible and the low-TMB barrier has been broken in analogous tumors, but there is no GIST-specific evidence, and it is only accessible through a trial. A dedicated autologous TIL + IL-2 study in GIST (NCT07647068) was among the trials in the dashboard shown earlier and is the single most direct route to this modality for her.

Professional opinion

Being direct rather than reassuring: **for a patient with active hepatic progression, a 32% ctDNA fraction, and only ~3 Muts/Mb, neither a custom mRNA vaccine nor TIL therapy is a realistic near-term answer, and both are weaker bets than the local-therapy-plus-continued-imatinib strategy already discussed.** The strongest immune signals in GIST specifically have come not from these modalities but from **ilixadencel** (intratumoral dendritic cells; 2/6 with SD/PR, both KIT exon 11 tumors) and **imatinib + interferon- α** , and even those are tiny studies. ^[19, 20] Checkpoint inhibitors alone in TKI-refractory GIST give a pooled ORR of only ~16%. ^[21]

If immunotherapy is to be pursued, the most rational sequence would be: (1) exploit the local-therapy window now to reduce the active tumor burden and lower ctDNA - which is also the state in which any vaccine or cell therapy performs best; then (2) enroll in a GIST-specific cell-therapy or immune trial (the TIL study, or a KIT-inhibitor-plus-immunotherapy combination), since combination with effective KIT inhibition is the recurring lesson from every positive GIST immune dataset. ^[2, 22] A vaccine or TIL product used *during*

uncontrolled progression, on this mutational background, is unlikely to change her trajectory.

Would it be useful to look specifically at the eligibility criteria and required tumor-tissue/biopsy logistics for the GIST TIL trial (NCT07647068), so you can judge whether harvesting viable TILs from her liver disease is even feasible in her timeframe?

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14. Other therapies, combinations and liver-directed approaches

Your question

gene therapy, arginine-depriving therapy, mitochondria-targeted therapies, neutrophil-targeted therapies, HSP90 Inhibitors, Chemo Immuno Precision Injections, Gleevec (imatinib) and Mektovi (binimetinib) and other Combination Therapies, combination of imatinib with either a MEK inhibitor or a PI3K inhibitor, Histotripsy, Transarterial Chemoembolization, Smart Nanoparticles, etc

OpenEvidence response

This is a request to appraise a long list of experimental and locoregional modalities against her specific biology - KIT exon 11 with secondary V654A + A829P, MSS, TMB 3, 32% ctDNA, oligoprogressive liver disease on imatinib. Let me research the GIST-relevant systemic combinations, the liver-directed options, and the metabolic/nanoparticle/gene-therapy frontier in parallel.

Being direct, as requested: of everything on this list, only the liver-directed therapies (TACE/TAE/TARE) are realistically actionable for her now, and none of the novel systemic or "frontier" modalities - gene therapy, arginine depletion, mitochondrial/neutrophil-targeting, smart nanoparticles, intratumoral chemo-immunotherapy - has clinical-stage evidence in GIST that would justify pursuing it over the local-therapy-plus-imatinib plan already discussed. Two items on your list (imatinib+Mektovi, imatinib+PI3K inhibitor) have actually been tested in her exact situation and *failed*. Below is a candid tier-by-tier appraisal, because the distinction between "biologically interesting" and "ready for this patient in September 2026" is the whole point.

The two combinations you named specifically have already been tested - and don't work in her setting

This is the most important correction, because Gleevec + Mektovi sounds rational but the trial data are unambiguous.

Imatinib + binimetinib (Mektovi). The biological rationale is real - MEK inhibition destabilizes ETV1, a GIST master regulator, and the combination caused complete regressions in mouse models. ^[1] But the clinical result splits sharply by setting. In **treatment-naïve** GIST the phase II trial was excellent (ORR 69%, median PFS 29.9 months). ^[2] In **imatinib-resistant KIT-mutant** GIST - her situation - the phase Ib trial showed **limited clinical activity**. ^[3] The reason is mechanistic and applies to every "imatinib + downstream inhibitor" idea: once KIT is imatinib-resistant, imatinib no longer suppresses KIT, so the combination collapses into MEK-inhibitor monotherapy against a fully active kinase. ^[4] Accordingly, NCCN v1.2026 lists binimetinib/imatinib only for **SDH-deficient** GIST (category 2B) - and her tumor is KIT-mutant, SDH-proficient, so even the guideline niche doesn't apply. ^[5]

Imatinib + PI3K/AKT/mTOR inhibitor. Same story, worse numbers. PI3K is the dominant KIT effector in GIST, and every combination has been tried: buparlisib (ORR 0%), alpelisib (ORR 2.9%, PFS 2.0 months), everolimus (ORR 0-2%, PFS 1.9-3.5 months), perifosine (ORR 0%). ^[6, 7, 8, 9] Uniformly disappointing, for the identical reason - the downstream drug becomes monotherapy against unopposed KIT. ^[4] Everolimus/TKI survives on the NCCN list only as "useful in certain circumstances" late-line, not as an active regimen. ^[5]

The general lesson: combining a drug she is *resistant to* with a pathway inhibitor does not rescue it. A downstream inhibitor would only make sense paired with a KIT inhibitor that still works against her V654A/A829P clones - and that pairing exists only in preclinical models (ripretinib + MEK inhibitor). ^[10]

Tier 1 - Actually available and appropriate now: liver-directed therapy

For her oligoprogressive, largely cystic liver disease with a few reactivating mural nodules, catheter-directed liver therapies are guideline-endorsed and have genuine GIST-specific data. NCCN v1.2026 explicitly endorses embolization, chemoembolization, and radioembolization for hepatic GIST progressing on TKI, continuing imatinib throughout. [5]

Modality	Best GIST-specific data	Median liver PFS	Notes	References
Thermal ablation (RFA/MWA)	Local recurrence 5-7%; median OS 90-100 mo in ablation series	14-28 mo	Best for a few small nodules <3 cm - matches her pattern	[11, 12]
DEB-TACE	ORR 94% (mRECIST), DCR 100% vs 4.7% control	11 mo vs 5 mo	For more diffuse/multifocal liver disease	[13]
Conventional TACE	PR 14-55%, DCR 77-88%	7-11 mo	Larger evidence base	[14, 15]
TARE (Y-90)	CR+PR ~89%; no PD in one series	15.9-16 mo	Worst H-PFS (7 mo) in KIT exon 11 + secondary mutations - <i>her genotype</i>	[16, 17]
Bland embolization (TAE)	60% radiographic response; GIST median OS 36 mo	~7 mo	Reasonable where chemo/radioembolization unavailable	[18, 19]
Modality	Status in GIST	Honest bottom line	References	
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Gene therapy (siRNA/shRNA KIT knockdown, exon-skipping, miRNA, CRISPR)	Preclinical only; adenoviral KIT-shRNA gave ~60% xenograft inhibition	Strong rationale (KIT-knockdown kills resistant GIST cells), but <i>delivery to all tumor cells in vivo is unsolved</i> ; no human trials	[20, 21, 22]	
Anti-KIT CAR-T	Preclinical only	Lyses GIST cells in mice, but on-target/off-tumor toxicity (KIT on stem cells, ICCs, melanocytes) is a serious barrier; no GIST trial	[23, 24]	
Smart nanoparticles (aptamer-drug conjugates, PLGA, lipid carriers, KIT-d-MMAE)	Preclinical only; KIT-d-MMAE effective in imatinib-resistant PDX	Creative platforms, none in clinical trials for GIST; manufacturing/EPR translation barriers	[25, 26]	

Modality	Best GIST-specific data	Median liver PFS	Notes	References
Arginine depletion (ADI-PEG20)	No GIST data; ASS1 status of GIST unknown	Would require confirming ASS1 deficiency on <i>her</i> tumor first; no GIST trial exists; GIST metabolic work has focused on glutamine, not arginine	[27, 28]	
Mitochondria-targeted (other than OXPHOS+imatinib)	Metabolic SDH-targeted drugs apply only to SDH-deficient GIST	Not applicable - she is SDH-proficient	[29, 30]	
Neutrophil-targeted therapy	Neutrophils are a <i>minor</i> component of the GIST microenvironment	Biologically the wrong target in GIST (dominated by M2 macrophages/Tregs); NLR is prognostic only, not a drug target	[31, 32, 33]	
Intratumoral chemo-immunotherapy "precision injections"	Only ilixadencel (intratumoral dendritic cells) tested: phase I, n=6, 2 responders (both KIT exon 11)	Tiny signal; intratumoral <i>chemo</i> has no rationale (GIST is chemoresistant, 0-15% response)	[34]	

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Conversation extension

Added 5 September 2026

New questions and responses, appended after page 86. The preceding archive remains unchanged. [Open the updated conversation.](#)

[15. Imatinib plus peginterferon: study setting and follow-up](#)

[16. Reports of TKI combinations with immunotherapy or antibodies](#)

15. Imatinib plus peginterferon: study setting and follow-up

Your question

can you elaborate on Imatinib + peginterferon-α2b?

was it 2nd stage, imatinib-resistant GIST? what happened after?

OpenEvidence response

Whether this was a treatment-naïve or imatinib-resistant population is the crux - let me pull the specific trial details and long-term follow-up on imatinib plus (peg)interferon-α in GIST.

The imatinib + peginterferon-α2b data come from a single small pilot/phase II study, and - importantly for your patient - it was done in *treatment-naïve* advanced GIST, not imatinib-resistant disease. This is the Chen LL et al. trial from MD Anderson (Benjamin group), and the distinction matters because the entire mechanistic rationale depended on imatinib *still working*.

What the study actually was

It was a small prospective pilot/phase II trial in **stage III/IV GIST**, with an interim analysis reported on **8 patients**. Imatinib and peginterferon-α2b were given **concurrently as the treatment regimen** - not sequenced, and not added after imatinib failure. ^[1] The design logic was immunological: imatinib's effective tumor killing supplies tumor antigens in vivo and undermines tumor-induced immune tolerance *without causing leukopenia*, while peginterferon-α2b acts as a "danger signal" driving dendritic-cell and cytotoxic-T-cell maturation toward a Th1 response. ^[1] So the combination was built on the assumption of a KIT-sensitive tumor that imatinib debulks - the opposite of your patient's current imatinib-resistant, V654A/A829P state.

What happened - results and follow-up

The reported outcomes were striking on their face:

- **Immunologic:** significant induction of IFN-γ-producing CD8+, CD4+, and NK cells, plus IFN-γ-producing tumor-infiltrating lymphocytes in blood and tumor beds - i.e., a genuine Th1/NK activation signal. ^[1, 2]
- **Clinical:** at a median follow-up of **3.6 years**, CR + PR = 100%, overall survival 100%; one patient died of an unrelated illness while in remission, and **six of seven evaluable patients** were in continuing PR/CR (five patients) or had a PFS (one patient) exceeding the upper 95% CI of genotype-specific PFS from the phase III imatinib-monotherapy trials (CALGB150105/SWOG S0033). ^[1]

What happened *after* - and why you've never seen it in guidelines

This is the key part of your question. The authors explicitly stated the trial was **"closed in preparation for a larger future trial."** That larger confirmatory trial never delivered practice-changing data - the combination was never validated in a randomized study, and it does ^[1] **not appear anywhere in the NCCN v1.2026 GIST regimens** (imatinib+IFN is not listed in first-line, later-line, or "useful in certain circumstances"). ^[3]

Several factors explain the stall, and they are candidly unfavorable:

- **Tiny, uncontrolled sample.** Eight patients with no randomized comparator; the "100% response" is hypothesis-generating, not confirmatory, and highly susceptible to selection. ^[1]
- **Interferon toxicity.** Across the analogous CML combination experience (SPIRIT, CML-study IV), imatinib+IFNα consistently produced **discontinuation rates >20%** driven by interferon-related adverse events, and in CML the combination improved molecular response but **did not improve overall survival.** That toxicity/benefit profile is the main reason the cytokine-plus-TKI concept faded. ^[2]
- **The field moved elsewhere** - to newer KIT inhibitors (ripretinib, avapritinib, bezucastinib) and to checkpoint/cell-based immunotherapy - rather than to interferon combinations. ^[4, 5]

Relevance to your patient - direct assessment

Being straightforward: **this regimen's encouraging data do not transfer to her situation.** The trial treated imatinib-*sensitive*, treatment-naïve tumors, where imatinib was actively killing cells and thereby feeding the immune response. Your patient has **imatinib-resistant disease** (V654A + A829P, 32% ctDNA, radiographic reactivation) - precisely the setting where imatinib no longer debulks effectively, so the "imatinib supplies antigen + relieves tolerance" mechanism largely collapses, exactly as it does for the imatinib + MEK/PI3K combinations discussed earlier. There is **no published imatinib + peginterferon-α2b data in imatinib-resistant GIST at all.** If interferon-based immune stimulation were to be revisited for her, it would only be rational paired with a KIT inhibitor that still suppresses her resistant clones, and it would be purely experimental.

Would it be helpful to look at whether any *currently active* trials are combining a next-generation KIT inhibitor (e.g., bezucastinib or ripretinib) with an immune-activating agent, which is the modern successor to this imatinib+interferon concept?

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16. Reports of TKI combinations with immunotherapy or antibodies

Your question

any other such reports with successful combination of TKI with immunotherapy/antibodies/etc for her case specifically?

OpenEvidence response

Let me pull the specific clinical trial data on TKI-plus-immunotherapy and TKI-plus-antibody combinations in GIST, focusing on what has actually been tested in the imatinib-resistant/refractory setting rather than treatment-naïve disease.

The honest bottom line up front: **there is no published report of a TKI + immunotherapy or TKI + antibody combination producing a clear success specifically in an imatinib-resistant KIT exon 11 tumor like hers - but two datasets contain the closest thing to a positive signal, and both happen to involve KIT exon 11 patients.** Every relevant trial has been done in the TKI-refractory setting (like her), and the recurring, mechanistically important lesson is that these combinations only work when the TKI still suppresses KIT - which is precisely the vulnerability created by her V654A/A829P resistance mutations. Below is the full landscape, ordered by how relevant each is to her.

The two closest-to-positive signals for her genotype

AXAGIST - avelumab (anti-PD-L1) + axitinib (VEGFR/KIT TKI). This 2025 phase II trial (n=56 evaluable) is the largest TKI + immunotherapy combination ever run in GIST and is the single most relevant dataset to your question. It enrolled pretreated GIST (53% had ≥3 prior TKI lines). Overall numbers were modest - ORR 8.9% (5 PRs), median PFS 4.6 months, median OS 14.2 months - **but the 5 responders had a median duration of response of 18.5 months**, i.e. a small subset achieved genuinely durable benefit, and the authors concluded the approach was "promising in a subset." Grade ≥3 AEs occurred in 30.4%. ^[1]

IMPRESARC - regorafenib + ICI (retrospective). In this French multicenter cohort, GIST was one of the *better*-responding histologies to checkpoint-based therapy: **46% of GIST patients had PFS >6 months and 25% exceeded 12 months.** Critically for her, the responders were mostly **KIT exon 11-mutant** (the rest KIT/PDGFR wild-type), all received ICI *in combination* with another agent, and 11 received regorafenib + ICI - exceeding regorafenib's expected 4.8-month third-line PFS. The single longest responder was on pembrolizumab + cyclophosphamide. ^[2] This is retrospective and uncontrolled, but it is the only signal suggesting her exact genotype (KIT exon 11) may enrich for immunotherapy benefit.

The combinations that were formally negative

Trial	Regimen	Setting	Result	References
PDR001 + imatinib (Ph Ib/II, n=36)	anti-PD-1 + imatinib	After imatinib+sunitinib +regorafenib	ORR 0% , 12-wk DCR 37.9% (endpoint not met), PFS 2.3 mo, OS 9.5 mo	[3]
Dasatinib + ipilimumab (Ph Ib, n=20 GIST)	KIT TKI + anti-CTLA-4	Median 3 prior lines	ORR 0% (RECIST); 54% PR by Choi; PFS 2.8 mo; "not synergistic"	[4]

Trial	Regimen	Setting	Result	References
Nivolumab + ipilimumab (Ph II, n=16)	anti-PD-1 + anti-CTLA-4 (no TKI)	Refractory	ORR 6.7% (1 CR), PFS 8.3 wk	[5]
Alliance A091401 GIST cohort (n=18)	nivo ± ipi (no TKI)	Heavily pretreated	ORR 0% both arms, PFS 1.5-2.9 mo	[6]

The pooled picture across 9 immunotherapy studies (169 GIST patients) is an ORR of ~13-16%, CBR ~42-46%, median PFS ~6 months. [7] Modest, with occasional durable outliers - not a reliable response.

Why these combinations underperform in *her specific situation*

This is the crux, and it is not a coincidence. Preclinical GIST models show anti-PD-1/PD-L1 alone does nothing, but **adding it to imatinib markedly suppresses tumor growth** - because imatinib itself is the immune driver (it activates CD8+ T cells, inhibits IDO1, apoptoses Tregs, and lowers tumor PD-L1). [8, 9] The problem: imatinib's immunostimulatory effect is **greatest in imatinib-sensitive disease and diminished once resistance emerges**. The dasatinib+ipilimumab failure was explicitly attributed to dasatinib being a poor immune-priming TKI in resistant disease. [4, 10]

Her tumor carries KIT exon 11 *plus* secondary V654A (exon 13) and A829P (exon 18) - she is imatinib-resistant. So the prerequisite for TKI+ICI synergy (effective ongoing KIT inhibition) is exactly what she lacks. The leading GIST group states this directly: these combinations "depend upon effective KIT/PDGFR α inhibition," argue for pairing immunotherapy with a *broad-spectrum* KIT inhibitor, and suggest doing so **earlier** in treatment rather than in the resistant setting. [11] Her MSS status and TMB of 3 Muts/Mb further place her outside the only NCCN-endorsed immunotherapy niche - ipilimumab+nivolumab is listed (category 2B) *only* for TMB-high GIST. [12]

Antibody-based approaches - all preclinical in GIST

To be complete, since you asked about antibodies specifically: none has clinical GIST data.

- **Anti-KIT monoclonal antibodies** (SR1, CK6, KITMAb): SR1 inhibited imatinib-resistant GIST xenografts, but the CK6 study - the one that actually tested combination - found **no synergy with imatinib or sunitinib**, sometimes inferior to TKI alone. [13, 14]
- **Antibody-drug conjugates**: the anti-GPR20 ADC **DS-6157a failed in phase I** (1 PR/34, program not advanced). Anti-KIT ADCs (LOP628, NN3201, SCF-DM1) have strong preclinical data - including complete remissions when combined with imatinib - but no published clinical results. [15, 16, 17, 18]
- **Anti-CD47, anti-CSF1R**: no GIST clinical data; CSF1R inhibition (pexidartinib) added nothing to KIT inhibition in GIST models. [19]

Professional opinion

None of these is ready to help her today, and pursuing an ICI/antibody combination *now*, on a resistant KIT background with MSS/low-TMB biology, is unlikely to work - the negative PDR001+imatinib and dasatinib+ipilimumab trials tested almost exactly this concept and returned 0% response. [3, 4] If immunotherapy is to be attempted at all, the evidence points to a specific rational construction rather than any off-the-shelf pairing: **pair the checkpoint inhibitor with a next-generation broad-spectrum KIT inhibitor that actually suppresses her V654A/A829P clones** (the logic the Heinrich group emphasizes), which is why the modern successor concept is bezuclastinib- or ripretinib-plus-immunotherapy rather than imatinib-plus-anything. [11, 20] The AXAGIST durable-responder subset and the IMPRESARC KIT exon 11 signal are the only threads suggesting a patient like her *can* benefit - both experimental, both requiring a trial. [1, 2]

Practically, this reinforces the earlier plan: use the local-therapy window to control the active liver disease, keep her on effective KIT suppression, and reserve immunotherapy for a trial that combines it with a KIT inhibitor covering her resistance mutations - not as standalone salvage.

Would it help to identify which currently recruiting trials are combining a next-generation KIT inhibitor (bezuclastinib or ripretinib) with a checkpoint inhibitor, since that specific pairing - not imatinib+ICI - is the construction the evidence actually supports for her resistance profile?

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